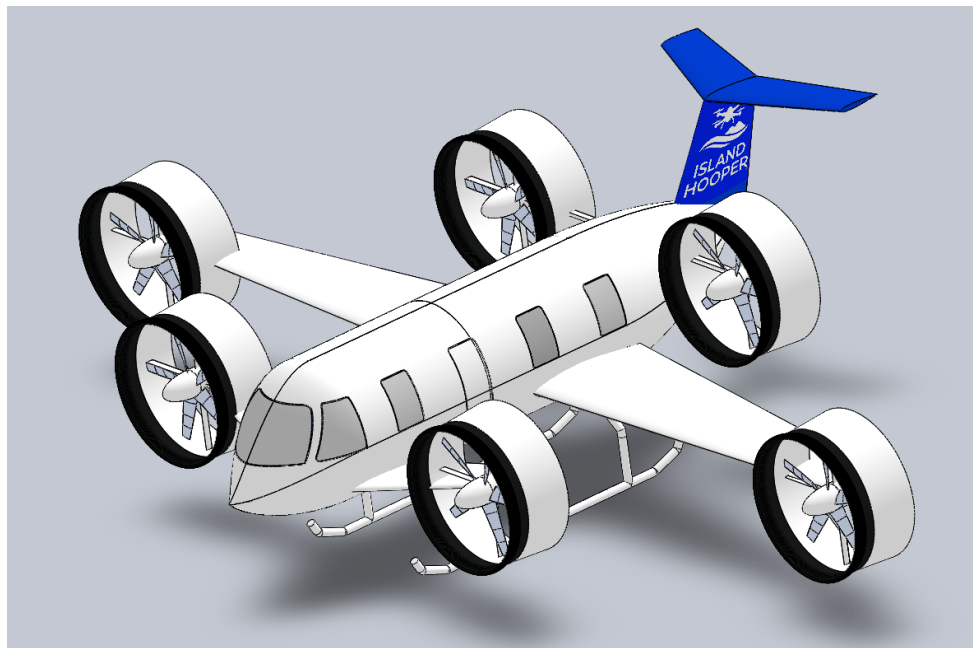


Conceptual Design of an Urban Air Mobility System for Inter-Island Transportation in the Balearic Archipelago



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Report elaborated for the Curricular Unit

Project in Aerospace Engineering

12 Jun, 2026

Executive Summary

This report presents the conceptual design of a possibly premium Urban Air Mobility (UAM) system for inter-island transportation in the Balearic Archipelago. The market study identifies a relevant opportunity supported by a resident population of approximately 1.26 million people and a seasonal tourism flow of about 19 million visitors in 2025. The proposed service targets passengers who value time, flexibility, comfort, and exclusivity, offering a faster and lower-emission alternative to conventional air, ferry, and helicopter connections.

The reference network uses existing aviation infrastructure and focuses on two routes: Ibiza-Palma de Mallorca, with a distance of approximately 140 km, and Palma de Mallorca-Menorca, with a distance of approximately 132 km. The mission is defined as a medium-range VTOL operation carrying one pilot and six passengers with baggage, corresponding to a preliminary payload of approximately 600 kg. The aircraft must therefore combine vertical take-off and landing capability, efficient cruise, over-water operational margins, low noise, and compatibility with the EASA small-category VTOL limits.

Four UAM architectures were benchmarked: Fixed-Wing VTOL, Tilt-Rotor, Tilt-Wing, and Tilt-Duct. An Analytic Hierarchy Process ranked these options against performance, emissions, comfort, cost, maintenance, and operational suitability. The Tilt-Duct configuration was selected because it offers a strong compromise between compact vertiport operation, enclosed-rotor safety, acoustic benefits, distributed electric propulsion, and efficient transition to cruise.

The resulting concept uses six tilting ducted rotors distributed across a canard-wing aircraft layout. Preliminary sizing with the Aircraft Design Tool and supporting calculations gives a maximum take-off mass of 2595 kg, remaining well below the regulatory limit. Aerodynamic sizing adopted a 13 m wingspan, aspect ratio of 9, and reference wing area of approximately 18.78 m², with the NACA 4412 selected as a robust airfoil for the disturbed flow and structural requirements of a tilt-duct aircraft. The fuselage was further developed with CAD and OpenVSP, giving an approximate length of 9.89 m and maximum diameter of 2.2 m.

The mission energy requirement is estimated at approximately 71.66 kWh, with 20.2 kWh required for the vertical phases, 21.6 kWh for climb, and 32.3 kWh for cruise. A solution relying entirely on battery was found to be highly sensitive to its specific energy, C-rate, and life cycle, with the optimistic Li-S case requiring about 329.89 kg of batteries. The retained solution is therefore a hybrid electric-fuel-cell architecture: a Li-S battery pack supplies VTOL, hover, and transient power, while a hydrogen PEMFC subsystem supplies cruise and climb energy. The preliminary hybrid system totals approximately 197.73 kg, representing a mass reduction of about 40.06% relative to the optimistic all-electric Li-S case.

Overall, the project supports the technical plausibility of a hybrid tilt-duct UAM aircraft for Balearic inter-island transport. The concept offers promising mission fit, lower direct operational emissions, improved range scalability, and a compact VTOL configuration. However, the design remains conceptual and requires further work in detailed rotor and duct design, CFD, structural validation, hydrogen integration, high-voltage safety, acoustic assessment, cost refinement, and certification planning.

List of Symbols

Symbol	Name / meaning
A	Conductor cross-sectional area; generic area
AR	Wing aspect ratio
b	Wing span
BPF	Blade pass frequency
c	Local chord; empirical exponent
$\bar{c}$	Mean aerodynamic chord
c_r, c_t	Root chord and tip chord
C_D, C_{D_0}	Drag coefficient and zero-lift drag coefficient
C_f	Skin-friction coefficient
C_L, C_l	Wing lift coefficient and airfoil lift coefficient
$C_{L,\max}$	Maximum wing and airfoil lift coefficients
C_M	Pitching moment coefficient
C_l	Rolling moment coefficient
C_n	Yawing moment coefficient
C_{M_α}	Pitching stability derivative
C_{l_β}	Rolling stability derivative
C_{n_β}	Yawing stability derivative
D	Drag force
d	Maximum fuselage diameter
E	Energy requirement or stored energy
F	Fuselage form factor
F_f	Fuselage drag force estimate
g	Gravitational acceleration
I	Electrical current
J	Allowable current density
k	Correction or loss factor
L	Lift force
$L_{\max}$	Maximum lift condition
l_f	Fuselage length
m	Aircraft or subsystem mass
M	Mach number

Symbol	Name / meaning
N	Number of blades or units
N_p, N_t	Production aircraft and test aircraft quantities
P	Power
q	Dynamic pressure
Q	Interference factor
R	Rotor radius
Re	Reynolds number
r/R	Non-dimensional rotor radial station
S	Reference wing area
S_{wet}	Wetted area
SM	Static margin
T	Temperature or thrust
t/c	Thickness-to-chord ratio
V	Flight speed or voltage
V_{cruise}	Cruise speed
V_{stall}	Stall speed
W	Aircraft weight
W_s	Structural weight in cost estimation
X_{CG}, X_{CT}	Center of gravity and center of thrust
X_{NP}	Neutral-point position
α	Angle of attack
β	Sideslip angle
γ	Flight-path angle
λ	Wing taper ratio
μ, ν	Dynamic viscosity and kinematic viscosity
ρ	Air density
σ	Rotor solidity

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1 Introduction

1.1 Motivation

This work is inserted in the Project in Aerospace Engineering of the respective Bachelor's Degree at Instituto Superior Técnico.

The Balearic archipelago offers a compelling opportunity for the study of Urban Air Mobility (UAM). Its geography creates recurrent transport needs between islands, while the strong seasonal tourism market generates periods of intense mobility demand. At the same time, the region faces increasing pressure to improve connectivity without amplifying noise, local emissions, and infrastructure congestion. In a period where aviation is increasingly required to transition toward cleaner and more efficient solutions, the Balearic case provides a relevant scenario for assessing whether a premium VTOL service can deliver meaningful emission reduction, time savings and operational flexibility.

The project is therefore not limited to the definition of a route or a vehicle concept. It combines market research, concept-of-operations definition, aircraft architecture selection, preliminary sizing, aerodynamic analysis, propulsion and energy assessment, CAD development, stability considerations, emissions, noise, and cost evaluation. This broader approach is necessary because the feasibility of an inter-island UAM aircraft depends on the interaction between mission requirements, aircraft mass, energy storage, vertical-flight power demand, infrastructure constraints, and certification-oriented safety considerations.

1.2 Objectives

The main objective of this report is to develop a coherent conceptual proposal for an inter-island UAM system tailored to the Balearic Islands. The project is guided by three main goals:

- To explore the electrification of aviation as a feasible pathway for short to medium-range regional mobility.
- To assess a cleaner alternative for short inter-island flights, where conventional jet-powered aircraft may be operationally excessive.
- To pursue a configuration with no direct in-flight CO₂ emissions, consistent with the long-term transition toward more sustainable aviation.

The report is therefore organized as a sequence of the work developed throughout the project, starting with the identification of the operational opportunity and progressing through concept selection, preliminary sizing, aerodynamic and propulsion analysis, CAD development, stability, emissions, noise, cost assessment, and final conclusions on the feasibility of the proposed goals.

2 Market Research: Balearic Islands

2.1 Population and Tourism

The Balearic Islands present a perfect environment for the introduction of Urban Air Mobility (UAM), namely the implementation of a luxury VTOL service. The archipelago is home to approximately 1.26 million people [9], already setting a decent baseline target demographic. However, the greatest attraction of the region is its massive tourism influx, which reached 19 million visitors in 2025 [10]. This volume represents a high concentration of high-net-worth individuals, especially in Ibiza and Mallorca.

It is important to note that this tourism is highly seasonal, being disproportionately larger from June to September. This characteristic could allow for a purely seasonal service model as a means of optimizing costs and streamlining maintenance operations during the off-season. Furthermore, aligning with the regional government's current values [11], the project should focus on a low-noise, zero-emission vehicle network to ensure long-term viability and social acceptance.

2.2 Distances and Routes

Following an analysis of the market's demographic and traffic patterns, both intra and inter-island, it was concluded that the optimal route is the link between Ibiza and Mallorca. Additionally, one linking to the island of Menorca is also desirable. To minimize costs, the proposed approach relies on utilizing existing aviation infrastructure. With an overview of private and public airports and helipads, coupled with an attempt to keep the two links close to each other to minimize the aircraft's range, the following route system was achieved (Figure fig. 2.1):

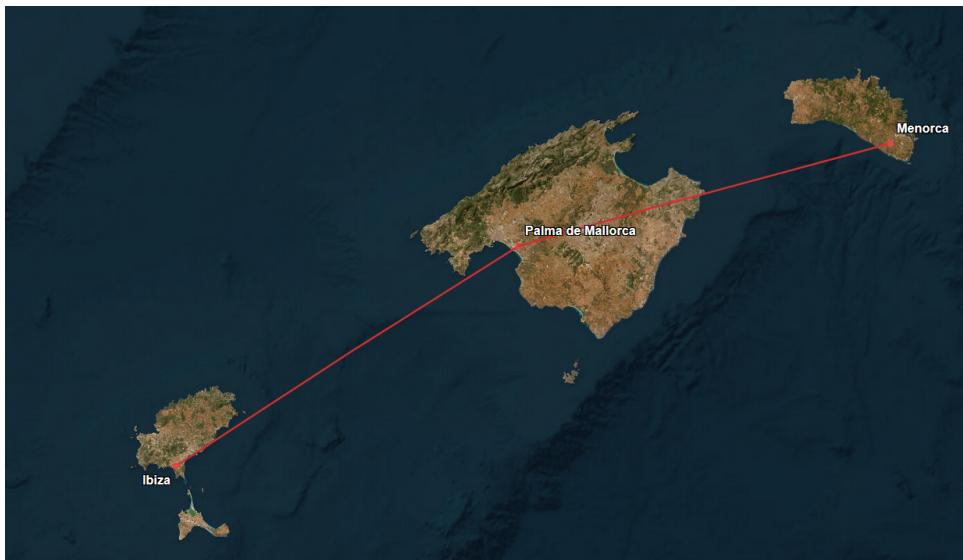


Figure 2.1: Proposed route system for the archipelago [1]

Route 1 (primary leg): Ibiza Airport (IBZ) to Palma de Mallorca Airport (PMI), distance: 140 km [1]

Route 2 (secondary leg): Palma de Mallorca Airport (PMI) to Menorca Airport (MAH), distance: 132 km [1]

2.3 Conditions

The conditions for flight in the region of the Balearic Islands are quasi-ideal.

The topography of the islands is exceptionally flat, with the main exception being the northern mountain range *Tramuntana*. Even so, this aspect does not interfere with the routes proposed for the project (Figure 2.2).

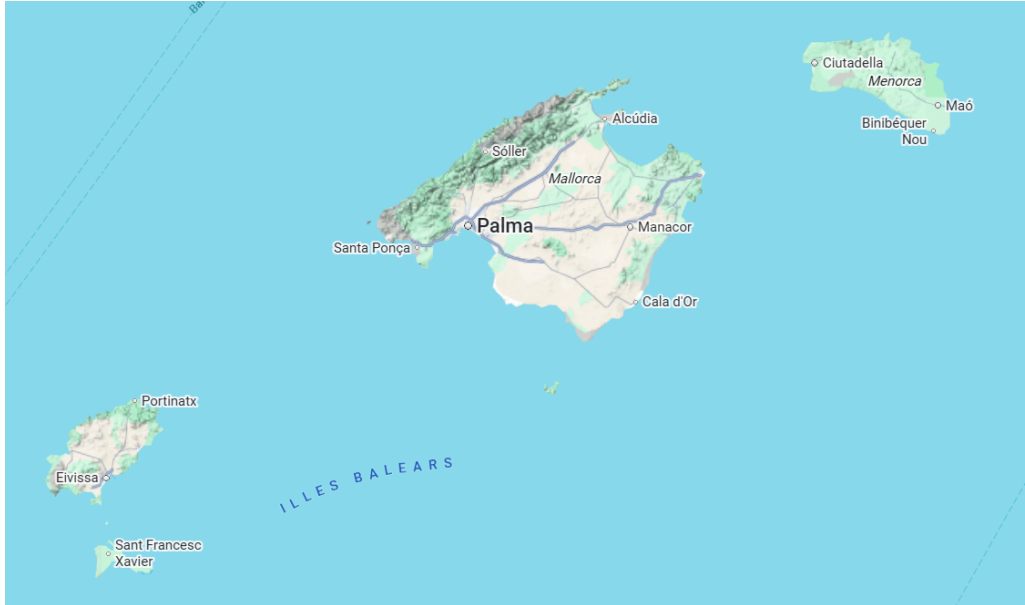


Figure 2.2: Topographic map of the Balearic Islands. Source: [2]

As exposed before, demand for this mode of transportation is expected to be strongly seasonal, so the highest concentration of flights would likely occur during the summer. Summer conditions in the region are highly favorable for flight, with average winds of 11.6 km/h, aside from occasional summer storms, and very limited cloud coverage.

The region is not heavily polluted in terms of air quality, so this transportation alternative would not dramatically improve that specific metric. Still, current transport methods have a significant environmental impact through both noise and maritime activity. The two main transport modes are ferries and conventional airplanes. Large ferries can significantly disturb the marine ecosystem, while conventional aircraft generate both noise and air pollution. A clean transportation system with reduced noise impact could help reduce the pressure created by the massive tourist influx on the Balearic ecosystem, both inland and offshore. It could also represent an important contribution to the European Union's objective of becoming a carbon-neutral continent by 2050.

2.4 Alternatives

The existing mobility alternatives to the proposed UAM service are conventional flights, helicopter flights and the most popular: ferries. A quick research was made to assess the generic estimatives for travel time, cost per passenger and emissions per passenger for the existing alternatives and a possible VTOL solution whose value will be assessed in more detail in Chapter 11.

Table 2.1: Comparison of transport modes.

Mode of transport	Travel time	Cost per passenger	Emissions per passenger
Airplane	35–40 min ($\approx$ 2–3 h incl. airport)	€45–140	$\approx$ 18–20 kg CO ₂
Boat (fast ferry)	1 h – 1 h 15 min	€70–95	$\approx$ 2–4 kg CO ₂
Helicopter	35–45 min (+10 min boarding)	€1,800–5,000	$\approx$ 27–85 kg CO ₂ + High noise
VTOL	35–40 min (+10 min boarding)	$\approx$ €300–1,000	$\approx$ 0 kg CO ₂ (if fully electric)

The VTOL solution posed as an option similar to the helicopter, though with a lower cost and emissions. It shows a significant reduction in travel time when compared to the airplane option, mainly due to the airport restriction, it also reduces the high cost in emissions per passenger. We see the fast ferry option as the best alternative, as it has a low travel time, low cost and low emission. With this in mind we emphasize that our option will lean, as a first approach, towards the premium or near luxury segment, accessing a market where the cost per passenger is not a huge restriction (as it seemed impossible to beat the ferry prices), and where the exclusivity is seen as a benefit. We must also bear in mind that the VTOL option is faster, and will not emit any exhaust gases, which is a strong point in favor of our transportation method.

3 Global UAM Solutions

3.1 Fixed Wing VTOL

The Fixed-Wing VTOL configuration, commonly referred to as the "Lift + Cruise" architecture, utilizes a decoupled propulsion strategy to separate the vertical and horizontal flight regimes. This design paradigm prioritizes mechanical simplicity and reliability by avoiding the complex actuating mechanisms required to physically rotate the propulsion units or the airframe.

3.1.1 Technical Characteristics

The defining hallmark of the Fixed-Wing VTOL architecture is the employment of two entirely independent sets of propulsion systems that remain statically fixed in their respective orientations. The vertical lift system typically consists of multiple electric rotors mounted on longitudinal booms or directly integrated into the wing structure, oriented exclusively to provide upward thrust. Conversely, horizontal thrust is generated by a separate, independent propulsion unit, usually a pusher or puller propeller mounted on the empennage or aft fuselage.

During the operational cycle, the vertical rotors are active exclusively during the takeoff, hover, and landing phases. Once the aircraft achieves a safe altitude, the cruise motor is engaged to provide forward acceleration. As the aircraft surpasses the stall speed of its fixed wings, aerodynamic lift fully supports the vehicle's weight. At this precise transition point, the vertical rotors are deactivated. To mitigate aerodynamic penalties, modern designs typically feature mechanisms that align and lock the stationary rotor blades in a low-drag configuration parallel to the freestream airflow. This transition relies entirely on software-driven thrust blending rather than mechanical articulation.

3.1.2 Analysis of Advantages and Disadvantages

The fundamental merit of the Lift + Cruise configuration lies in its exceptional mechanical simplicity and inherent safety. By eliminating the heavy and complex tilting hinges, gearboxes, and cross-shafting found in tilt-rotor or tilt-wing designs, this architecture significantly reduces maintenance requirements and simplifies the type certification process. The static nature of the lift rotors also facilitates a highly redundant Distributed Electric Propulsion (DEP) network, ensuring that the failure of a single lifting rotor does not critically compromise the hover capability.

However, these operational dividends are counterbalanced by significant aerodynamic and mass penalties during the cruise phase. The most profound drawback is the parasitic drag induced by the exposed vertical rotors and their supporting boom structures, even when the blades are locked in an optimal alignment. Furthermore, the vertical propulsion system becomes essentially "dead weight" during forward flight, as it contributes nothing to aerodynamic lift or horizontal thrust. This dual penalty of increased empty weight and elevated cruise drag invariably restricts the maximum range and payload capacity of the aircraft compared to fully convertible architectures. Consequently, while highly effective for shorter urban hops, its application for extended inter-island maritime routes requires careful optimization of the battery mass fraction.

3.1.3 Existing Implementations and Concept Designs

Notable implementations of the Fixed-Wing VTOL architecture include the Wisk Aero air taxi (Generation 6), the Beta Technologies ALIA-250, and the Archer Midnight. These platforms utilize dedicated vertical lifting rotors combined with an independent pusher propeller for cruise flight. Specifically, the Archer Midnight exemplifies this with its rear lifting rotors that lock into place during forward flight to reduce parasitic drag. Together, they serve as prime examples of prioritizing mechanical simplicity and certification viability in modern UAM design.

3.2 Tilt-Rotor

The tilt-rotor configuration represents a sophisticated hybrid class of aircraft designed to synthesize the VTOL capabilities of a helicopter with the high-speed, long-range cruise performance of a fixed-wing airplane. Indeed, this architectural paradigm is defined by its ability to reconfigure its propulsion system during flight, allowing it to navigate the distinct physical requirements of both urban hovering and regional transit cruise.

3.2.1 Technical Characteristics

The primary hallmark of the tilt-rotor is its dual-mode propulsion system. Unlike conventional rotorcraft, which are aerodynamically constrained by the physical limitations of retreating blade stall and high drag at speed, the tilt-rotor employs pivotable nacelles. During the hover phase, the proprotors operate in a horizontal plane to generate lift directly. Upon reaching a sufficient transition speed, the nacelles rotate forward, transforming the proprotors into propellers. In this mode, the aerodynamic lift is transferred from the rotors to the fixed wings, significantly increasing operational efficiency. This transition is governed by complex flight control laws that shift aerodynamic responsibility from the proprotors to the fixed-wing surfaces, thereby enabling operational speeds and altitudes that exceed the capabilities of traditional VTOL platforms. The inherent versatility of this design allows for the utilization of localized vertiports within urban centers while maintaining the transit great distances in a relatively fast manner.

3.2.2 Analysis of Advantages and Disadvantages

While the tilt-rotor offers unparalleled advantages in terms of mission radius and cruise velocity, its implementation is accompanied by significant engineering challenges and performance trade-offs. The primary benefit lies in its superior range and the ability to fly at altitudes up to 25,000 feet. Furthermore, the reduction in proprotor RPM during the cruise phase significantly diminishes noise pollution, which is a vital metric for maintaining a social license to operate near urban environments.

However, these merits must be weighed against the inherent mechanical complexity of the tilting mechanisms and the structural weight penalties associated with reinforced wing designs required to support rotating mass at the extremities. These factors often result in a reduced payload-to-weight ratio and higher maintenance requirements compared to simpler multicopter or fixed-wing VTOL configurations. Additionally, the high disc loading characteristic of tilt-rotors generates intense downwash during hover operations, which necessitates specialized vertiport infrastructure to mitigate ground-level environmental disturbances and ensure the safety of ground personnel. Consequently, while the tilt-rotor remains one of the most capable configurations for long-range transport, its adoption is contingent upon overcoming the fiscal and technical hurdles.

3.2.3 Existing Implementations and Concept Designs

Joby S4 Designed by Joby Aviation, the Joby S4 is an all-electric vertical take-off and landing aircraft based on a tilt-rotor configuration, in which multiple electrically driven propellers are mounted on tilting nacelles to enable the transition between vertical and forward flight regimes. The aircraft is intended for urban air mobility applications, particularly for short-range passenger transport conducted at high operational frequency, and its design places strong emphasis on reducing noise emissions while maintaining efficient propulsion and ensuring a high level of operational safety.



Figure 3.1: Joby S4, by Joby Aviation [3]

Leonardo AW609 Developed by Leonardo, the AW609 is a civil tilt-rotor aircraft designed to combine the vertical lift capability of rotary-wing systems with the cruise performance of fixed-wing aircraft. Featuring large proprotors mounted on tilting nacelles at the wingtips, the configuration enables efficient operation across both hover and forward flight conditions. The platform is intended for a wide range of missions, including executive transport, search and rescue, and offshore support.



Figure 3.2: AW609, by Leonardo [4]

3.3 Tilt-Wing

The Tilt-Wing configuration represents an advanced propulsive architecture that bridges the gap between rotary-wing VTOL capability and high-efficiency fixed-wing cruise. The distinguishing feature of this architecture is that the entire wing assembly, including all integrated propulsion units and their associated nacelles, rotates around a transversal, span-wise axis at the wing root.

3.3.1 Technical Characteristics

Unlike standard tilt-rotor configurations, where only the engine nacelles rotate at the wingtips, the tilt-wing designs the wing chord to always align with the resulting force vector of the propellers. The primary aerodynamic advantage of the tilt-wing lies in the effective management of the propeller-wing aerodynamic coupling, known as the propeller slipstream interaction. By pivoting the whole lifting surface, the high-energy, turbulent airflow generated by the propellers (the slipstream) is forced directly over the entire wing chord. This effect guarantees that the airflow remains attached to the aerodynamic surfaces throughout the critical transition phase. Even at exceptionally high angles of attack during takeoff, where a fixed wing would immediately stall, the forced slipstream maintains aerodynamic control and lift production. This interaction results in a smoother, more continuous lift hand-off between the propellers and the wing as the aircraft gains forward velocity.

3.3.2 Analysis of Advantages and Disadvantages

The tilt-wing configuration is often selected for missions requiring high aerodynamic cruise efficiency and moderate ranges. The critical performance benefits include aerodynamic transition efficiency and the elimination of hover download. Because the wing structure is not placed beneath the rotor downwash in hover, nearly 100% of the propeller's thrust is utilized for vertical lift. Furthermore, the vertical wing profile results in a more compact aerodynamic ground footprint.

Conversely, the largest disadvantage is the extreme vulnerability of the aircraft to lateral wind gusts (crosswinds) during the transition phase. With the large surface area of the wing facing the relative wind, significant yaw and roll moments are generated. Rotating the entire wing assembly also requires highly complex, high-torque, and redundant actuation systems at the wing-fuselage joint, adding substantial weight. Lastly, as the wing rotates, the center of pressure shifts significantly, creating a massive pitching moment that the aircraft must counter with its flight surfaces, such as a T-tail, or through thrust vectoring.

3.3.3 Existing Implementations and Concept Designs

While historically limited by mechanical complexity, the advent of Distributed Electric Propulsion (DEP) has made the tilt-wing architecture a viable option for modern UAM applications.

Airbus Vahana Developed by Airbus, this autonomous VTOL demonstrator utilized a unique tandem tilt-wing configuration (both front and rear wings rotated). Equipped with eight independent electric motors, it successfully proved the viability of software-driven electric tilt-wing transitions, proving that DEP could replace heavy mechanical linkages.

Dufour Aerospace Aero3 Serving as a benchmark for contemporary applications, the Aero3 utilizes a single tilt-wing approach integrated with a hybrid-electric powertrain for extended regional flights. It features electric propellers distributed along the wing's leading edge to maximize the slipstream effect, and employs a tail rotor integrated into its T-tail to actively counter the massive pitching moments generated during the transition phase.

3.4 Tilt-Duct

The tilt-duct configuration represents an advanced development of vectored-thrust propulsion, in which the propulsive units are enclosed within aerodynamically refined ducts or shrouds. In this arrangement, the rotors are integrated into movable housings capable of transitioning between vertical and horizontal orientations, thereby enabling the generation of both lift and forward thrust within a unified system. Consequently, this approach reduces the aerodynamic inefficiencies typically associated with open-rotor configurations, while also contributing to a more compact and spatially efficient airframe design. Within the broader context of contemporary aerial mobility, the tilt-duct concept emerges as a particularly suitable solution for high-frequency operations. This is especially relevant in scenarios where ground-level safety and noise containment are of primary concern, as the ducted structure inherently supports improved acoustic management and enhanced operational safety.

3.4.1 Technical Characteristics

The defining feature of the tilt-duct architecture lies in its reliance on ducted-fan propulsion, most commonly implemented through electrically driven fans, which exploit the aerodynamic effect of pressure recovery at both the inlet and outlet of the duct to enhance static thrust generation. In contrast to the open-rotor arrangement of tilt-rotor systems, the presence of the duct serves to confine the airflow, thereby reducing tip-vortex losses and enabling higher disc loading within a comparatively smaller rotor diameter. As the aircraft transitions from vertical to forward flight, these ducted units rotate to align with the flight direction and subsequently operate as efficient propulsion pods during cruise conditions.

Moreover, many contemporary configurations incorporate DEP systems, whereby multiple smaller and independently controlled ducted fans are employed, frequently positioned along the trailing edge of the wing. This approach not only provides redundancy in lift generation but also allows for refined control through differential thrust. In addition, the integration of these ducted elements with the wing structure enables the ducts themselves to contribute to lift production during forward flight, which in turn improves the overall lift-to-drag ratio and aerodynamic efficiency of the airframe.

3.4.2 Analysis of Advantages and Disadvantages

The tilt-duct configuration offers a distinct set of operational attributes, though it introduces some specific mechanical and physical constraints. The primary merits of the tilt-duct reside in its augmented safety and compact spatial profile, as the shrouds insulate ground personnel from rotating blades and facilitate a reduced total airframe span, a necessity for navigating the restricted geometries of urban vertiports. Furthermore, the duct assembly provides acoustic attenuation, effectively sequestering noise and refracting it away from receptors, thereby fostering greater public receptivity. From a performance perspective, the architecture enables optimized cruise proficiency at moderate velocities, as the duct geometries diminish tip-vortex losses and can be engineered to generate supplemental

aerodynamic lift during the translational flight phase.

Conversely, these dividends are countered by substantial impediments, most notably the intensified mass and parasitic drag inherent in the duct structures themselves, where the additional weight of the shrouds frequently results in a higher empty mass. The configuration also needs intricate thermal regulation protocols, as the encased electric motors and power electronics are deprived of the passive cooling flow enjoyed by exposed propellers. Additionally, while maintaining efficiency during cruise, tilt-ducts frequently demonstrate diminished hover performance due to the elevated power demands necessitated by high disc loading, which can precipitate rapid energy depletion during protracted takeoff or landing sequences. Finally, the mechanical intricacy involved in synchronizing a multitude of tilting units constitutes a profound barrier to type certification, particularly regarding the verification of autonomous flight control architectures and the mitigation of system failures.

3.4.3 Existing Implementations and Concept Designs

Lilium Jet Created by the German aerospace company Lilium, the Lilium Jet is an all-electric vertical take-off and landing aircraft employing a tilt-duct distributed propulsion architecture, in which multiple ducted fans are integrated within movable wing and canard surfaces to enable thrust vectoring between vertical and horizontal flight. Intended for regional air mobility, it is designed to deliver high-speed and zero-emission intercity transport while maintaining vertical lift capability (Figure 3.3).

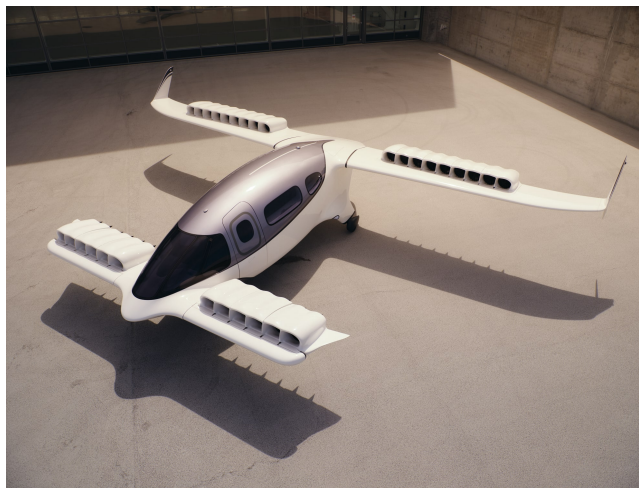


Figure 3.3: Lilium Jet, by Lilium [5]

Bell Nexus 4EX Developed by the American manufacturer Bell Textron, the Bell Nexus 4EX is an urban air mobility platform based on a tilt-duct configuration with four rotating ducted fans providing both lift and forward thrust. Designed for short-range piloted operations, it accommodates multiple passengers and incorporates a modular propulsion system supporting electric and hybrid-electric solutions (Figure 3.4).



Figure 3.4: Bell Nexus 4EX, by Bell Textron [6]

4 Concept of Operations

4.1 Overview and Requirements

The definition of the Concept of Operations (ConOps) is based on the objective of establishing a fast connection between the Balearic Islands, reducing travel time compared to existing transport modes while simultaneously alleviating congestion in current infrastructure. According to the analysis presented in section 2.2, the reference routes correspond to Ibiza-Palma de Mallorca (approximately 140 km) and Palma de Mallorca-Menorca (approximately 132 km). Consequently, the aircraft must be designed to perform short- to medium-range inter-island missions with high operational frequency and vertical take-off and landing (VTOL) capability.

Based on the preliminary study, the reference mission consists of transporting six passengers and one pilot, including baggage, resulting in an estimated payload near the 600kg. The vehicle is expected to operate typically between an altitude of 1500 to 3000 m, representing a compromise between aerodynamic efficiency, energy consumption, and operational flexibility. Importantly, mission performance must not be assessed solely under ideal conditions; instead, it must incorporate margins to account for adverse winds, holding patterns, missed approaches, and potential diversions.

4.1.1 Payload

The payload is one of the most influential parameters in the system definition, as it directly affects energy requirements, battery sizing, and propulsion power. From a regulatory standpoint, the aircraft must comply with EASA requirements for small-category VTOL aircraft; in particular, the maximum certificated take-off mass (MCTOM) should not exceed 5700 kg [12]. At this preliminary stage, an allowable center-of-gravity variation of approximately 7% is also assumed, ensuring that different passenger and baggage configurations do not compromise stability or controllability.

Beyond its nominal value, payload variability plays a critical role in operational feasibility. Each mission may involve different combinations of passengers, baggage, and initial energy levels. Therefore, payload should be treated not only as a commercial parameter but as a key design variable. An increase in payload generally leads to higher power demand during hover and climb phases, while simultaneously reducing the energy margin available for reserves and contingencies.

4.1.2 Flight Phase Parameters

The most demanding route identified is approximately 140 km. To complete this mission within 30 to 40 minutes, the required average block speed lies between 210 and 280 km/h. However, this value includes all flight phases-hover, transition, climb, cruise, descent, approach, and landing-implying that the cruise speed must be higher, realistically within the range of 250-350 km/h.

Energy-intensive phases, particularly hover, initial climb, and terminal descent, require careful consideration. These segments typically account for a significant portion of total energy consumption, even if their combined duration is on the order of 10 minutes. As a result, aircraft performance cannot be evaluated solely based on cruise efficiency or range; a complete block-time energy breakdown is essential.

The selected operational altitude range of 1500-3000 m reflects a trade-off between competing effects. Higher altitudes generally reduce aerodynamic drag and improve cruise efficiency, but require additional energy during climb, which may offset overall gains. Therefore, the optimal altitude must result from a joint optimization of mission time, energy consumption, and operational margins.

4.1.3 Mission Profile

With these parameters in mind, a set mission profile is established:

Vertical climb: 0-91m (300ft) at 4m/s, enough to clear any obstacles and ensure a safe and smooth transition

Hover: 10 seconds at 91m, during which the aircraft transitions from vertical to forward flight configuration

Climb: 91-2286m (7500ft) at 60m/s and 5° climb angle

Cruise: 2286 m at 77m/s

Descent: 2286-91m at 60m/s and -5° climb angle

Hover: 10 seconds at 91m, during which the aircraft transitions from forward to vertical flight configuration

Vertical descent: 91m at 4m/s, with a soft touchdown

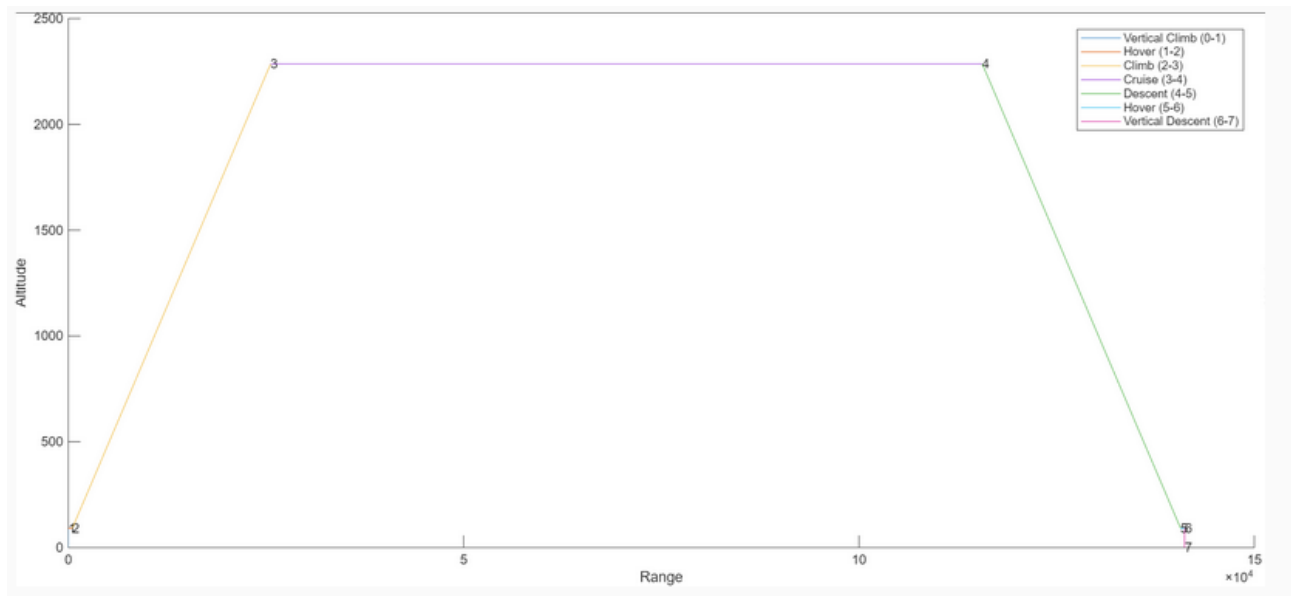


Figure 4.1: Defined mission profile

By choosing comfortable climb and descent angles, the aircraft covers 25km of horizontal distance during each of these segments. Consequently, the cruise phase only needs to cover 90km to complete the total 140km range. Following this mission profile, the flight takes approximately 33 minutes to complete.

4.1.4 Safety Strategies

Given the inter-island nature of the operation, safety considerations must extend beyond nominal mission execution. The aircraft must retain sufficient energy reserves to accommodate degraded scenarios, including missed approaches, diversions to alternate landing sites (*divert*), holding patterns, and extended hover operations. In this study, a preliminary energy reserve of 30-40% of the total mission energy is considered appropriate to ensure operational robustness.

This requirement has direct implications for aircraft sizing. Increased payload reduces the fraction of energy available for reserves, requiring a careful balance between commercial capacity and safety margins. Accordingly, the ConOps must be defined based on realistic mission scenarios rather than ideal conditions, ensuring compliance with minimum operational requirements under disturbances or environmental uncertainties.

4.2 Concept Sketches

Aware of all the conditions and equipment that already exist, the team started to draw initial designs for the proposed VTOL aircraft. In order to obtain the best results possible, different configurations were put in perspective. The following sections detail the technical characteristics and operational mechanics of the four primary architectures selected for evaluation: the Tilt-Rotor, Tilt-Wing, Tilt-Duct, and Fixed-Wing + VTOL.

4.2.1 Tilt-Rotor Configuration

The Tilt-Rotor architecture is defined by a stationary fixed-wing structure equipped with articulating nacelles mounted at the wingtips. These nacelles house the primary power plants and propellers. During vertical takeoff and landing operations, the nacelles are oriented perpendicular to the longitudinal axis of the fuselage, allowing the rotors to generate vertical lift.

The transition to forward flight involves a mechanized rotation of the nacelles through a 90-degree arc. During this conversion phase, the aircraft gradually shifts the lift requirement from the rotors to the fixed-wing assembly as forward airspeed increases. Once the nacelles reach a fully horizontal orientation, the propellers act as traditional propellers, providing forward thrust while the wings provide the necessary aerodynamic lift for cruise. Control during hover is typically managed through differential collective pitch and nacelle tilt, while conventional flight surfaces take over during high-speed flight (Figure 4.2).

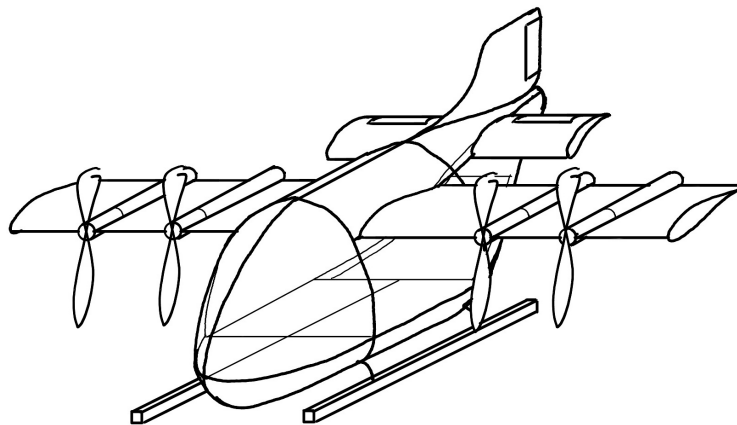


Figure 4.2: Tilt-Rotor Configuration

4.2.2 Tilt-Wing Configuration

The Tilt-Wing configuration utilizes a design where the entire wing assembly, including all integrated power plants and propellers, rotates around a central span-wise axis. For vertical flight operations, the wing is tilted to a vertical position so that the chord line is aligned with the vertical axis of the aircraft. This directs the full thrust of the propellers downward to facilitate liftoff and hovering maneuvers.

As the aircraft transitions to cruise, the entire wing is incrementally lowered to a horizontal position. A defining characteristic of this mechanism is that the propellers remain aligned with the wing's leading edge throughout the entire rotation. This ensures that the propeller slipstream always flows over the wing surface, which helps maintain attached airflow (Figure 4.3).

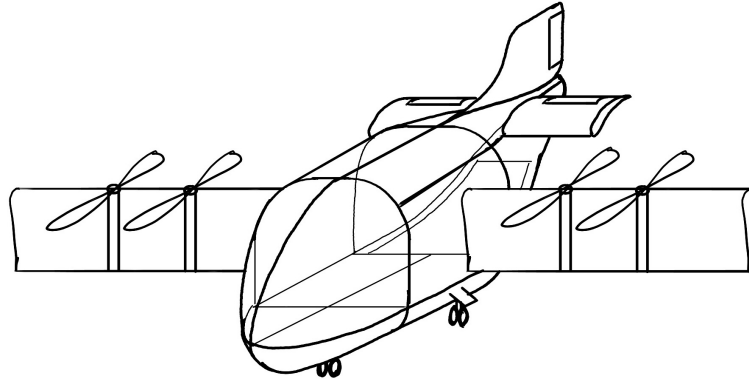


Figure 4.3: Tilt-Wing Configuration

4.2.3 Tilt-Duct Configuration

The Tilt-Duct architecture utilizes ducted fans — rotors enclosed within aerodynamic shrouds — mounted on pivoting mechanisms. These units are distributed across the airframe to provide balanced thrust. For vertical maneuvers, the ducts are oriented to exhaust air directly toward the ground. The duct geometry is designed to facilitate a high-pressure differential, contributing to the total lift generated by the propulsion unit.

In the transition to forward flight, the duct assemblies pivot forward to provide horizontal propulsion. The shrouds serve as physical containment for the rotor blades and function as aerodynamic surfaces that can generate a degree of lift and stability in forward flight. Control is typically maintained through a combination of motor RPM modulation, thrust vectoring via the pivoting ducts, and internal stator vanes that deflect the exhaust flow to manage yaw and roll during low-speed flight. This configuration prioritizes a compact footprint and reduced acoustic signature.

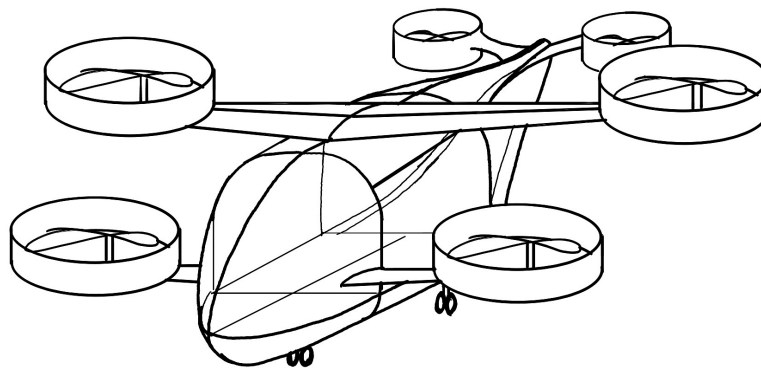


Figure 4.4: Tilt-Duct Configuration

4.2.4 Fixed-Wing + VTOL Configuration

The Fixed-Wing + VTOL configuration utilizes a decoupled propulsion strategy. This design employs two separate sets of motors that remain fixed in their respective orientations. The vertical lift system consists of multiple rotors mounted on longitudinal booms or integrated into the wing structure, oriented solely for upward thrust. A separate,

independent propulsion unit — usually a pusher or puller propeller — is mounted to the airframe to provide horizontal thrust.

During the flight cycle, the vertical rotors are activated exclusively for takeoff, hover, and landing. Once the aircraft reaches a safe altitude, the cruise motor provides forward acceleration. When the aircraft surpasses the stall speed of the fixed wing, the wing assumes the full lift load, and the vertical rotors are deactivated. In many implementations, these lift rotors are then locked in a low-drag alignment to streamline the aircraft for efficient long-distance cruise. This architecture relies on software-driven transitions rather than mechanical tilting hinges.

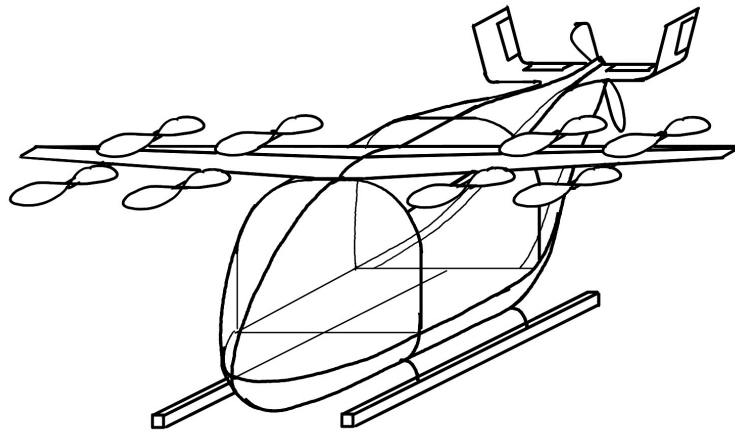


Figure 4.5: Fixed-Wing + VTOL Configuration

4.3 AHP

Given the great set of concepts already proposed it is fairly subjective to choose one over the others purely on aesthetics. There is, therefore, the need for some objective method able to choose, based on established criteria, the concept that best fits the proposed mission.

A great tool for that effect is the Analytic Hierarchy Process, furthermore referenced simply as AHP. This tool is already built into the Aircraft Design Tool and therefore does not need to be discussed further.

In simple terms, the tool provides a structured approach to decision making by assigning a score to each concept. To make use of this tool, the most important step is to define the criteria and subcriteria according to what the mission requires or prioritizes. The adopted hierarchy is shown in the sideways tree chart below (Figure 4.6).

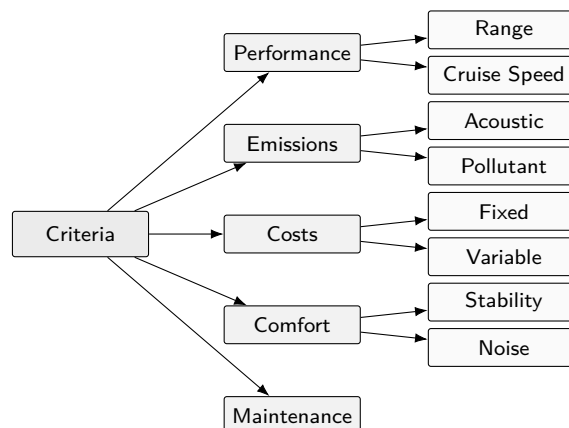


Figure 4.6: AHP criteria tree for UAM concept selection.

Each criterion and subcriterion is weighted using square reciprocal matrices, that is, pairwise comparison matrices whose diagonals are filled with 1's and whose remaining entries follow the rule $a_{ij} = 1/a_{ji} \forall i \neq j$. Overall, there are three levels: the priorities among the main criteria, the priorities among the subcriteria, and the performance of each concept within each subcriterion. The values used for each first row are positive integer relations being the remaining rows automatically defined. A suggestive example regarding the weighting on the main criteria is given below (Table 4.1), though a more exhaustive exposition is left to the appendix (A.1).

C/C	Performance	Emissions	Costs	Comfort	Maintenance
Performance	1	2	3	1	3
Emissions	0.5	1	1.5	0.5	1.5
Costs	0.333	0.667	1	0.333	1
Comfort	1	2	3.003	1	3
Maintenance	0.333	0.667	1	0.333	1
Final Ratios	0.316	0.158	0.105	0.316	0.105

Table 4.1: Main Criteria Relative Scores

The manually computed matrices were fed to the AHP which returned a value between 0 and 1 for each concept. This value represents how acceptable is that given concept under the mission priorities defined earlier. Those results can be seen in the following table (4.2):

Final Scores	
Tilt Rotor	0.265
Tilt Wing	0.195
Tilt Duct	0.279
Fixed Wing + VTOL	0.2556
Winner Concept	Tilt-Duct

Table 4.2: Concept Choosing - AHP Final Scores

Therefore the concept was set to be *Tilt-Duct* and the following chapter works exclusively on making this concept fly and perform exactly as envisioned.

5 Aircraft Design Tool

5.1 Study MTOW

One of the crucial steps on designing an aircraft is making a good estimation of the aircraft weight. Given the targeted market we must follow the *European Union Aviation Safety Agency* (EASA). This establishes some boundaries that can't be crossed. One of them concerns the MCTOM (*Maximum Certified Take-Off Mass*) that can't exceed 5700kg as of 2024 regulations [12]. The maximum passenger capacity is too fixed and can't exceed 9 passengers.

This gives a baseline for the design iterations. The initial plan sat on 6 passengers with respective baggage and a single pilot. Furthermore the design iterations will prove if this is viable or whether is to ambitious or to conservative.

A rough estimate points to a minimum 600kg of payload without any margin at this point. Finding the correspondent MTOM is trickier. A known rule of thumb to find it is to quadruple the payload value giving, for the case we are dealing with, about 2400kg. This value, though leaning to the lower side, is quite acceptable and aligned with other know solutions already flying.

An important note concerns the fact of augmenting the payload or adding a safety margin. For each additional passenger, an additional 400kg need to be added to this estimation. Nevertheless this is just the entry point for the iterative design process and a metric to a more conscious evaluation of the results given by the *Aircraft Design Tool* to be used further.

5.2 Preliminary Concept and Parametric Analysis

To initiate the design synthesis, a baseline configuration for the Mega-Duct aircraft was established utilizing a canard-wing architecture with an aft-mounted main wing. While the conceptual design employs a V-tail assembly to support the structural booms, current modeling constraints within the Aircraft Design Tool (ADT) necessitated the implementation of a high-mounted horizontal stabilizer (H-tail) as a first-order approximation. Aerodynamically, this high placement is advantageous as it isolates the tail from the wakes generated by both the canard and the main wing, thereby preserving control surface efficiency [13].

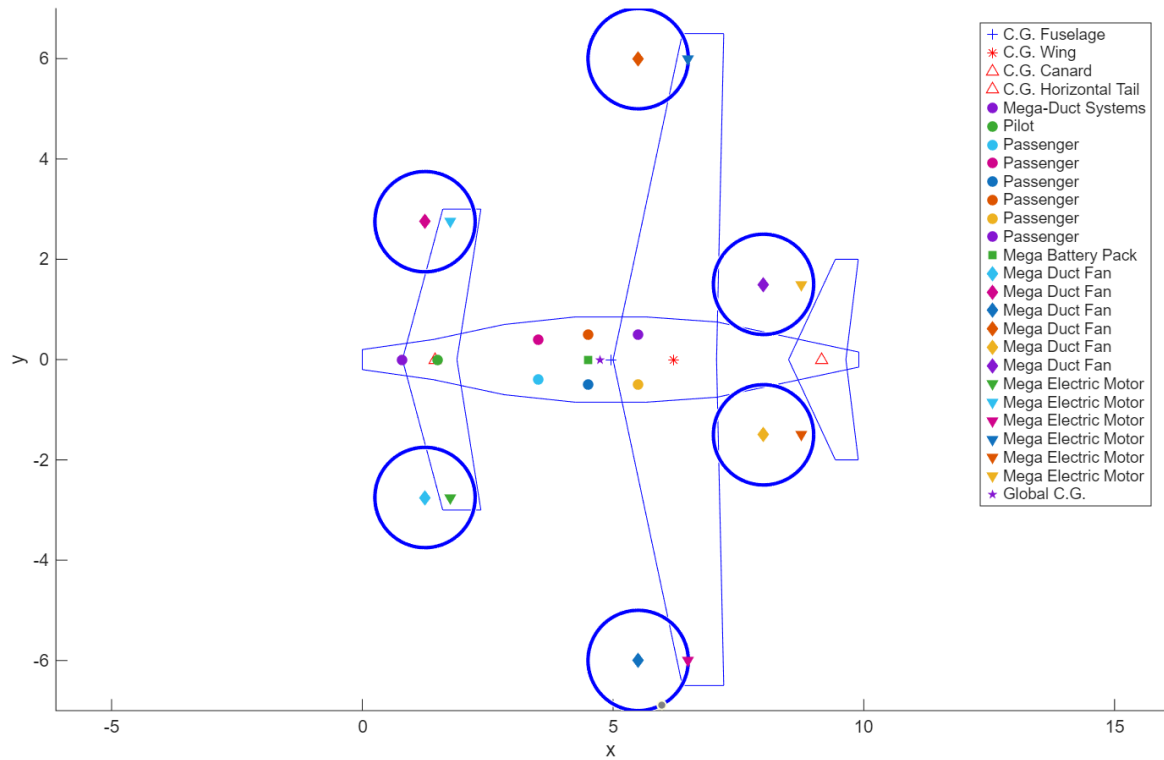


Figure 5.1: 2D Configuration model indicating CG points for each component and the overall CG.

5.2.1 Propulsion System

The propulsion system comprises six tilting ducted rotors, each featuring a 1 m radius, symmetrically distributed with two units per lifting surface across the canard, main wing, and tail assemblies. Primary power is delivered by a hybrid network consisting of a Proton Exchange Membrane Fuel Cell (PEMFC) and a high-efficiency (95%) Lithium-Sulfur (Li-S) battery pack, which drives the electric motors and their respective tilting mechanisms. In terms of internal layout, the cabin is configured to accommodate one pilot and six passengers, while the navigation and flight control systems are housed in the forward fuselage. All remaining subsystems, including the powertrain, are fully integrated within the main fuselage structure to optimize volume and weight distribution.

Table 5.1: Key propulsive design parameters post-iteration.

Parameter	Value	Unit
Number of Motors/Rotors	6	-
Rotor Radius	1.00	m
Max Power per Motor	260.00	kW
Total Installed Power	1560.00	kW

While the optimization of heterogeneous rotor sizing presents a potential path for future performance gains, all six rotors and motors are currently modeled as identical components to maintain analytical simplicity in this design phase.

5.2.2 Aerodynamic Sizing and Iterative Refinement

Following the definition of the baseline configuration, an iterative refinement of the aircraft sizing was conducted, focusing primarily on the lifting surfaces and mass allocation. The main objective was to ensure that the design

points for both the vertical take-off and landing (VTOL) and forward flight phases converged feasibly within the respective design space.

During the convergence iterations, the reference wing area was increased to reduce the wing loading, thereby improving low-speed lift capabilities and cruise efficiency. Concurrently, the nominal power of the electric motors was scaled up to satisfy the stringent power loading requirements necessary for the purely propulsive lift phases (VTOL). The coupling of vertical and forward design points regarding power loading stems from the dual-use nature of the tilt-duct rotors. Unlike configurations with dedicated propulsion systems for each flight phase (such as lift-plus-cruise), the Mega-Duct architecture requires a unified power dimensioning that satisfies both the high-thrust demands of VTOL and the aerodynamic efficiency required for forward flight [14] (Figure 5.2).

Table 5.2: Key aerodynamic design parameters post-iteration.

Parameter	Value	Unit
Total Reference Area	24.32	m ²
Main Wing Area	18.78	m ²
Canard Area	5.54	m ²
H-Tail Area	3.20	m ²

5.2.3 Design Space and Design Points

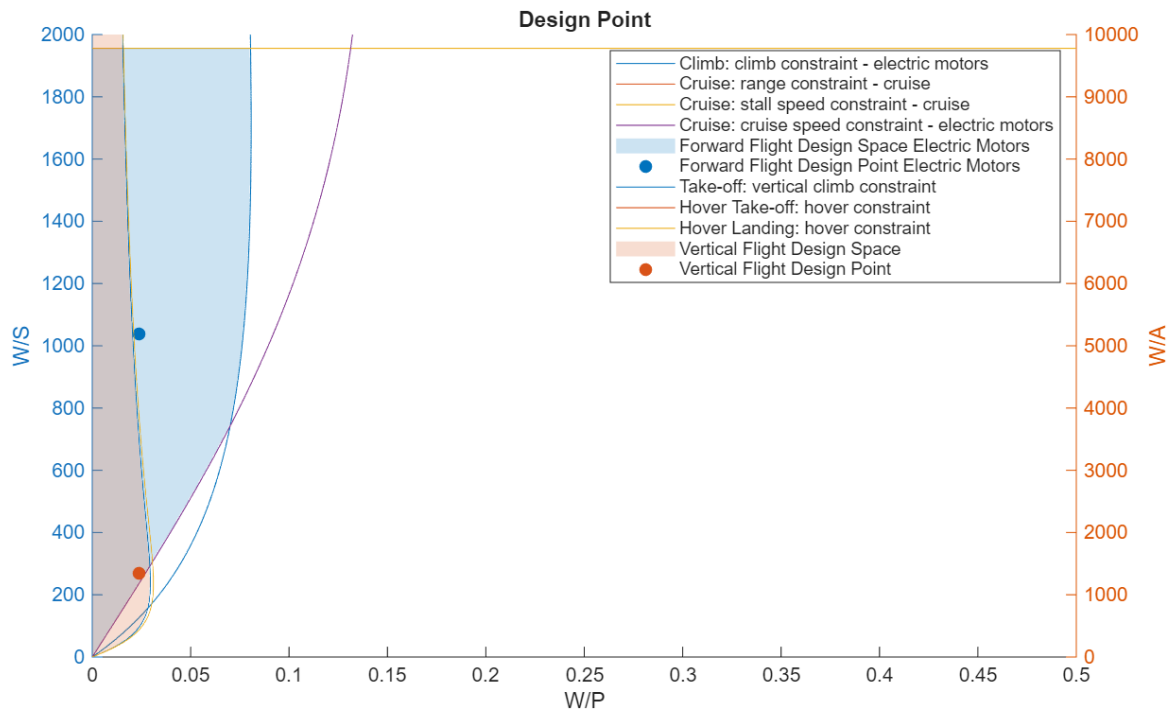


Figure 5.2: Design points graph for VTOL and forward flight phases.

The sizing matrix plot establishes the aircraft's design space by intersecting the performance constraints for both forward and vertical flight modes. For the forward flight regime, evaluated against the wing loading (W/S) on the left axis and power loading (W/P) on the horizontal axis, the feasible design region is primarily bounded by the stall speed and climb constraints. The selected "Forward Flight Design Point", situated at approximately $W/P = 0.0241$ kg/W and $W/S = 1047$ N/m², represents an optimized compromise that satisfies cruise speed targets while remaining safely within the allowable aerodynamic and structural limits modeled in ADT.

Concurrently, the vertical flight constraints dictate the disk loading parameters (W/A) featured on the right vertical axis. The "Vertical Flight Design Point" is positioned at a significantly lower disk loading threshold, around $W/A = 1351 \text{ N/m}^2$, falling squarely within the restricted vertical flight design space. This specific allocation is heavily constrained by hover and vertical take-off/landing requirements, ensuring that the six tilting ducted fans possess sufficient control authority and thrust-to-weight margin during transition phases without oversizing the electric motors.

Table 5.3: Design point parameters for the Mega-Duct aircraft.

Flight Regime	Parameter	Value
Forward Flight	Power Loading (W/P)	$\approx 0.0241 \text{ kg/W}$
	Wing Loading (W/S)	$\approx 1047 \text{ N/m}^2$
Vertical Flight	Power Loading (W/P)	$\approx 0.0241 \text{ kg/W}$
	Disk Loading (W/A)	$\approx 1351 \text{ N/m}^2$

5.2.4 Mission Profile

Finally, the aircraft sizing was validated against the established mission profile (Figure 4.1), with specific adjustments implemented to enhance the model's realism. Although the operational parameters for transition phases were previously defined, limitations within the ADT concerning transition phases required their exclusion from the current analysis. Consequently, the time and energy consumption associated with these specific phases are not accounted for in this iteration.

Table 5.4: Defined mission profile segments for the Mega-Duct.

Segment	Type	Altitude [m]	Velocity [m/s]	Time [s]	Range [km]
1. Take-off	Vertical Climb	0 $\rightarrow$ 91	4.0	22.75	0.00
2. Hover Take-off	Hover	91	0.0	15.00	0.00
3. Climb	Climb (5°)	91 $\rightarrow$ 2286	60	419.75	25.18
4. Cruise	Cruise	2286	85.0	1176.47	100.00
5. Descent	Descent (-5°)	2286 $\rightarrow$ 91	70.0	359.78	25.18
6. Hover Landing	Hover	91	0.0	15.00	0.00
7. Landing	Vertical Descent	91 $\rightarrow$ 0	4.0	22.75	0.00
Total				$\approx 34 \text{ min}$	150.36

5.2.5 Mass Allocation and Weight Distribution

By utilizing the empty mass component in ADT, the Maximum Take-Off Weight (MTOW) was initially estimated at 2511.77 kg, which is consistent with early empirical approximations. However, with the consolidation of the subsystem specifications, including the optimized Lithium-Sulfur (Li-S) battery pack, fuel cell integration, and the updated ducted fan and motor assemblies, the total weight increased to 2595.18 kg. This final distribution reflects the iterative resizing of the systems, ensuring that the aircraft meets both performance objectives and stringent regulatory safety margins without compromising mission feasibility (Figure 5.3).

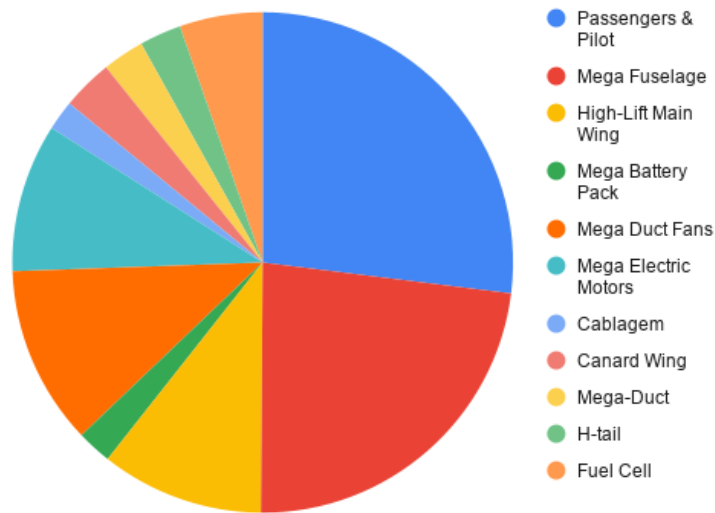


Figure 5.3: Mass distribution breakdown of the Mega-Duct aircraft (2595.18 kg).

It should be noted that the fuselage mass incorporates a high strength-to-weight ratio through the use of composite materials (300.45 kg for fiber) reinforced with structural metal (312.3 kg). Additionally, the rotor mass accounts for the complete duct assembly surrounding the propellers (including tilt and pitch mechanisms), and the electric motor mass includes the respective power units for the six stations.

Table 5.5: Mass distribution of the Mega-Duct aircraft components.

Component	Mass [kg]	Details / Remarks
Passengers & Pilot	700.00	7×100 kg each, including baggage
Mega Fuselage	612.75	300.45 kg (Fiber) + 312.3 kg structural metal
High-Lift Main Wing	273.00	$13 \text{ kg/m}^2 \times 21 \text{ m}^2$
Mega Battery Pack	58.37	Lithium-Sulfur (Li-S)
Mega Duct Fans	300.00	$6 \times (20 \text{ kg Propeller} + 15 \text{ kg Duct} + 15 \text{ kg Tilt} + \text{Pitch})$
Mega Electric Motors	249.00	6×41.5 kg each
Wiring	37.7	Cable weight
Canard Wing	85.00	
Mega-Duct Systems	70.00	Control and navigation systems
T-tail	70.00	
Fuel Cell	139.36	Fuel cell system
Total MTOW	2595.18	

6 Aerodynamic Considerations

6.1 Wing Aerodynamic and Structural Design

6.1.1 Design Requirements and Mission Constraints

The preliminary sizing of an aircraft's wing requires a rigorous derivation of geometric and aerodynamic properties based on established operational parameters and constraints. To ensure robust performance margins and accommodate mid-cycle upgrades or payload variations, a conservative MTOW of 2750 kg was adopted as first order approach. Later iterations of the whole concept would converge to the MTOW value seen in the previous chapter.

The wing planform architecture is defined by an aspect ratio (AR) of 9 and a total wingspan (b) of 13 m. From these fundamental parameters, the reference wing area (S) is calculated as follows:

$$S = \frac{b^2}{AR} = \frac{13^2}{9} \approx 18.778 \text{ m}^2 \quad (6.1)$$

The target flight envelope isolates two critical velocity boundaries: a cruise speed (V_{cruise}) of 85 m s^{-1} to meet mission range and endurance criteria, and a stall speed (V_{stall}) capped at 40 m s^{-1} to ensure low-speed controllability and acceptable landing characteristics.

To bridge the gap between idealized two-dimensional (2D) airfoil performance and realistic three-dimensional (3D) finite wing aerodynamics, an efficiency correction factor (k) is implemented [7]. This factor accounts for three-dimensional flow roll-up, fuselage interference, and localized downwash distributions. It is evaluated as:

$$k = 0.95 \times 0.9 = 0.855 \quad (6.2)$$

6.1.2 Determination of Target Aerodynamic Coefficients

The operating lift coefficients required during the primary phases of flight are derived by balancing the aerodynamic lift with the gravitational force acting on the aircraft ($W = MTOM \cdot g$). The ambient atmospheric conditions correspond to an operational temperature $T = 273.291 \text{ K}$ ($\approx 0.141^\circ\text{C}$), yielding a standard kinematic viscosity (ν) of $0.0000171534 \text{ m}^2\text{s}^{-1}$.

Cruise Condition Sizing

The 3D ideal lift coefficient ($C_{L,\text{ideal}}$) represents the required lifting capability during steady, level cruise at the design altitude and velocity ($V_{\text{cruise}} = 85 \text{ m s}^{-1}$). Assuming sea-level air density ($\rho_0 = 1.225 \text{ kg m}^{-3}$), the 3D lift coefficient is calculated as:

$$C_{L,\text{ideal}} = \frac{MTOW \cdot g}{0.5 \cdot \rho_0 \cdot V_{\text{cruise}}^2 \cdot S} = \frac{2750 \cdot 9.80665}{0.5 \cdot 1.225 \cdot 85^2 \cdot 18.778} = 0.406475 \quad (6.3)$$

To map this 3D wing lift coefficient to the equivalent 2D section lift coefficient ($C_{l,\text{ideal}}$) required for airfoil database comparisons, the efficiency factor k is applied:

$$C_{l,\text{ideal}} = \frac{C_{L,\text{ideal}}}{k} = \frac{0.406475}{0.855} = 0.475409 \quad (6.4)$$

Stall and Maximum Lift Sizing

To satisfy the safety and regulatory mandates associated with a maximum stall speed of 40 m s^{-1} , the wing must be capable of generating a high maximum lift coefficient ($C_{L,\text{max}}$). Under un-flapped configurations, this represents the ultimate aerodynamic limit before boundary layer separation dominates the suction surface:

$$C_{L,\text{max}} = \frac{\text{MTOW} \cdot g}{0.5 \cdot \rho_0 \cdot V_{\text{stall}}^2 \cdot S} = \frac{2750 \cdot 9.80665}{0.5 \cdot 1.225 \cdot 40^2 \cdot 18.778} = 1.465874 \quad (6.5)$$

The corresponding maximum 2D section lift coefficient ($C_{l,\text{max}}$) required from the chosen airfoil section is:

$$C_{l,\text{max}} = \frac{C_{L,\text{max}}}{k} = \frac{1.465874}{0.855} = 1.71459 \quad (6.6)$$

Operational Reynolds Number

The dynamic behavior of the boundary layer, including transition and separation points, is governed by the operational Reynolds number (Re). Using the mean aerodynamic chord ($\bar{c} = 1.532880 \text{ m}$) as the characteristic length scale, the Reynolds number at cruise velocity is established as:

$$Re = \frac{V_{\text{cruise}} \cdot \bar{c}}{\nu} = \frac{85 \cdot 1.532880}{0.0000171534} = 7595858.547 \quad (6.7)$$

This high Reynolds number confirms that the flow over a significant portion of the wing will be turbulent, which increases skin friction drag but offers greater resistance to flow separation compared to lower Reynolds number regimes.

6.1.3 Airfoil Selection and Comparative Analysis

The selection of the primary airfoil section involves a trade-off study across multiple NACA families to balance high cruise efficiency, benign stall characteristics, and structural suitability for a complex tilt-duct propulsion architecture. Five primary candidate profiles were selected (based on the ideal lift coefficient and on the maximum lift coefficient - Figure 6.1) for numerical evaluation: NACA 65₂-415, NACA 64₁-412, NACA 63₁-412, NACA 4412, and NACA 4415 (A.3).

tilt-duct aircraft.

First of all, geometrical comparison reveals that the NACA 64-412 section features a significantly thinner trailing edge compared to the NACA 4412. Structurally, the wing must support the ducts mechanisms mounted at the tip stations. A thin trailing edge compromises the torsional stiffness and localized shear strength required to safely transfer these massive concentrated loads along the wing.

Furthermore, the presence of a tilt-duct propulsion unit introduces severe aerodynamic interactions. The physical duct housings and rotating blades generate high levels of localized turbulence and non-uniform inflow velocity fields across the span. Natural laminar flow profiles, such as the NACA 6-series, rely on highly undisturbed pressure gradients to maintain a laminar boundary layer [7]. The turbulent wake and vibration induced by the ducts would trigger premature boundary layer transition, completely destroying the low-drag benefits of the NACA 64-412 and potentially leading to localized flow separation or lift deficits.

Based on these combined aerodynamic and structural integration criteria, the NACA 4412 (A.3.4) was selected as the final airfoil profile for the primary wing structure.

6.1.4 Planform Geometry and Sizing Calculations

Taper Ratio Optimization

The chordwise distribution of the wing is tailored using a taper ratio (λ) of 0.4, since it provides a linear approximation of an ideal parabolic/elliptical lift distribution for an unswept wing [8, 15]. By matching the lift distribution closely to an ellipse, the spanwise variations in downwash are minimized, which directly reduces the induced drag coefficient without needing to rely on more complex methods.

Chord Distributions

Using the established parameters ($b = 13$ m, $AR = 9$, $\lambda = 0.4$), the precise geometric chords are derived using standard planform sizing formulations [8]:

- Root Chord (c_r): The chord length at the centerline junction is computed as:

$$c_r = \frac{2 \cdot b}{AR \cdot (1 + \lambda)} = \frac{2 \cdot 13}{9 \cdot (1 + 0.4)} = \frac{26}{12.6} = 2.063492 \text{ m} \quad (6.8)$$

- Tip Chord (c_t): The chord length at the outermost spanwise station is determined via the taper ratio relationship:

$$c_t = \lambda \cdot c_r = 0.4 \cdot 2.063492 = 0.825397 \text{ m} \quad (6.9)$$

- Mean Aerodynamic Chord ($\bar{c}$): The longitudinal reference length for static stability and aerodynamic center positioning is evaluated as:

$$\bar{c} = \frac{2}{3} \cdot c_r \cdot \left(\frac{1 + \lambda + \lambda^2}{1 + \lambda} \right) = \frac{2}{3} \cdot 2.063492 \cdot \left(\frac{1 + 0.4 + 0.16}{1 + 0.4} \right) = 1.532880 \text{ m} \quad (6.10)$$

Sweep Determination

As a result of the taper ratio being set at 0.4, the quarter-chord line sweep ($\Lambda_{c/4}$) is set to exactly 0° (Figure 6.2). This ensures that the primary structural spar, typically aligned with the locus of aerodynamic centers at 25% chord,

runs straight perpendicular to the fuselage centerline [16].

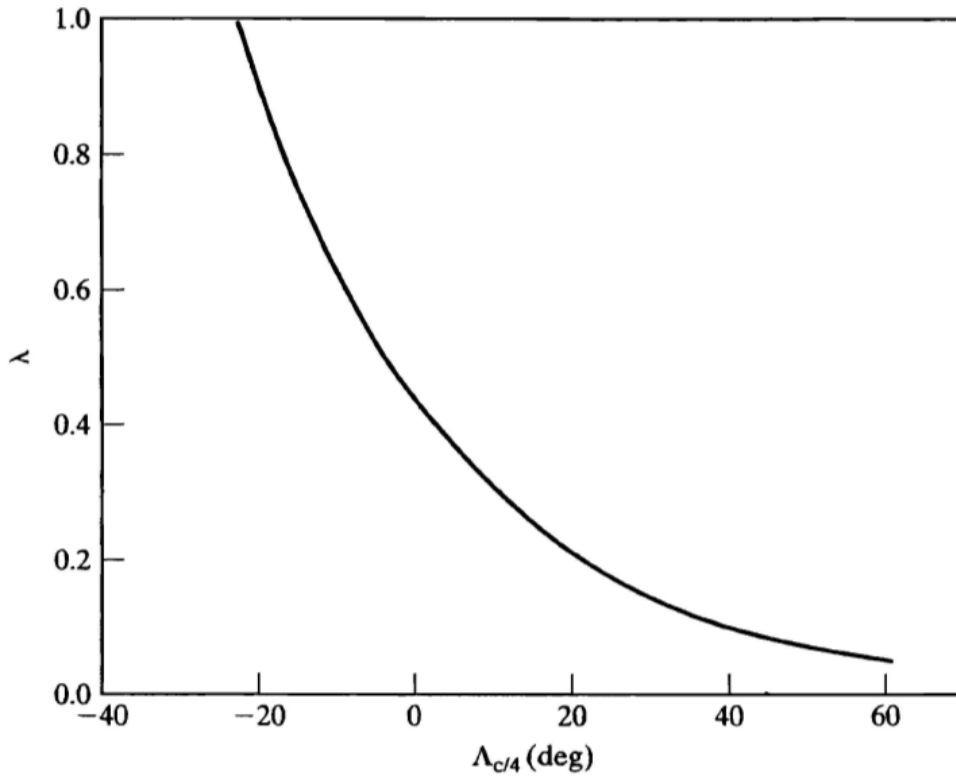


Figure 6.2: Taper ratio of trapezoidal wings as a function of the quarter-chord line sweep [8].

However, because the wing features a significant taper ($\lambda = 0.4$), the closing geometric lines naturally create a positive sweep at the leading edge (Λ_{LE}). This relationship is quantified using the standard planform transformation formula [15]:

$$\tan(\Lambda_{LE}) = \tan(\Lambda_{c/4}) + \frac{1 - \lambda}{AR \cdot (1 + \lambda)} \quad (6.11)$$

Substituting the known parameters into the expression yields:

$$\tan(\Lambda_{LE}) = \tan(0^\circ) + \frac{1 - 0.4}{9 \cdot (1 + 0.4)} = 0 + \frac{0.6}{12.6} = 0.047619 \quad (6.12)$$

$$\Lambda_{LE} = \arctan(0.047619) = 2.726311^\circ \quad (6.13)$$

This minor forward sweep of the leading edge has negligible impacts on high-speed compressibility drag. However, it should still be accounted for in the final geometry of the wing.

Dihedral Angle Selection

The geometric dihedral angle (Γ) is defined as exactly 0° across the entire span. While positive dihedral is traditionally utilized in conventional aircraft to provide passive lateral stability (roll-subsidence via slip effects), it introduces a severe geometric penalty for a tilt-duct VTOL configuration.

During vertical takeoff, landing, and hover transition phases, the tip-mounted propulsion units must rotate into

a vertical orientation to generate pure upward lift. If the wing possessed a non-zero dihedral angle, the thrust vectors of the tip-mounted rotors would be canted outwards relative to the main aircraft vertical axis during vertical flight modes. This misalignment would create parasitic horizontal force components, inducing severe control cross-coupling, reducing net vertical lifting efficiency, and causing control system instability during ground-effect hovering. A dihedral angle of $\Gamma = 0^\circ$ guarantees that the propulsive rotors remain in a perfectly synchronized horizontal plane during vertical operations, maximizing vertical force and streamlining control effort.

6.2 Canard

Following the preliminary sizing of the main wing, a similar methodology was applied to the aerodynamic and geometric sizing of the canard. This forward surface plays an important role in the overall aircraft configuration, not only as an auxiliary lifting surface, but also as a structural element for supporting and integrating the two front rotors. Therefore, its design must take into account not only aerodynamic requirements, but also structural strength.

Unlike a conventional horizontal stabiliser, the position of the canard means that its flow field may influence the aerodynamic behaviour of the main wing, making a careful selection of its geometry, aerofoil and position essential.

Based on the first iterations carried out in ADT, a canard span of $b_c = 6\text{m}$ and an aspect ratio of $AR_c = 6.5$ were defined. From these two parameters, the reference area of the canard can be calculated as:

$$S_c = \frac{b_c^2}{AR_c} = \frac{6^2}{6.5} \approx 5.538\text{m}^2 \quad (6.14)$$

The geometry was defined with a taper ratio of $\lambda_c = 0.7$, allowing the tip chord to be reduced without introducing an excessive variation in the lift distribution along the span.

The root chord is given by:

$$c_{r,c} = \frac{2S_c}{b_c(1 + \lambda_c)} = \frac{2 \cdot 5.538}{6.0(1 + 0.7)} \approx 1.086 \text{ m} \quad (6.15)$$

The tip chord is obtained directly from the taper ratio:

$$c_{t,c} = \lambda_c c_{r,c} = 0.7 \cdot 1.086 \approx 0.760 \text{ m} \quad (6.16)$$

The mean aerodynamic chord of the canard is calculated as:

$$\bar{c} * c = \frac{2}{3} c * r, c \left(\frac{1 + \lambda_c + \lambda_c^2}{1 + \lambda_c} \right) = \frac{2}{3} \cdot 1.086 \left(\frac{1 + 0.7 + 0.7^2}{1 + 0.7} \right) \approx 0.933 \text{ m} \quad (6.17)$$

A leading-edge sweep angle of $\Lambda_{LE,c} = 15^\circ$ was also applied. The longitudinal position of the leading edge was defined as $X_c = 0.8 \text{ m}$, in order to reduce the direct interference of the flow generated by the canard on the main wing. In addition, the vertical position was defined as $Z_c = -0.3\text{m}$.

Furthermore, a positive dihedral angle of $\Gamma_c = 13^\circ$ was adopted, contributing to a slight increase in the lateral stability of the aircraft. The dihedral also raises the canard tips, favouring the integration of the front ducts and increasing the geometric clearance from the ground, which is useful during operations close to the ground.

Regarding the selection of the airfoil, the NACA 0015 (A.4) was chosen. The use of a symmetrical aerofoil is justified by the need to maintain predictable aerodynamic behaviour across different operating regimes. The relative thickness of 15% represents a significant structural advantage, since the canard is required to support the two front

rotors. It is therefore necessary to ensure sufficient internal volume for the installation of structural reinforcements and possible mounting interfaces for the propulsion systems.

6.3 V-Tail

After the sizing of the main wing and the canard, the preliminary definition of the tail was carried out. For this aircraft, a V-tail configuration was selected, in which the functions of the horizontal and vertical stabilisers are combined into two inclined surfaces. The objective of this configuration was to reduce the number of aerodynamic surfaces and junctions with the fuselage, thereby decreasing the wetted area and interference drag.

However, this configuration introduces greater complexity at the control level, since the movable surfaces of the V-tail must combine the functions of elevator and rudder.

Based on the first geometric iterations, a V-tail span of $b_{VT} = 4m$ and an aspect ratio of $AR_{VT} = 5$ were defined. From these two parameters, the reference area of the V-tail can be calculated as:

$$S_{VT} = \frac{b_{VT}^2}{AR_{VT}} = \frac{4^2}{5} \approx 3.2m^2 \quad (6.18)$$

The geometry was defined with a taper ratio of $\lambda_{VT} = 0.4$, allowing the tip chord to be reduced and consequently reducing mass and moments of inertia.

The root chord is given by:

$$c_{r,VT} = \frac{2S_{VT}}{b_{VT}(1 + \lambda_{VT})} = \frac{2 \cdot 3.2}{4.0(1 + 0.4)} \approx 1.143 \text{ m} \quad (6.19)$$

The tip chord is obtained directly from the taper ratio:

$$c_{t,VT} = \lambda_{VT} c_{r,VT} = 0.4 \cdot 1.143 \approx 0.457 \text{ m} \quad (6.20)$$

The mean aerodynamic chord of the V-tail is calculated as:

$$\bar{c} * VT = \frac{2}{3} c * r, VT \left(\frac{1 + \lambda_{VT} + \lambda_{VT}^2}{1 + \lambda_{VT}} \right) = \frac{2}{3} \cdot 1.143 \left(\frac{1 + 0.4 + 0.4^2}{1 + 0.4} \right) \approx 0.849 \text{ m} \quad (6.21)$$

A leading-edge sweep angle of $\Lambda_{LE,VT} = 25^\circ$ was adopted. This value is higher than the one used for the canard, as its purpose is to delay the loss of aerodynamic effectiveness at higher-speed regimes. The position of the leading edge was defined as $X_{VT} = 9.5m$, in order to increase the moment arm of the tail relative to the centre of gravity, contributing to longitudinal stability and increasing the static margin of the aircraft. To minimise the upwash and downwash caused by the canard and the main wing, the vertical position was defined as $Z_{VT} = 1.6m$, measured from the top of the fuselage, in order to ensure adequate structural clearance and aerodynamic efficiency.

Furthermore, a dihedral angle of $\Gamma_{VT} = 45^\circ$ was adopted, defining the V-shaped geometry. This choice allows each surface to contribute simultaneously to longitudinal and directional stability. At this preliminary stage, an angle of 45° represents a balanced solution between the horizontal stabiliser function and the vertical stabiliser function, avoiding excessive prioritisation of either component.

Regarding the selection of the aerofoil, the NACA 0015 (A.4) was chosen for the same reasons previously presented regarding the canard.

7 Propulsion System and Energy Analysis

The propulsion system is a natural follow-up of the aerodynamic analysis. Although the present assessment remains conceptual, it is central to the feasibility of the proposed aircraft, since it determines whether the reference mission can be achieved with current or near-future technology while preserving the objective of a cleaner inter-island transport solution.

The key design decision is the selection of the propulsion and energy-storage architecture. The propulsion can be roughly categorized into combustion and electric solutions. Given the main goals of this project, a combustion engine is not desirable. Regarding the energy storage system, there are broader solutions including aviation fuel, batteries, fuel cells and their specific subcategories.

The analysis therefore considers the energy required by the mission, the mass and performance implications of different technological setups, and the extent to which the final configuration remains viable in terms of power, weight and operational endurance.

7.1 Energy Requirements

As any conceptual approach, the process is inherently iterative and tangled, requiring information from other affected systems. Since many of the weights were approximated we gave a significant margin on the first estimate of MTOW to use. Further convergence lead us to the mass value of approximately 2413.32kg that already includes the final choice of the energy system solution.

Below are the calculations for a possible last iteration of both energy system and MTOW, at least to what propulsion is concerned. For the VTOL segments, the analysis assumes six fully electric rotors, a rotor radius of 1 m, and sea-level hover conditions with an air density of 1.225 kg/m^3 , representing a conservative operating case.

For hover and vertical flight, the aircraft weight is estimated as

$$W = mg \approx 23675 \text{ N.} \quad (7.1)$$

A 20% lift margin is considered for maximum thrust conditions, giving $L_{max} \approx 28410 \text{ N}$. Dividing the load across six rotors results in a nominal hover thrust of approximately 3946 N per rotor and a maximum thrust of approximately 4735 N per rotor. Using momentum theory, the ideal induced power is then corrected by a factor of $k = 1.5$, which combines a safety margin and non-ideal losses. This gives an estimated real hover power of 134 kW per rotor and a peak VTOL power of 176 kW per rotor.

The resulting energy demand for the vertical phases is summarized in Table 7.1. The mission includes two hover periods and two vertical climb/descent periods, so the total energy associated with hover and VTOL operation is estimated at approximately 20.2 kWh.

Table 7.1: Estimated power and energy requirements for hover and VTOL phases.

Parameter	Value	Unit
Hover thrust per rotor	3946	N
Maximum thrust per rotor	4735	N
Real hover power per rotor	134	kW
Real maximum VTOL power per rotor	176	kW
Total hover energy	6.7	kWh
Total VTOL energy	13.5	kWh
Total vertical-phase energy	20.2	kWh

The forward-flight phases were assessed using the aerodynamic parameters obtained from the current design iteration. Cruise is evaluated at 85 m/s for 1176.47 s, with a cruise air density of 0.978 kg/m^3 , a drag coefficient of 0.00672581, and a wetted/reference area value of 41.61375 m^2 . This gives a drag force of approximately 989 N and, assuming an overall propulsive efficiency of 0.85, a cruise power of approximately 98.9 kW. The corresponding cruise energy is therefore 32.3 kWh.

For climb, the aircraft is evaluated at 60 m/s and a 5° climb angle, giving an estimated climb power of 185.2 kW and an energy requirement of 21.6 kWh. During descent, the drag power is lower than the gravitational power component, allowing a small amount of energy recovery. Assuming a regenerative efficiency of 25%, the descent phase contributes approximately -2.14 kWh . Combining the vertical, climb, cruise, and descent estimates gives a total mission energy of approximately 71.66 kWh, as shown in Table 7.2.

Table 7.2: Estimated mission energy breakdown for the fully electric case.

Mission phase	Energy	Unit
Hover and VTOL	20.2	kWh
Climb	21.6	kWh
Cruise	32.3	kWh
Descent regeneration	-2.14	kWh
Total	71.66	kWh

These results indicate that the vertical phases account for a significant fraction of total mission energy despite their short duration. This confirms that rotor sizing and disc loading are critical design drivers. Increasing rotor radius reduces disc loading and therefore lowers induced power during hover, improving the viability of the fully electric configuration. However, due to geometrical constraints it is not possible to increase further the rotor radius without effect on structures, pilot vision, obstruction of emergency exits, etc.

For a detailed account of the numerical values and step-by-step calculations underpinning the expressions presented in this section, the reader is referred to Appendix A.2.

7.2 All Electric

The all-electric option is attractive because it eliminates direct in-flight CO_2 emissions and is consistent with the environmental positioning of the proposed service. However, its feasibility depends strongly on battery specific energy, discharge capability, cycle life, and production impact. The technologies considered at this stage are lithium-ion (Li-ion), lithium-sulfur (Li-S), and lithium-air (Li-air) batteries.

The reference comparison is summarized in Table 7.3. Conventional high-performance Li-ion batteries offer mature

technology and relatively high cycle life, but their specific energy remains low for this mission class. Li-S batteries offer a meaningful improvement in specific energy and are therefore more attractive for short regional VTOL missions, although they remain less mature and present limitations related to cycle life and discharge rate. Li-air batteries show the best mass potential, but they are still experimental and cannot be considered a near-term solution for UAM certification.

Table 7.3: Battery technology comparison for the all-electric configuration.

Technology	Specific energy	Estimated MTOW	Battery mass	Main limitation
Li-ion	0.90–1.08 MJ/kg	Outside design space	–	Low specific energy for this mission
Li-S	1.08–1.62 MJ/kg	3028.4–2456.6 kg	555.6–303.4 kg	Low maturity, C-rate, and cycle life
Li-air	~ 3.60 MJ/kg	2022.4 kg	114.35 kg	Experimental technology

The ADT-based energy study confirms the same trend. For Li-S batteries, the estimated maximum take-off mass varies between 3028.4 kg and 2456.6 kg across the assumed specific-energy range. The corresponding battery mass varies from 555.6 kg to 303.4 kg. As expected, lower specific energy increases battery mass, which in turn increases wing loading, disc loading, and power loading. Near the lower end of the Li-S range, the design points move toward the boundary of the feasible design space.

Li-air batteries appear very favorable from a purely mass-based perspective, with an estimated battery mass of only 114.35 kg and an MTOW near 2022.4 kg. Nevertheless, this option is not suitable for the current design horizon because the technology remains experimental, with limited demonstrated cycle life and insufficient maturity for aviation certification. Advanced Li-ion batteries are more mature, but the baseline Li-ion specific-energy range places the design points outside the feasible design space. Even optimistic advanced Li-ion variants remain challenging unless significant improvements in pack-level energy density are achieved.

Consequently, Li-S is the most plausible all-electric candidate among the considered options, but it still introduces important operational constraints. The expected cycle life is limited when compared with mature Li-ion packs, and the relatively low C-rate is problematic because VTOL aircraft require high discharge power during take-off, hover, and landing. This problem can fortunately be solved using a smart architecture of the battery pack. In addition, the nominal mission is expected to finish with approximately 20–35% state of charge. Under a pessimistic assumption of 20% final state of charge, five flights would require approximately four full charging cycles. If a Li-S pack is assumed to withstand only 200 cycles in the worst case, one battery set would support only about 250 flights before replacement.

Overall, a fully electric architecture is technically attractive and environmentally coherent, but it remains highly sensitive to battery assumptions. For the present aircraft, the all-electric solution is feasible only if high-specific-energy batteries can also provide sufficient discharge rate, acceptable cycle life, and manageable replacement costs. This motivates the parallel consideration of hybrid electric-fuel-cell solutions, which may combine high energy density with improved endurance and lower weight.

7.3 Hybrid Electric-Fuel Cell

As shown in the previous section, the fully electric concept is technically attractive but remains strongly constrained by the mass, cycle life, and discharge capability of the battery system. The natural next step is therefore to consider a hybrid architecture that preserves electric propulsion at the rotors while reducing the energy-storage burden placed

entirely on the battery pack.

Three broad hybrid propulsion architectures were considered: turboelectric, serial hybrid, and parallel hybrid. A turboelectric system is the least attractive option for the present mission, since it is formally hybrid but remains very close to a combustion-based architecture in practice. This conflicts with the low-emission objective of the project and does not make full use of the distributed electric propulsion layout. A parallel hybrid arrangement was also considered, but it introduces major integration difficulties for a six-duct configuration. Transmitting mechanical power to each duct would either require six combustion engines, one per duct, or a complex shaft-and-gearbox system. Both options would increase mass, noise, maintenance costs, and failure points.

The preferred architecture is therefore a serial hybrid electric system. In a conventional serial hybrid, a combustion engine drives a generator, and the generated electrical power is either supplied directly to the electric motors or used to recharge the batteries. In the present design, this concept is adapted into a hydrogen fuel-cell architecture: the combustion engine and generator are replaced by an electrochemical reactor, while the fuel tank is replaced by a compressed hydrogen storage system. The rotors remain fully electric, so duct control is decoupled from any mechanical shaft power and can be managed directly through the motor controllers.

From an energy-management perspective cruise is the longest segment but has a relatively low and steady power requirement. Climb has moderate power needs due to some of the weight being, at this point, compensated by the lift generated in the respective surfaces. Finally vertical take-off, hover, and landing are shorter but much more power-intensive. For this reason, the proposed hybrid strategy uses the fuel-cell system as a steady power source sized primarily for cruise and climb, while the battery supplies the high transient power required during VTOL and hover. This division allows each subsystem to operate closer to its most favorable regime.

For the fuel-cell technology, proton exchange membrane fuel cells (PEMFC) and solid oxide fuel cells (SOFC) were compared. The main trade-offs are summarized in Table 7.4.

Table 7.4: Comparison between PEMFC and SOFC technologies for the hybrid architecture.

Characteristic	PEMFC	SOFC
Start-up behavior	Fast start-up	Slow start-up; must reach operating temperature
Power density	High, typically suitable for compact aviation systems	Lower system-level density and generally heavier installation
Operating temperature	Up to approximately 80°C	Around 800°C
Cooling and thermal integration	Easier to cool with atmospheric airflow	More difficult to heat, maintain, and thermally isolate
Hydrogen/fuel tolerance	Requires very pure H ₂ ; sensitive to CO contamination	More fuel-flexible
Reactor stability	Requires humidity control	Sensitive to sulfur poisoning and thermal cycling
Typical lifetime	Approximately 5000 h	Potentially long, but less direct data for this application

Considering the mission profile and operational constraints, PEMFC is selected as the most suitable fuel-cell option. This decision is consistent with the direction followed by major aerospace programmes, such as Airbus ZEROe. Its rapid start-up, lower operating temperature, and simpler thermal integration make it more compatible with a compact VTOL aircraft, despite the need for high-purity hydrogen and humidity control.

The preliminary sizing divides the energy storage between a Li-S battery pack and a hydrogen PEMFC subsystem. The battery is sized for the high-power VTOL and hover phases. Using the previously estimated 20.2 kWh for these phases and adding a 30% safety margin gives an energy requirement of approximately 26.26 kWh.

Several companies are currently improving Li-S battery technology. *Theion*, for example, is developing aviation-oriented batteries with specific energies close to 0.5 kWh/kg and with the stated objective of reaching 1.0 kWh/kg in the coming years. This value is already above the conservative assumption adopted in this sizing. Using 0.45 kWh/kg for the Li-S pack gives a battery mass of approximately 58.35 kg [17].

The hydrogen subsystem is assigned to cruise and climb, with an estimated energy requirement of 51.77 kWh.

Adding the same 30% safety margin gives approximately 67.30 kWh. In terms of power, this subsystem is sized by the climb requirement, which is the limiting steady-flight condition at 185.2 kW. The American Department of Energy hydrogen program states system-level hydrogen storage values of approximately 1.71 kWh/kg, including tank, valves, and hydrogen, the storage mass is estimated at 39.36 kg [18]. For the electrochemical reactor and associated conversion plant, a representative target set by Strategic Research and Innovation Agenda (SRIA) 2.0 kW/kg gives a fuel-cell system mass of approximately 100 kg [19][20].

The complete hybrid energy system is therefore estimated at 197.73 kg, as summarized in Table 7.5. Compared with the most optimistic all-electric Li-S case previously considered (303.4 kg), this corresponds to a mass reduction of approximately 34.8%. The architecture is also more scalable: increasing range mainly affects the cruise energy demand. With the adopted 30% margin, the cruise energy requirement is approximately 0.421 kWh/km, corresponding to only 0.246 kg/km of additional hydrogen storage mass. Thus, an extra 10 km of range would in theory add only about 2.5 kg to the total energy system mass, which is a major selling point of this design.

Table 7.5: Preliminary hybrid electric-fuel-cell energy-system sizing.

Subsystem	Sizing basis	Estimated mass	Role
Li-S battery pack	26.26 kWh at 0.45 kWh/kg	58.35 kg	VTOL, hover, and transient power
Hydrogen storage	67.30 kWh at 1.71 kWh/kg	39.36 kg	Energy storage for cruise and climb
PEMFC and conversion plant	185.2≈200 kW at 2.0 kW/kg	100.0 kg	Steady electrical power generation
Total hybrid energy system	–	197.73 kg	Combined architecture

This hybrid solution is therefore more promising than a purely battery-based architecture from a mass perspective. It also reduces the depth of discharge imposed on the battery, which may improve cycle life and operational availability. Nevertheless, it introduces additional complexity through hydrogen storage, fuel-cell thermal management, power electronics, and safety certification. For the present conceptual design, the hybrid electric-fuel-cell configuration is retained as the most balanced solution between low emissions, energy density, and VTOL power capability.

7.4 Rough Avionics and Motor Choices

The architecture selection has direct implications for the avionics and control system. Since the preferred configuration is a series hybrid electric architecture, the six ducted rotors remain electrically driven and independently controllable. This is a major advantage over a parallel hybrid configuration, because no mechanical shafting or gearbox network is required to distribute power to the ducts. Each motor can be commanded individually, preserving the control authority needed for hover, transition, and low-speed flight.

For the propulsion units, the EMRAX 348 Medium Voltage motor with Combined Cooling (CC) was selected. This choice is primarily driven by its high operational limit of 800 Vdc, which reduces voltage drops, improves overall electrical efficiency, and allows for a reduction in cable cross-section, saving wiring mass. The motor delivers a peak power of 260 kW, safely exceeding the 176 kW worst-case requirement for the critical 30-40 seconds of sea-level hover. Furthermore, its ability to generate up to 1000 Nm of peak torque enables a direct-drive configuration. By eliminating the gearbox, the design drastically reduces mechanical complexity and structural weight. Finally, the compact physical dimensions of the motor, 348 mm in outer diameter and 107 mm in axial width, allow for seamless aerodynamic integration within the duct's central spinner.

Technical data	EMRAX 348 Medium Voltage		
Cooling options	AC	LC	CC
Ingress protection	IP21	IP65	IP21
Cooling specification	AF=20 m/s ; AA=25°C	WF=8 l/min at 50°C ; AA=25°C	WF=8 l/min at 50°C ; AA=25°C
Weight [kg]	41	42	41,5
Diameter / width [mm]	348 / 107		
Maximal battery voltage [Vdc] and max load RPM	800 Vdc (2800 RPM)		
Peak motor power at max RPM [kW]	260		
Continuous motor power at load RPM [kW]	103	110	129
Maximal rotation speed [RPM]	4000 (4500 for a few seconds with magnetic field weakening)		
Maximal motor current [Arms]	450		
Continuous motor current [Arms]	210		
Maximal motor torque [Nm]	1000		
Continuous motor torque [Nm]	400	425	500
Torque / motor current [Nm/1Aph rms]	2,5		
Cogging torque [Nm]	5		
Maximal temperature of copper windings and magnets [°C]	120		
Motor efficiency [%]	92 – 98 %		
Internal phase resistance at 25 °C [mΩ]	12,3		
Input phase wire cross-section [mm²]	17,0		
Wire connection	star		
Induction Ld/Lq [μH] of 1 phase	180 / 195		
Controller / motor signal	sine wave		
AC voltage between two phases [Vrms/1RPM]	0,1520		

Figure 7.1: EMRAX348 Medium Voltage Specifications.

7.5 Rotors

The rotor design is heavily influenced by the demanding aerodynamic environment of the ducted fan configuration, particularly the presence of transonic flow regimes. Operating at the maximum nominal speed of 2800 RPM, the 1.0 m radius rotor reaches a tip speed of approximately 293.2 m/s, corresponding to Mach 0.86. During the 85 m/s cruise phase, the helical Mach number at the tip increases to approximately 0.88. Consequently, all blade sections from $r/R \approx 0.65$ outwards operate in a transonic regime, well above the drag divergence Mach number of conventional profiles. To mitigate severe wave drag, the design utilizes state-of-the-art NASA SC(2) supercritical airfoils in the outer span.

A 5-blade configuration was selected based on a trade-off between aerodynamic performance and acoustic constraints. This blade count yields an optimal solidity ($\sigma = \frac{N \cdot c_{mid}}{2\pi R} \approx 0.76$) that falls perfectly within the recommended 0.6–1.0 range for ducted fans. Acoustically, an odd number of blades decorrelates pressure pulses, effectively reducing the tonal annoyance of the Blade Pass Frequency (BPF), which operates at 233 Hz. Furthermore, pairing a 5-blade rotor with 3 stator vanes yields a greatest common divisor of 1 ($GCD(5, 3) = 1$), effectively preventing destructive tonal lock-in and minimizing rotor-stator interaction noise. The blades are separated by a 72° angle, providing good dynamic balance and assembly clearance. The rotor features a 400 mm spinner diameter, with the blades spanning towards a duct throat radius of 1005 mm. A rigorous 5 mm tip clearance (0.5% of the radius) is maintained, which aligns with literature recommendations to suppress tip vortices and maximize thrust efficiency.

The aerodynamic profile of the blade varies radially to accommodate the changing local Mach numbers and structural loads. The root sections (from the 200 mm hub up to 25% span) utilize NACA 65A-series airfoils with a maximum thickness-to-chord ratio (t/c) ranging from 14% to 12% to withstand high bending moments. From

Table 7.6: Radial blade design parameters and aerodynamic profile distribution.

Station	Radius (mm)	Airfoil	t/c (%)	Chord (mm)	Twist (°)	Local Mach
Root	200	NACA 65A-415	14	220	~ 42	0.25
25%	350	NACA 65A-412	12	205	~ 28	0.35
50%	600	NASA SC(2)-0410	10	185	~ 18	0.52
75%	800	NASA SC(2)-0408	8	165	~ 12	0.70
Tip	1000	NASA SC(2)-0406 / OA-206	6**	140	~ 8	0.86

the mid-span (50%) to the tip, the blade transitions to NASA SC(2) supercritical airfoils. ** The thickness ratio is strictly tapered down to 6% at the tip; this is a hard aerodynamic constraint, as every additional 1% of thickness decreases the critical Mach number by approximately 0.02, which would inadvertently push the tip into a highly inefficient supersonic regime. The blade chord tapers from 220 mm at the root to 140 mm at the tip, accompanied by an aggressive twist distribution ranging from roughly 42° at the root to 8° at the tip, ensuring an optimal angle of attack is maintained across the entire span.

7.6 Powertrain Architecture Overview

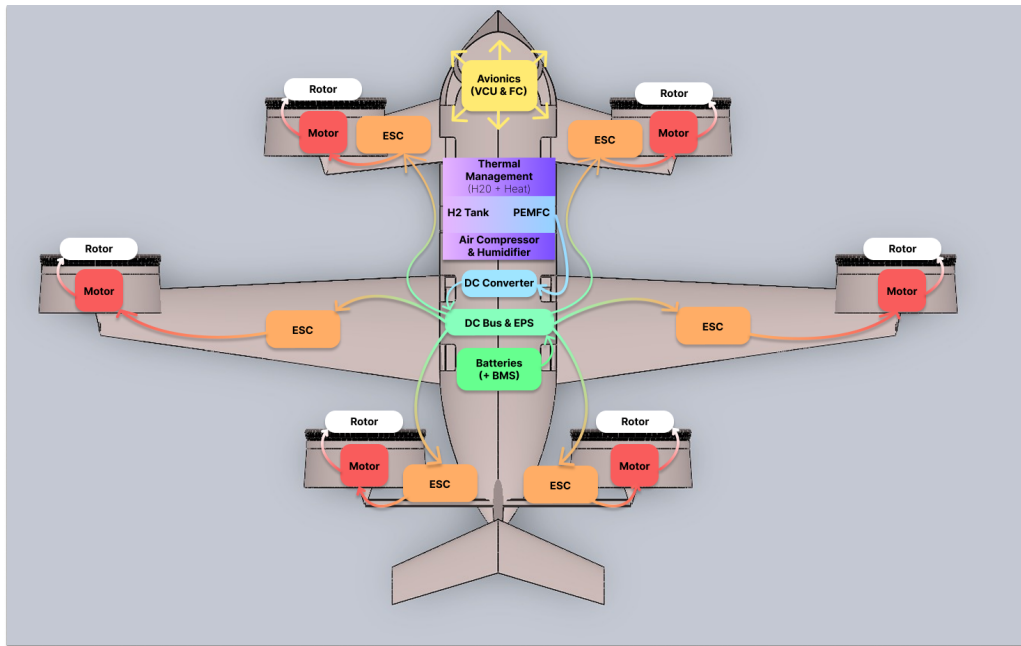


Figure 7.2: 2D Powertrain configuration.

The aircraft employs a hybrid hydrogen-electric propulsion architecture designed to combine the high specific energy of hydrogen with the high specific power capability of advanced Lithium-Sulfur (Li-S) batteries. This approach addresses one of the primary challenges of regional Urban Air Mobility operations: providing sufficient energy for a 150 km inter-island mission while maintaining acceptable aircraft mass and preserving the high power levels required during vertical flight.

The architecture is organized around an 800 V DC high-voltage distribution network that serves as the electrical backbone of the aircraft. Energy is supplied by two complementary sources: a 200 kW Proton Exchange Membrane Fuel Cell (PEMFC) system fed by a 700 bar compressed hydrogen tank, and a Li-S battery pack sized to satisfy peak-

power demands and emergency reserve requirements. Both systems are connected to a common DC bus, enabling dynamic power sharing according to mission requirements. The resulting configuration combines the endurance advantages of hydrogen propulsion with the rapid response and redundancy offered by battery storage, thereby providing a balanced solution for the demanding operational profile of a tilt-duct UAM aircraft.

7.6.1 Energy Flow and Operating Philosophy

The energy management strategy is based on the complementary operation of the fuel-cell and battery subsystems throughout the mission. During vertical take-off, hover, and transition phases, propulsion power demand reaches its maximum value. Under these conditions, the battery pack provides the majority of the instantaneous power required by the six propulsion units, while the fuel cell contributes continuous baseline power. This strategy avoids excessive fuel-cell sizing, reduces overall system mass, and allows the electrochemical conversion system to operate near its optimal efficiency point.

As the aircraft transitions into forward flight and reaches cruise conditions, power demand decreases substantially and the fuel cell becomes the primary energy source. The battery then assumes a supporting role, supplying transient loads, stabilizing the electrical network, and preserving sufficient energy reserves for contingency operations. During descent, power requirements decrease further, allowing the fuel-cell system to sustain the majority of onboard electrical loads while maintaining battery availability for the final VTOL segment. This operating philosophy maximizes overall efficiency while ensuring adequate redundancy and reserve energy under both nominal and off-design operating conditions.

7.6.2 Hydrogen Energy Subsystem

The hydrogen subsystem is responsible for providing sustained power during cruise flight. Hydrogen is stored in a lightweight composite pressure vessel operating at approximately 700 bar, selected as a compromise between storage density, technological maturity, and aerospace integration requirements.

The stored hydrogen is supplied to a 200 kW PEMFC stack, where it reacts electrochemically with oxygen extracted from ambient air. This process generates electrical power with water and heat as the only by-products. Compared with conventional combustion-based propulsion systems, the fuel cell eliminates direct carbon emissions while achieving higher conversion efficiencies and substantially lower acoustic signatures.

To ensure stable operation across the full flight envelope, the fuel-cell subsystem incorporates an air-compression and humidification unit responsible for maintaining cathode pressure and membrane hydration. In addition, a dedicated thermal management system regulates operating temperatures and dissipates waste heat generated during electrochemical conversion. The electrical output of the fuel-cell stack is conditioned through a DC/DC converter before being delivered to the high-voltage bus. Since fuel-cell voltage varies with load demand and operating conditions, this converter ensures a stable electrical interface between the hydrogen subsystem and the aircraft power network.

7.6.3 Electrical Energy Storage Subsystem

A Li-S battery pack is integrated directly into the high-voltage distribution network. Li-S technology was selected due to its superior theoretical specific energy compared with conventional lithium-ion chemistries, offering significant mass reductions for a given energy capacity and making it particularly attractive for aerospace applications.

The battery system performs multiple functions within the powertrain architecture. Beyond supplying peak power during VTOL operations, it acts as an electrical buffer capable of absorbing rapid load fluctuations and supporting transient operating conditions. Furthermore, the battery provides a critical layer of redundancy by ensuring continued propulsion capability in the event of fuel-cell degradation or temporary power deficits.

Battery operation is supervised by a dedicated Battery Management System (BMS), which continuously monitors cell voltages, temperatures, current flow, and state-of-charge. Through continuous health monitoring and protective functions, the BMS ensures safe operation throughout all phases of flight while maximizing battery performance and service life.

7.6.4 Power Distribution and Control Architecture

Power distribution throughout the aircraft is achieved through a centralized 800 V DC bus, which serves as the primary electrical interface between the energy sources and propulsion system. The adoption of a high-voltage architecture is primarily motivated by efficiency considerations, as it enables the transmission of large amounts of electrical power while minimizing conductor mass and resistive losses.

Supervision of the entire powertrain is performed by the Vehicle Control Unit (VCU), which functions as the central energy management controller. The VCU continuously determines the optimal power split between the fuel-cell and battery subsystems while monitoring system health, thermal conditions, fault states, and available energy reserves. This centralized control philosophy enables efficient operation of the hybrid architecture while ensuring compliance with safety and reliability requirements.

7.6.5 Propulsion System

Propulsive power is delivered to six distributed tilt-duct propulsion units through dedicated Electronic Speed Controllers (ESCs) and EMRAX 348 Medium Voltage electric motors. The ESCs convert DC electrical power from the high-voltage bus into controlled three-phase AC power while providing precise torque and rotational speed regulation.

The EMRAX 348 motor was selected due to its exceptional power density, high efficiency, compact dimensions, and proven suitability for aerospace applications. Each motor directly drives a tilt-duct propulsion unit, eliminating the need for complex gearbox arrangements and thereby improving mechanical reliability while reducing maintenance requirements.

The distributed propulsion architecture provides both vertical lift and forward thrust depending on rotor orientation. In addition to enhancing operational flexibility, the use of six independent propulsion units improves fault tolerance and system redundancy, key requirements for future certified UAM operations.

7.6.6 Powertrain Integration and Component Placement

The physical arrangement of the powertrain components was primarily driven by four design objectives: minimization of high-current cable lengths, optimization of aircraft center-of-gravity (CG) location, efficient thermal management and proper electromagnetic compatibility.

The battery pack, fuel-cell system, DC/DC converter, and Vehicle Control Unit were therefore concentrated within the central fuselage section, close to the aircraft center of gravity. Since the battery and hydrogen subsystems represent the largest masses within the powertrain, locating them near the CG minimizes stability variations resulting

from fuel consumption and battery state-of-charge changes throughout the mission. Furthermore, central placement reduces structural bending moments and facilitates maintenance access.

The hydrogen storage tank was positioned immediately aft of the passenger compartment and adjacent to the fuel-cell system. This arrangement minimizes hydrogen feed-line lengths, reduces pressure losses, and simplifies the integration of the thermal management subsystem. Locating the fuel-cell stack close to the hydrogen tank additionally reduces system complexity and improves packaging efficiency.

The high-voltage DC bus and associated power electronics were also integrated within the central fuselage. Since the propulsion motors are distributed across the canard, main wing, and tail surfaces, a centralized power distribution architecture minimizes the number of power conversion stages while simplifying fault detection and electrical isolation.

The six EMRAX 348 electric motors are installed directly within the tilt-duct assemblies. This configuration eliminates the need for mechanical transmissions, gearboxes, and long rotating shafts, thereby reducing mechanical losses and improving overall system reliability. Positioning the motors at the propulsion units also minimizes low-voltage, high-current motor cable lengths and simplifies thermal integration with the surrounding airflow.

Overall, the adopted arrangement concentrates the heaviest energy-generation and storage components near the aircraft center of gravity while distributing only the propulsion units to the lifting surfaces. This architecture reduces wiring complexity, limits structural penalties, and improves both maintainability and fault tolerance.

7.6.7 Powertrain Wiring and Weight Estimation

The electrical distribution network constitutes a non-negligible fraction of the total powertrain mass and therefore requires a preliminary sizing analysis. In a worst-case scenario, assuming a maximum peak power requirement of 176 kW per motor, the total installed power for the six distributed propulsion units reaches approximately 1056 kW. Since the propulsion system operates at a nominal voltage of 800 V DC, the peak current in the main distribution bus can be estimated as

$$I_{max} = \frac{P}{V} = \frac{1056 \times 10^3}{800} = 1320 \text{ A} \quad (7.2)$$

Assuming equal power sharing among the six propulsion units during this critical flight phase, the maximum current supplied to each motor branch becomes

$$I_{motor} = \frac{1320}{6} = 220 \text{ A} \quad (7.3)$$

Copper was selected as the conductor material due to its high electrical conductivity, excellent fatigue resistance, established aerospace certification history, and relatively low contact resistance. Although aluminium offers lower density, the larger conductor cross-sections required to achieve equivalent electrical performance often offset a significant portion of the mass advantage while introducing additional connection and corrosion challenges.

The required conductor cross-sectional area can be estimated using an allowable current density of

$$J = 5 \text{ A/mm}^2 \quad (7.4)$$

which is representative of continuously loaded aerospace power cables with appropriate thermal management.

The corresponding required conductor area for each propulsion branch is therefore

$$A = \frac{I_{motor}}{J} = \frac{220}{5} = 44 \text{ mm}^2 \quad (7.5)$$

To accommodate this worst-case requirement while maintaining adequate thermal and operational margin, a standard metric conductor size of 50 mm² was selected. Because the required cross-section falls within the capacity of this standard cable size, the subsequent weight estimations remain identical to those of nominal operating conditions.

The density of copper is $\rho_{Cu} = 8960 \text{ kg/m}^3$ which yields a linear conductor mass of

$$m' = \rho_{Cu} A_{cable} = 8960 \times 50 \times 10^{-6} = 0.448 \text{ kg/m} \quad (7.6)$$

Assuming an average routing distance of 5 m from the central power electronics bay to each propulsion unit, the total conductor length becomes

$$L_{tot} = 6 \times 5 \times 2 = 60 \text{ m} \quad (7.7)$$

where the factor of two accounts for the supply and return conductors.

The estimated bare conductor mass is therefore

$$m_{cable} = m' L_{tot} = 0.448 \times 60 = 26.9 \text{ kg} \quad (7.8)$$

Additional insulation, shielding, connectors, and mounting hardware typically increase cable-system mass by approximately 30–50%. Applying a conservative factor of 1.4 yields

$$m_{HV} = 1.4 \times 26.9 = 37.7 \text{ kg} \quad (7.9)$$

Consequently, the complete high-voltage wiring network is estimated to contribute approximately 38 kg to the aircraft mass budget.

This value is consistent with current distributed-electric-propulsion demonstrators operating in the MW power class and confirms the importance of adopting an 800 V architecture. Operating at lower nominal voltages would necessitate stepping up to much heavier wire gauges (e.g., 70 mm² or 95 mm²), which would substantially increase conductor mass and associated installation penalties.

For comparative purposes, if a low-voltage architecture of 400 V DC were adopted instead, the current per motor branch would double to 440 A ($I_{max} = 2640 \text{ A}$). Maintaining the same current density of 5 A/mm² would require a cross-sectional area of 88 mm², dictating the use of a standard 95 mm² cable gauge. With a linear mass of 0.851 kg/m, the total system mass (applying the same 1.4 installation factor) would escalate to approximately 71.5 kg, imposing a substantial weight penalty of nearly 34 kg.

8 CAD's Overview

The CAD models were developed as soon as the respective parts studies were done. However in this conceptual approach a few items are still based on reference values, rough calculations and aesthetics that would be improved further in a preliminary design.

8.1 Fuselage CAD Model

The fuselage CAD model was developed as the geometric baseline for the cabin, cockpit, landing-gear integration, wing attachment region, and energy-storage installation. The initial sizing followed a conceptual design approach and was later refined with an OpenVSP model. OpenVSP was used as an additional parametric CAD tool because it allows the fuselage cross-sections, longitudinal stations, and global proportions to be adjusted quickly while preserving a smooth aerodynamic outer mould line considered, at this point, as already optimal. This makes it useful at the present stage, where the aircraft geometry is still being iterated together with the propulsion and stability analyses.

The preliminary fuselage length was estimated using the empirical relation

$$l_f = aW^c, \quad (8.1)$$

where the coefficients for a general aviation twin aircraft (though far it's still the most similar model) are taken as $a = 0.366$ and $c = 0.42$. Using the current maximum take-off mass estimate gives a fuselage length of approximately 9.89 m. The maximum fuselage diameter was set initially to 1.63 m but the required space for all the storage and systems was too low. The value was increased to 2.2 m, resulting in a fineness ratio of

$$\frac{l_f}{d} = \frac{9.89}{1.63} \approx 4.50. \quad (8.2)$$

This value provides a bulky but aerodynamically acceptable shape for the proposed VTOL aircraft.

Table 8.1: Main fuselage sizing parameters used for the CAD model.

Parameter	Value	Unit
Fuselage length	9.89	m
Maximum diameter	1.63	m
Fineness ratio	6.07	–
Approximate habitable length	4.3–5.0	m
Seat arrangement	1+1	–
Number of passengers	6	–

The cabin layout was defined around a compact 1 + 1 seat arrangement. A representative seat width of 0.5 m was adopted, with a longitudinal seat pitch close to 0.8–1.0 m. The aisle width was set at approximately 0.5 m, above the minimum value considered in the preliminary notes, while the aisle height was taken as approximately 1.5 m to maintain a usable internal volume. These values support a six-passenger cabin while leaving space for the cockpit, emergency exits, structural frames, and energy-system integration.

Table 8.2: Cabin layout assumptions for the fuselage CAD model.

Parameter	Value	Unit
Seat width	0.45	m
Seat pitch	0.8–1.0	m
Aisle width	0.50	m
Aisle height	1.90	m
Internal seated height target	1.70	m
Emergency-exit width	0.50	m
Emergency-exit height	0.70	m

The fuselage is divided into three main geometric regions. The nose and cockpit region use a rounded and progressive shape to avoid abrupt curvature changes and to preserve pilot visibility. The central region contains the passenger cabin, doors, battery modules, additional interfaces and is therefore kept nearly cylindrical to maximize usable volume. Between the cockpit and the seats is an additional cargo area that balances the weight. The rear cone closes the fuselage with a gradual reduction in cross-section, limiting flow separation and reducing unnecessary wetted area. As a preliminary guideline, the upper rear-cone angle should remain below approximately 10° – 12° , the lateral convergence angle below approximately 30° , and the lower rear angle below approximately 25° .

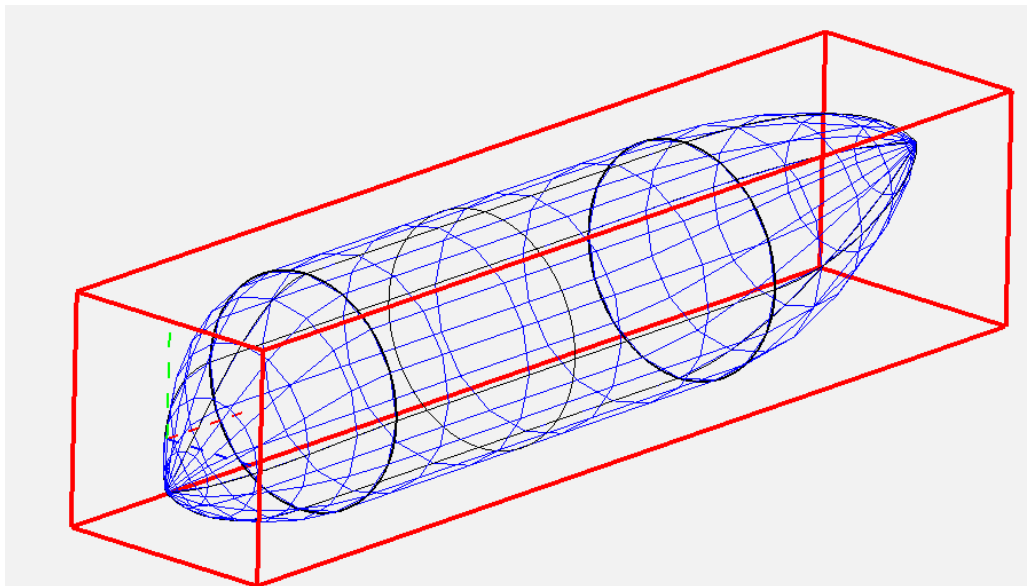


Figure 8.1: OpenVSP fuselage model.

Emergency constraints were also considered during the CAD layout. The aircraft is treated as a small-category VTOL aircraft, so the cabin must allow rapid evacuation after an emergency landing [12]. The preliminary arrangement therefore assumes one emergency exit on each side of the cabin. These exits must remain clear of the tilt-ducts and wing structure, must not force passengers to evacuate toward the rotor plane, and should remain usable after an inclined landing or a limited over-water emergency scenario. The battery compartment should also be kept away from the primary evacuation path. Regarding this, the rotor placement is still the most difficult task that would most surely need another iteration.

The cockpit geometry was reserved at the front of the fuselage, with an approximate cockpit length of 1.8 m. The windscreen and nose shape should support the required pilot field of view, especially for vertical and low-speed operations. Based on reference angle values, the cockpit curvature was formulated but further iterations should converge to a more decisive geometry.

A first aerodynamic estimate was made for the fuselage contribution in cruise. This additional study is an append to the previous aerodynamic analysis that leaves a good baseline value for future optimization. Using a cylindrical wetted-area approximation gives

$$S_{wet} \approx \pi dl_f = 50.64 \text{ m}^2. \quad (8.3)$$

At the cruise condition, with $\rho = 0.97787 \text{ kg/m}^3$ and $V = 85 \text{ m/s}$, the dynamic pressure is $q = 3532.6 \text{ N/m}^2$. Taking a Reynolds number of approximately 4.13×10^7 where the fuselage length was taken as linear reference dimension, the turbulent skin-friction coefficient is then estimated as $C_f = 2.42 \times 10^{-3}$. With a fuselage form factor of $F = 1.284$ and interference factor $Q \approx 1$, the resulting fuselage drag force is approximately 555 N. Referenced to a wing area of 13 m^2 , this corresponds to an estimated fuselage zero-lift drag contribution of

$$C_{D_0} \approx 1.21 \times 10^{-2}. \quad (8.4)$$

Overall, the OpenVSP fuselage model should be treated as a working geometric baseline rather than a final concept shape. The fuselage model was then imported to SolidWorks where the rest of the components were designed and assembled.

8.2 Ducts CAD Model

The integration of the enclosing circular duct performs three vital functions in system performance: it provides thrust augmentation of approximately 25% to 35% compared to a free rotor due to the pressure distribution at the inlet lip; it executes static pressure recovery along the diffuser; and it confines the flow, severely mitigating blade tip vortices. The duct is designed to operate efficiently under vertical flight conditions (VTOL), where axial flow predominates, and in horizontal cruise flight at a speed of 85 m/s.

The CAD modeling of the duct assemblies was driven by strict aerodynamic and geometric continuity requirements to ensure optimal thrust augmentation and minimal drag during cruise. The macro-geometry of the completed duct dictates a maximum outer diameter of 2.275 m and a total axial height of 1.100 m.

The inlet lip profile was modeled in the CAD software using spline curves with a G2 geometric continuity constraint. This methodological requirement is critical to prevent surface curvature discontinuities that would otherwise cause local peaks in the static pressure coefficient (C_p) and, consequently, induce premature boundary layer transition or flow separation at the inlet. The maximum thickness of the lip was strategically positioned between 15% and 20% of the chord, and the lip radius was defined at 4.7% of the internal diameter to optimize the suction pressure distribution before the throat, a configuration that maximizes the experimental figure of merit as validated by Abrego and Bulaga [21].

The outer wall of the duct was designed with a thin, supercritical aerodynamic contour, strictly maintaining a thickness-to-chord ratio of $t/c \leq 6\%$. The geometry incorporates a severely sharp trailing edge to suppress base aerodynamic wake generation and minimize parasitic drag during the horizontal cruise phase.

Internally, the diffuser's semi-opening angle was rigidly constrained to approximately 4° , and the expansion ratio was set to 1.097. This expansion ratio falls perfectly within the optimal 1.05 to 1.15 range established by Zhang and Barakos [22]. This geometric limit was enforced in the CAD model to prevent the internal expansion angle from approaching the critical 7° to 8° threshold, at which massive adverse pressure gradients and complete flow separation would occur, nullifying the duct's pressure recovery benefits.

Finally, downstream of the rotor plane and integrated directly into the diffuser's internal walls, three stationary guide vanes featuring a NACA 0010 profile were modeled. The integration of these stators serves a dual purpose: they provide the necessary structural mounting interface for the central electric motor hub, and they perform aerodynamic swirl recovery. By being positioned exclusively aft of the rotor, these vanes effectively attenuate the rotational momentum of the induced airflow, converting that wasted kinetic energy into a beneficial static pressure gain at the exhaust.

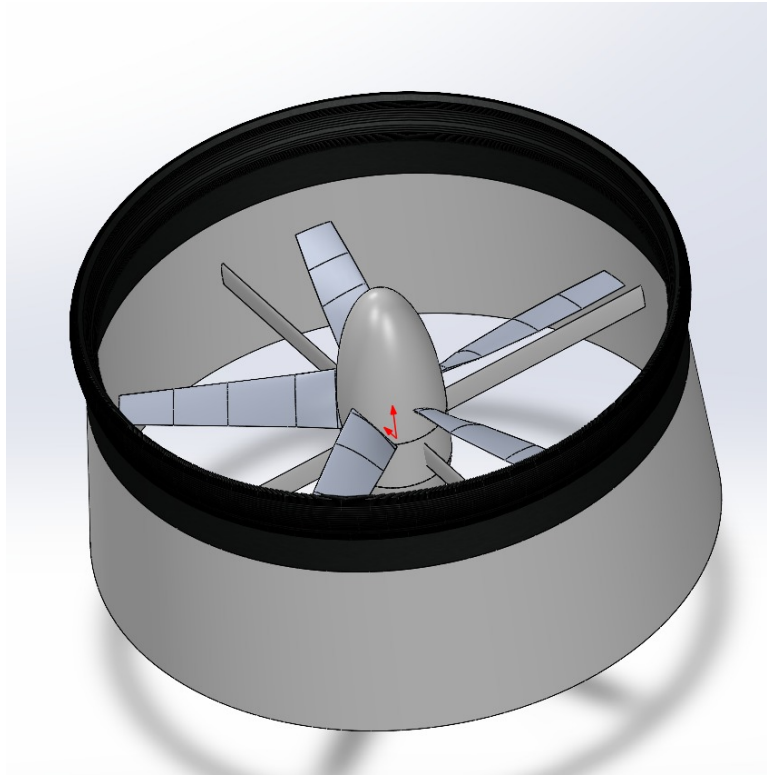


Figure 8.2: Duct CAD model

8.3 Seats CAD Model

The passenger cabin is equipped with lightweight, ergonomically designed seats tailored to the spatial constraints and comfort requirements of the aircraft. As illustrated in the CAD model, the seat features a composite-style outer shell with integrated padding to minimize overall mass while maintaining structural integrity and passenger support. Dimensionally, the seat is defined by an outer width of 0.43 m and a seating depth of 0.5 m, optimizing the available cabin volume. The main backrest structure stands at a height of 0.9 m, mounted on a robust lower support base with a height of 0.5 m, ensuring proper ergonomic positioning and adequate sightlines for the occupants throughout the flight.

8.4 Landing Gear CAD Model

A skid-type landing gear was adopted for the aircraft. This solution was selected due to its structural simplicity, robustness and suitability for a ducted VTOL configuration, where it is essential to ensure a minimum clearance between the ground and the ducts.

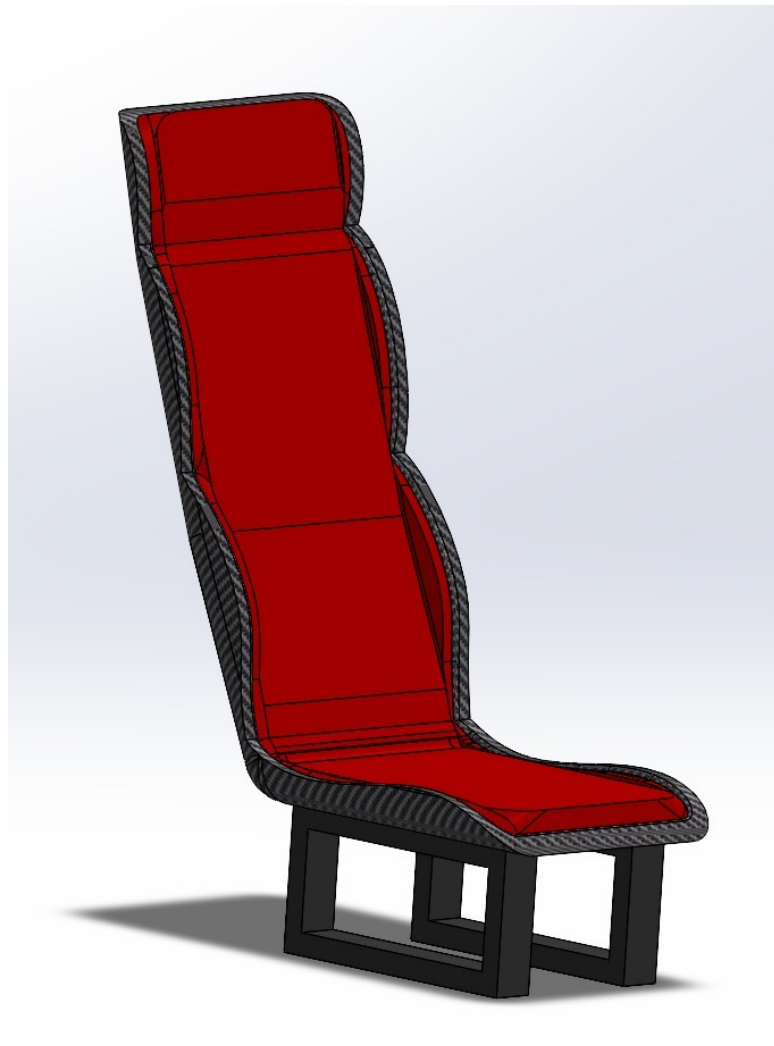


Figure 8.3: Seat CAD model

Each side of the skid has a longitudinal length of approximately 5150 mm, with a lateral spacing of around 2300 mm. The maximum height of the support is approximately 1400 mm, which is sufficient to ensure that the ducts do not come into contact with the ground.

One of the main criteria for the sizing of the skid landing gear was to include the CG within its support area with a large safety margin. This is a fundamental condition for ensuring the static stability of the aircraft when supported on the ground, preventing it from tipping forwards, backwards or laterally.

The longitudinal extent of the skid also ensures an adequate margin with respect to possible contact between the nose or tail and the ground.

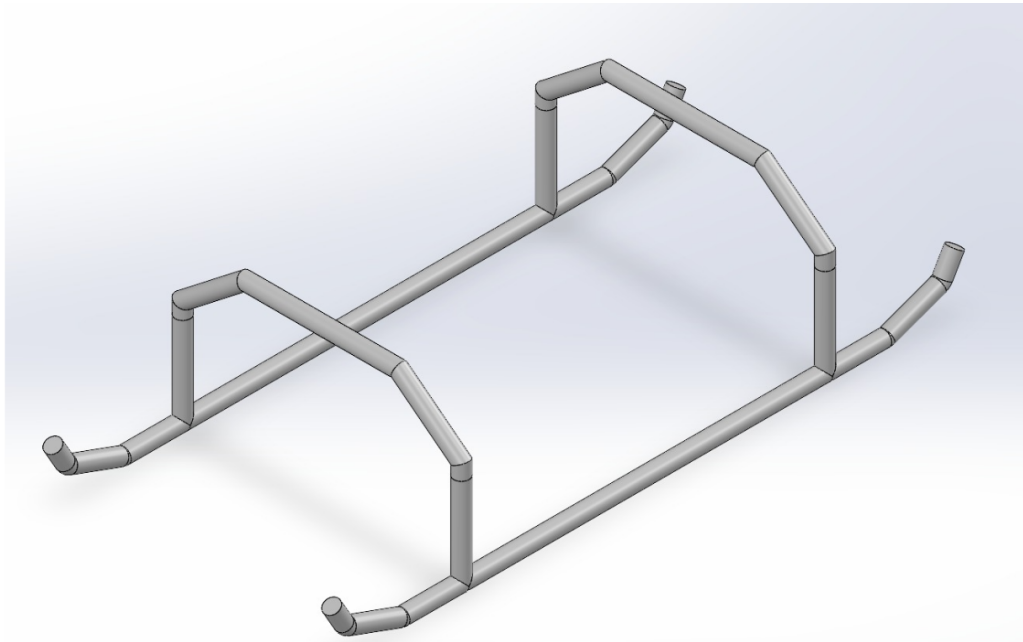


Figure 8.4: Landing gear CAD model

8.5 Tail CAD Model

Following preliminary aerodynamic assessments, significant concerns arose regarding the initial V-tail configuration. Due to the positioning of the propulsion system, the aft rotor group would inevitably generate highly turbulent downwash and wake characteristics directed into the V-tail surfaces, severely compromising their aerodynamic effectiveness.

To mitigate these interference effects, an alternative empennage layout was developed in CAD, as illustrated in Figure 8.5. The revised design transitions to a T-tail configuration, maintaining comparable geometric sizing and surface areas to the baseline calculations. By elevating the horizontal stabilizer atop a robust vertical stabilizer, the primary lifting and control surfaces are positioned well above and in between the rotor discs, operating in relatively unperturbed freestream airflow during forward flight. Consequently, a comprehensive aerodynamic and structural reevaluation of this T-tail configuration is designated as a high-priority objective for the next design phase.

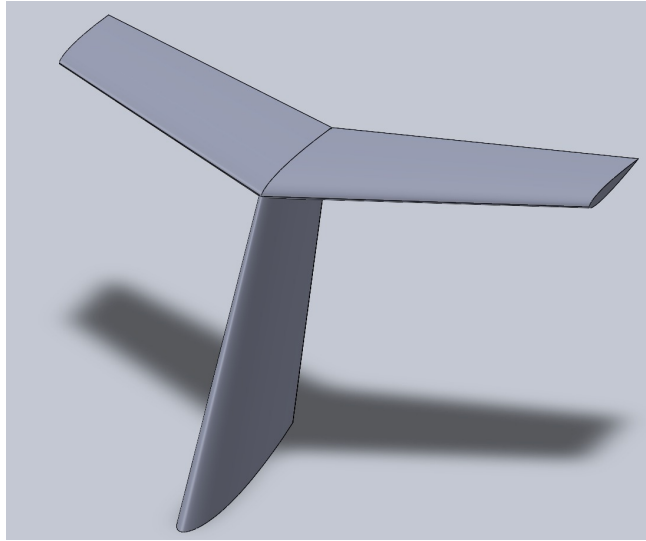


Figure 8.5: T-tail CAD model.

9 Stability

The evaluation of the static stability (longitudinal, lateral, and directional components) was based on the analysed aircraft configuration, with the modeled V-tail, taking into account the position of lift-generating surfaces and other key internal components, such as batteries, hydrogen systems, motors, rotors and passengers. It will only be valid for cruise conditions, for which the analysis was made. Finally, an evaluation of stability in vertical phases was made.

9.1 Longitudinal Stability

Longitudinal stability represents the inherent tendency of an aircraft to return to its equilibrium angle of attack (α) following an external disturbance, such as wind gusts or turbulence [23]. If a perturbation alters the angle of attack, the aircraft must automatically generate a restoring pitching moment to recover its original steady flight condition without pilot intervention. Mathematically, static longitudinal stability requires that the partial derivative of the pitching moment coefficient (C_M) with respect to the angle of attack is negative:

$$\frac{\partial C_M}{\partial \alpha} \equiv C_{M_\alpha} < 0 \quad (9.1)$$

Furthermore, it is recommended that this value remains within the interval:

$$-1.5 \leq C_{M_\alpha} \leq -0.16 \quad (9.2)$$

Figure 9.1 shows a portion of the relationship between C_M and α based on XFLR5 simulations at 85 m/s. It clearly confirms that the desired derivative is negative for all relevant values of α .

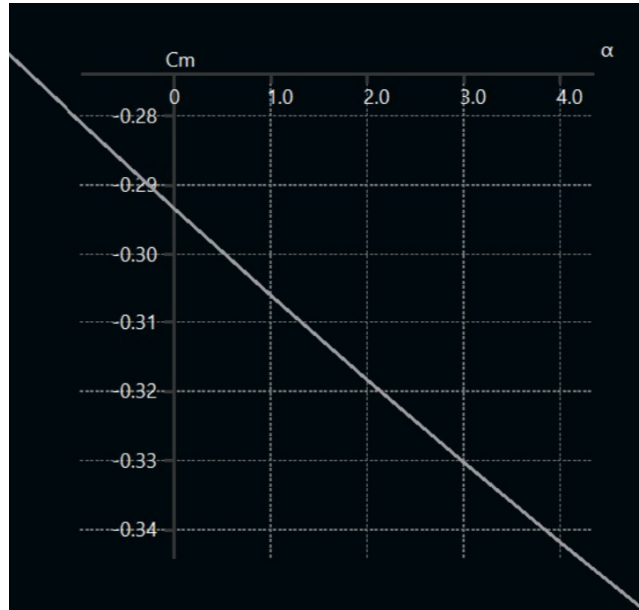


Figure 9.1: C_M vs α .

The software also presents the value of C_{M_α} to be equal to $-0.75/^\circ$, well within the recommended margins.

Also, a direct quantification of this longitudinal stability is provided by the static margin (SM), which measures the relative distance between the neutral point and the center of gravity and is defined as:

$$SM = \frac{X_{NP} - X_{CG}}{\bar{c}}, \quad (9.3)$$

where X_{NP} is the position of the neutral point, X_{CG} is the center of gravity position, and $\bar{c}$ is the mean aerodynamic chord of the wing.

To ensure stability, SM should be positive, meaning that the center of gravity must be located ahead of the neutral point. Resorting to the XFLR5 geometrical simulation of the model (A.6), a value of 11.2% was obtained for SM ($X_{CG} = 4.537m$ and $X_{NP} = 4.709m$), indicating a highly stable aircraft configuration.

9.2 Lateral Stability

Lateral stability is related to the aircraft's rotation along its longitudinal axis, roll [23]. It requires that the derivative of the aircraft's rolling moment coefficient with respect to the sideslip angle (β) is negative, in order for the aircraft to generate a restoring rolling moment that opposes the sideslip, i.e.:

$$\frac{\partial C_l}{\partial \beta} \equiv C_{l_\beta} < 0 \quad (9.4)$$

Additionally, it is recommended that this value remains within the interval:

$$-0.9 \leq C_{l_\beta} \leq -0.05 \quad (9.5)$$

Through the XFLR5 simulation, the curve between C_l and β is obtained (Figure 9.2), yielding a derivative value of $C_{l_\beta} = -0.14/^\circ$, which falls safely within the set bounds. The graph displays a negative slope as desired, confirming both criteria for lateral stability.

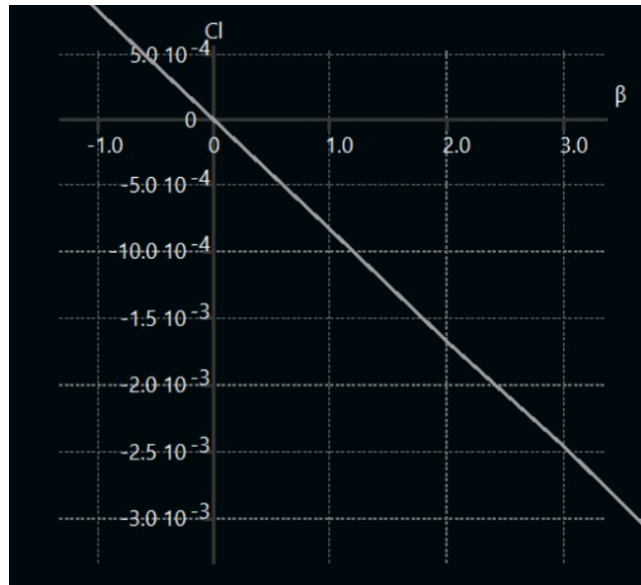


Figure 9.2: C_l vs β .

9.3 Directional Stability

Lastly, directional stability concerns the aircraft's rotational behavior about its vertical (yaw) axis [23]. It dictates that the derivative of the directional moment coefficient with respect to the sideslip angle (β) must be positive, ensuring that the airframe generates a restoring yawing moment to counter the sideslip, specifically:

$$\frac{\partial C_n}{\partial \beta} \equiv C_{n\beta} > 0 \quad (9.6)$$

Furthermore, conventional design guidelines suggest that this parameter fall within the following range:

$$0.08 \leq C_{n\beta} \leq 0.28 \quad (9.7)$$

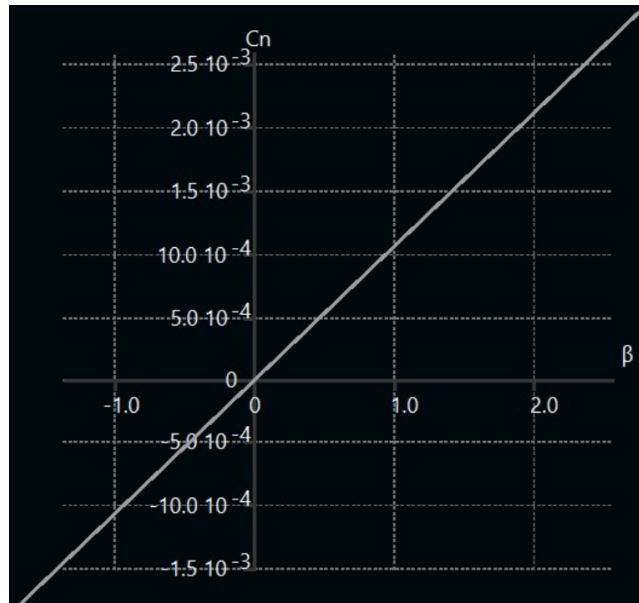


Figure 9.3: C_n vs β .

Utilizing the XFLR5 simulation, the relationship between C_n and β was established (Figure 9.3), yielding a derivative value of $C_{n\beta} = 0.07$. Although this result lies marginally below the recommended lower bound, the minor deviation is deemed acceptable for this preliminary design phase. Crucially, the plot displays a strictly positive slope, thereby satisfying the primary criterion for directional static stability.

9.4 Vertical Stability in VTOL Configuration

Vertical stability during the VTOL and Hover phases is primarily governed by the longitudinal alignment between the aircraft's center of gravity (X_{CG}) and the total center of thrust (X_{CT}) generated by the rotors. For an aircraft to exhibit inherent static stability in a hover, these two points should ideally coincide to eliminate any unbalanced moments.

The longitudinal thrust centre X_{CT} is defined as the thrust-weighted average of the individual rotor positions along the fuselage axis. Since all six rotors produce equal thrust $T_i = T/6$, the weighting coefficients are uniform and the expression reduces to a simple arithmetic mean of the rotor station coordinates:

$$X_{CT} = \frac{\sum_{i=1}^6 T_i x_i}{\sum_{i=1}^6 T_i} = \frac{1}{6} \sum_{i=1}^6 x_i = \frac{2x_1 + 2x_2 + 2x_3}{6} = \frac{x_1 + x_2 + x_3}{3} \quad (9.8)$$

where x_1 , x_2 , and x_3 denote the longitudinal distances from the fuselage nose to each pair of symmetrically opposed rotors. Substituting the rotor station locations:

$$X_{CT} = \frac{2300 + 5300 + 7925}{3} = \frac{15525}{3} = 5175 \text{ mm} = 5.175 \text{ m} \quad (9.9)$$

From the preliminary mass and propulsion layout analyses of the proposed aircraft, a significant longitudinal discrepancy undoubtedly exists. This offset generates a continuous pitch moment, rendering the baseline configuration inherently unstable without active control intervention.

To mitigate this instability and ensure equilibrium, the following control strategy is proposed for future optimization beyond this preliminary design phase: Active attitude control must be achieved by implementing a dynamic thrust differential between the forward and aft rotor groups. By modulating the RPM or collective pitch of specific rotors, the propulsion system can generate a continuous counter-moment to balance the offset. Alternatively, a structural redesign shifting the longitudinal coordinates of the rotor hubs could be explored to achieve passive equilibrium and eliminate the offset entirely.

10 Emissions & Noise

To rigorously assess the environmental profile of the proposed hybrid eVTOL aircraft, this chapter evaluates its lifecycle Global Warming Potential (GWP) and acoustic footprint. Given that the propulsion system operates with zero direct emissions during the flight phase [24], the environmental assessment shifts toward a cradle-to-grave framework, capturing the upstream carbon debt and the localized operational impacts [25, 26].

10.1 Emissions for Production and Replacement

The primary environmental trade-off of hybrid-electric configurations lies in the shifting of emissions from the operational phase to the material extraction and manufacturing stages. Therefore, quantifying the production and replacement footprints is critical to determining the true net sustainability of the aircraft. The analysis is framed around the following technical boundaries:

- **System Boundaries (LCA):** Due to the lack of consolidated aerospace-grade data regarding the manufacturing and End-of-Life (EOL) decommissioning of novel composite airframes, the scope of this assessment is strictly delimited to the primary energy storage and conversion architectures [25, 26], namely, the Lithium-Sulfur (Li-S) battery pack and the Proton Exchange Membrane Fuel Cell (PEMFC).
- **Technological Constraints:** While the hybrid architecture mitigates the low specific energy constraints inherent to pure battery systems [27], the production of high-power electrical components and complex thermal management systems [28] introduces a significant initial environmental burden that must be amortized over the aircraft's commercial operation.
- **Amortization and Lifespan Metrics:** Because these energy systems degrade over time due to cyclic thermal and electrical stress, their manufacturing emissions cannot be treated as a permanent capital expense. However, different technologies exhibit distinct degradation mechanisms and operational lifetimes. Therefore, the manufacturing emissions associated with each subsystem are amortized individually according to their respective useful lives. For this study, a useful life of **1000 cycles** is assumed for the Lithium-Sulfur (Li-S) battery pack. For the Proton Exchange Membrane Fuel Cell (PEMFC) system, considering an estimated lifespan of 5000 flight hours and an average flight duration of 35 minutes, it is assumed to operate for approximately **8571 cycles**. This approach provides a more realistic emission-per-flight allocation and better reflects the long-term durability advantages of fuel cell technology.

10.1.1 Methodology for Emissions Estimation (Li-S Batteries and Fuel Cells)

To evaluate the GWP of the project, the total impact is mathematically decoupled into two core dimensions: the **Fixed Production/Replacement Component** and the **Variable Operational/Refueling Component** [26].

Step 1: Emissions from Energy Systems Production

The manufacturing phase entails the highest initial carbon investment per unit of nominal capacity or power. The Diluted Production Impact Equation is defined as:

$$\text{Diluted Emission} = \frac{\text{System Size [kWh or kW]} \times \text{Emission Factor} \left[\frac{\text{kgCO}_2\text{-eq}}{\text{kWh or kW}} \right]}{\text{Useful Life Cycles } (N)} \quad (10.1)$$

The baseline emission factors used to compute this fixed overhead are derived from current lifecycle inventories and are compiled in Table 10.1. Note that for the PEMFC, the emission factor is based on the stack's power rating rather than energy capacity.

Table 10.1: Baseline emission factors for the manufacturing of energy storage and conversion systems.

Energy Technology	Emission Factor	Base Reference
Lithium-Sulfur Battery (Li-S)	105.0 $\left[\frac{\text{kgCO}_2\text{-eq}}{\text{kWh}} \right]$	Teah et al. [29]
Fuel Cell (H ₂ - PEMFC)	24.17 $\left[\frac{\text{kgCO}_2\text{-eq}}{\text{kW}} \right]$	Mori et al. [30]

* Note: The PEMFC emission factor is derived from a 48 kWel stack emitting 1160 kgCO₂-eq during manufacturing.

Step 2: Emissions from the Use and Refueling Phase

Although the aircraft generates zero direct greenhouse gases in flight [24], the indirect operational carbon footprint depends exclusively on the carbon intensity of the electricity grid used to charge the battery pack and the industrial pathway chosen for hydrogen production.

(A) Li-S Battery Recharge (Power Grid Footprint)

The operational impact varies according to the energy mix of the geographical region where the aircraft plugs into the vertiport charging infrastructure [31]. The applicable reference values based on the European Environment Agency (EEA) data are as follows:

- **Sweden (Predominantly Renewable):** 8 gCO₂-eq/kWh
- **Portugal:** 119 gCO₂-eq/kWh
- **Spain (Continental):** 158 gCO₂-eq/kWh (Note: Other local sources report approximately 131 gCO₂-eq/kWh [32]).
- **European Union Average (EU-27):** 210 gCO₂-eq/kWh
- **Italy:** 225 gCO₂-eq/kWh
- **Germany:** 329 gCO₂-eq/kWh
- **Insular Regions / Isolated Systems (e.g., Balearic Islands):** Due to a heavy historical reliance on localized fossil fuel thermal generation, the carbon intensity is high, typically fluctuating between 400–500 gCO₂-eq/kWh. For conservative modeling within this project, a baseline value of 500 gCO₂-eq/kWh is adopted [33].

(B) Fuel Cell Refueling (Hydrogen Production)

For the Liquid Hydrogen (LH₂) supply chain, the Well-to-Tank carbon footprint is highly sensitive to the primary energy source and chemical extraction process [27]:

- **LH₂ via Electrolysis (Green Hydrogen):** 0.03 kgCO₂-eq/kWh (derived from renewable electricity) [34].
- **LH₂ via Natural Gas Reforming (Grey Hydrogen):** 0.330 kgCO₂-eq/kWh (derived from fossil fuel steam reforming) [35].

10.1.2 Project Emissions Calculation

Based on the definitive sizing parameters of the developed eVTOL platform, the aircraft incorporates a 26.26 **kWh** Lithium-Sulfur (Li-S) battery pack for production assessment and a 200 **kW** Proton Exchange Membrane Fuel Cell (PEMFC) stack, which supplies an estimated 51.77 **kWh** of energy per typical flight mission. During flight, the battery effectively consumes 20.2 **kWh**.

Fixed Calculation (Component Manufacturing)

The absolute upfront carbon debt embedded in the vehicle's manufacturing phase is calculated as follows:

- **Li-S Battery Pack Production Overhead:**

$$26.26 \text{ kWh} \times 105 \text{ kgCO}_2\text{-eq/kWh} = 2757.30 \text{ kgCO}_2\text{-eq}$$

- **PEMFC Stack Production Overhead (Proportional to 200 kW):**

$$200 \text{ kW} \times \left(\frac{1160 \text{ kgCO}_2\text{-eq}}{48 \text{ kW}} \right) = 4833.33 \text{ kgCO}_2\text{-eq}$$

The emissions associated with each subsystem are amortized according to their respective expected lifecycles.

$$\text{Li-S Manufacturing Contribution} = \frac{2757.30}{1000} = 2.757 \text{ kgCO}_2\text{-eq/flight} \quad (10.2)$$

$$\text{PEMFC Manufacturing Contribution} = \frac{4833.33}{8571} = 0.564 \text{ kgCO}_2\text{-eq/flight} \quad (10.3)$$

Therefore, the total fixed manufacturing burden allocated to each flight becomes:

$$2.757 + 0.564 = \mathbf{3.321 \text{ kgCO}_2\text{-eq/flight}} \quad (10.4)$$

This value represents the lifecycle manufacturing impact of the onboard energy systems distributed over their expected operational lifetimes.

Variable Calculation (Flight Operation)

The model was applied to the Balearic Islands utilizing the local grid emission factor of 500 gCO₂-eq/kWh (0.500 kg/kWh). The analysis assumes a useful life of 1000 cycles for the Li-S battery pack and 8571 cycles for the PEMFC system.

Assuming a baseline mission scenario that consumes 20.2 kWh from the battery and 51.77 kWh of hydrogen energy for the fuel cell, the lifecycle carbon footprint per flight is calculated and compared across two distinct hydrogen infrastructure pathways:

1. Scenario A: Local Insular Grid + Green Hydrogen (Electrolysis)

$$\text{Flight Footprint}_{\text{Green}} = 3.321 + (20.2 \times 0.500) + (51.77 \times 0.03) \quad (10.5)$$

$$= 3.321 + 10.100 + 1.553 \quad (10.6)$$

$$= 14.974 \text{ kgCO}_2\text{-eq/flight} \quad (10.7)$$

2. Scenario B: Local Insular Grid + Grey Hydrogen (Natural Gas Reforming)

$$\text{Flight Footprint}_{\text{Grey}} = 3.321 + (20.2 \times 0.500) + (51.77 \times 0.330)$$

$$= 3.321 + 10.100 + 17.084$$

$$= 30.505 \text{ kgCO}_2\text{-eq/flight}$$

10.1.3 Emissions - Conclusion

The Life Cycle Assessment (LCA) applied to this hybrid UAM configuration highlights that direct zero-emission flight does not automatically guarantee absolute environmental sustainability. In isolated or fossil-dependent energy ecosystems, such as the Balearic Islands, the operational carbon intensity dramatically reshapes the aircraft's ecological profile.

When realistic component lifetimes are considered, the manufacturing contribution remains relatively low. Although the energy systems require significant upfront carbon investments, evaluating the manufacturing footprint over the estimated operational flights limits the amortization penalty. As a result, the total manufacturing burden allocated to each flight represents 3.321 kgCO₂-eq/flight.

Using the local Balearic electricity grid (0.500 kgCO₂-eq/kWh), battery recharging contributes approximately 10.100 kgCO₂-eq per flight. The hydrogen production pathway then becomes one of the dominant factors influencing overall lifecycle emissions. Green hydrogen produced via electrolysis contributes approximately 1.553 kgCO₂-eq per flight, whereas grey hydrogen produced through natural gas reforming contributes approximately 17.084 kgCO₂-eq per flight.

Consequently, the total lifecycle footprint reaches 14.974 kgCO₂-eq/flight for the green hydrogen scenario and 30.505 kgCO₂-eq/flight for the grey hydrogen scenario. This represents an increase of approximately 104% when fossil-derived hydrogen is utilized.

These results demonstrate that the environmental performance of hybrid hydrogen-electric eVTOL aircraft is driven predominantly by the carbon intensity of the supporting energy infrastructure. Therefore, the successful deployment of such aircraft in insular regions depends not only on advances in vehicle technology but also on the availability of low-carbon electricity and dedicated green hydrogen production pathways.

10.2 Noise Emissions

Acoustic impact represents a primary barrier to public acceptance and regulatory certification in urban centers. Unlike conventional localized emissions, the noise signature of an eVTOL aircraft directly correlates with high-population public health indicators, where continuous exposure to aeroacoustic profiles above 75 dB is linked to heightened cardiovascular stress and hypertension [26]. The following subsections detail the rotor tip speed optimizations, sound

pressure level (SPL) boundary distributions, and the mitigation strategies implemented in the vehicle's flight control laws to meet stringent municipal aeroacoustic constraints.

10.2.1 Analytical Aeroacoustic Framework and Modeling

To model the vehicle's acoustic signature during conceptual design without incurring heavy computational fluid dynamics (CFD) expenses, a distributed, noncoherent analytical architecture was implemented in MATLAB. Following the classic formulations of Gutin (1948) and Deming (1940), the overall noise profile is parsed into discrete harmonic loading (p_{mL}) and thickness (p_{mT}) pressure components.

The root-mean-square (RMS) loading noise pressure (p_{mL}) for a given harmonic (m) is driven by aerodynamic forces acting on the blades and is calculated as follows:

$$p_{mL} = \frac{mN_B\Omega}{2\sqrt{2\pi a\Delta S}} \left(T \cos \theta - \frac{Qa}{\Omega r_e^2} \right) J_{mN_B} \left(\frac{mN_B\Omega}{a} r_e \sin \theta \right) \quad (10.8)$$

The thickness noise pressure (p_{mT}), resulting from the periodic displacement of air by the physical volume of the advancing blade element, is defined by:

$$p_{mT} = -\frac{\rho(mN_B\Omega)^2 N_B}{3\sqrt{2\pi\Delta S}} c t_{max} r_e J_{mN_B} \left(\frac{mN_B\Omega}{a} r_e \sin \theta \right) \quad (10.9)$$

Where:

- Ω is the rotor's angular speed;
- θ is the angle between the source and the observer, as described by the model;
- ΔS is the distance between the source and the observer;
- T and Q represent the thrust and torque per rotor, respectively;
- N_B is the number of blades per rotor ($N_B = 5$);
- $m = 1$ isolates the primary fundamental blade passage frequency (BPF) harmonic ($n = m \times N_B = 5$);
- $a = 343$ m/s is the speed of sound, assumed constant;
- ρ is the local atmospheric density;
- $c = 0.185$ m and $t_{max} = 0.02$ m represent the chord length and the maximum blade thickness, respectively;
- $r_e = 0.8$ m is the effective aerodynamic rotor radius (80% of total radius);
- J_5 represents the first-kind Bessel function of order 5, capturing the directional focus of the harmonic energy.

Assuming noncoherent phase interactions between the aircraft's propulsive groupings, the total sound pressure level (SPL) is accumulated across the multi-rotor system ($N_R = 6$) relative to the standard human auditory baseline ($p_{ref} = 20 \mu\text{Pa}$):

$$\text{SPL} = 10 \log_{10} \left(N_R \frac{p_{mL}^2 + p_{mT}^2}{p_{ref}^2} \right) \quad (10.10)$$

The Table 10.2 below shows all the parameters used for the SPL calculations, regarding the two different operating conditions.

Table 10.2: Operating conditions used for the SPL calculations

Parameter	Symbol	Maximum Load	Cruise
Thrust per rotor	T [N]	4735	988.71
Torque per rotor	Q [N m]	750	500
Angular velocity	Ω [rad/s]	293.2	157.1
Air density	ρ [kg/m ³]	1.225	0.978

Because a detailed vortex-noise analysis is highly complex, it is not included in the present conceptual-level assessment.

10.2.2 Sound Pressure Level Distribution Analysis

Under maximum loading conditions (Figure 10.1), the SPL distribution displays a pronounced peak near the aircraft associated with elevated thrust generation and rotors' angular speed during hover and vertical flight. The increased aerodynamic loading amplifies the loading-noise component, producing the highest acoustic footprint encountered throughout the mission profile. This result is consistent with operational experience from both conventional rotorcraft and modern eVTOL demonstrators, where take-off and landing phases generally represent the critical acoustic conditions. Furthermore, for the most part, the SPL is over the 75 dB threshold mentioned earlier, which is not the case for some specific angles and distances.

Conversely, the cruise-flight results shown in Figure 10.2 indicate a substantial reduction in overall noise emissions. During forward flight, the required rotor thrust decreases considerably as aerodynamic lift generated by the wing supports the majority of the vehicle weight. The consequent reduction in rotor loading lowers both harmonic pressure components and decreases the resulting SPL throughout the observation field.

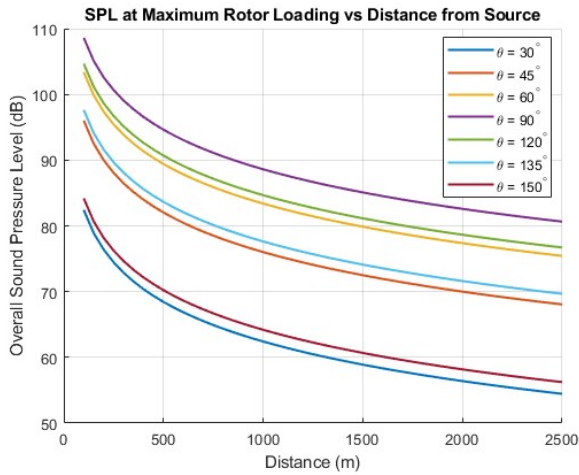


Figure 10.1: SPL for maximum rotor loading [A.7]

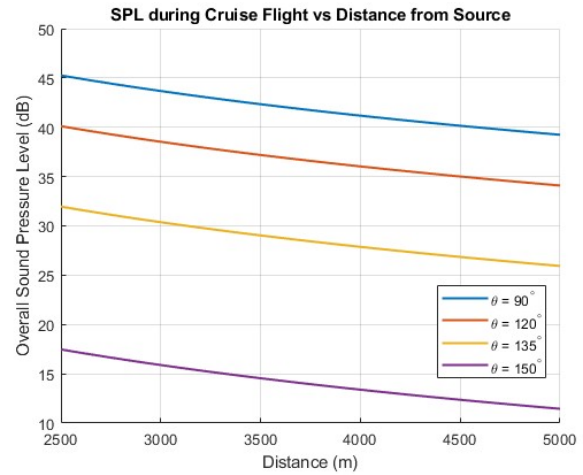


Figure 10.2: SPL for cruise flight conditions [A.7]

Overall, in both operating conditions, the maximum sound pressure level occurs near the rotor plane ($\theta \approx 90^\circ$), where the Bessel function term reaches its highest magnitude and the acoustic radiation is concentrated. The results indicate that the aircraft's acoustic signature remains concentrated within the VTOL phases, while cruise operations generate significantly lower community noise exposure.

10.2.3 Effect of Ducted Rotors on Noise Reduction

The selected tilt-duct architecture provides several inherent aeroacoustic advantages compared with equivalent open-rotor configurations. The primary benefit arises from the suppression of blade-tip vortex formation. In conventional propellers, the pressure difference between the pressure and suction sides of the blade generates strong tip vortices that contribute significantly to broadband noise production. The presence of the duct limits this cross-flow motion, which literature indicates can reduce the broadband noise component by 8 to 10 dB during approach phases [36]. In this case, doing a vortex noise analysis is highly complex; traditional semi-empirical models, like the one defined by Schlegel (1966), fail to provide useful insights as they neglect duct acoustic cut-off frequencies and boundary layer ingestion (BLI) effects.

Additionally, the duct acts as a partial acoustic shield, attenuating the direct propagation of rotor-generated pressure fluctuations toward external observers. Acoustic simulations of integrated urban ducted systems show that a portion of the acoustic energy is reflected, absorbed, or redirected by the duct geometry, which can contribute to an overall noise level reduction outside the propulsion system.

Furthermore, the duct also enables the use of smaller rotor diameters operating at lower tip speeds for a given thrust requirement due to the pressure-recovery effects generated inside the shroud. Since aerodynamic noise scales strongly with blade tip velocity following the $50 \log_{10}(N_2/N_1)$ scaling law, even a moderate 20% reduction in rotational speed produces a noticeable 4 dB acoustic benefit (growing to 8 dB at a 30% speed reduction).

These characteristics explain why ducted-fan concepts such as the Lilium Jet and Bell Nexus have consistently been proposed as low-noise solutions for urban air mobility applications. Although the additional structure introduces mass and drag penalties, the resulting reduction in community noise represents a significant advantage for operations near populated areas where public acceptance is a critical design driver.

10.2.4 Potential Noise Mitigation Strategies

There are no internationally finalized ICAO noise certification standards for UAM and eVTOL aircraft, and EASA is currently addressing noise requirements on an individual, case-by-case basis. In this section, only loudness was studied in a semi-empirical way. However, it is of extreme importance to reduce acoustic pollution as much as possible. To achieve this, future analyses must evaluate sound frequency alongside loudness. Human auditory perception varies significantly across different frequencies, meaning that the pitch and tonal characteristics of eVTOL rotors can greatly impact public annoyance, even at lower decibel levels.

Although the predicted noise levels exceed the desired community-noise target during maximum-load operation, several mitigation measures may be considered in future design iterations. The primary objective of these strategies is to reduce ground-level sound pressure levels during take-off and transition, which represent the acoustically critical phases of the mission.

A promising approach consists of implementing noise-aware flight control laws. Adaptive rotor-speed scheduling can minimize rotor RPM whenever full thrust is unnecessary, while optimized transition maneuvers can reduce the duration of high-power operating conditions. Similarly, trajectory optimization may be used to increase separation from populated areas during the noisiest phases of flight.

Additional reductions may be achieved through passive acoustic treatments. Acoustic liners integrated into the duct structure can attenuate tonal pressure fluctuations before they radiate to the surrounding environment, while localized fuselage insulation can improve passenger comfort by reducing the transmission of rotor-induced noise and

vibration into the cabin.

Ultimately, the results indicate that acoustic performance is strongly influenced by the relatively high rotor tip speed required during maximum-load operation. Future design refinements should therefore focus on reducing rotor rotational speed while combining advanced flight-control strategies and acoustic treatments to further mitigate the environmental impact of the vehicle.

11 Costs Analysis

The life-cycle cost (LCC) of an aircraft can be divided into three main categories: research, development, test, and evaluation (RDT&E) costs; acquisition cost (A); and operation and maintenance (O&M) costs [8].

$$C_{LCC} = C_{RDT\&E} + C_A + C_{O\&M} \quad (11.1)$$

For this study, it was decided to consider the first two components (RDT&E and A), which, when added together, represent the aircraft's selling price. We therefore chose not to carry out a detailed assessment of the operation and maintenance (O&M) costs, which consist of expenses related in this case with electricity and hydrogen, oils and lubricants, as well as those associated with the crew and maintenance team.

For this study, the primary focus of the engineering cost model is placed on the first two components (RDT&E and A), which, when added together, determine the aircraft's manufacturing selling price. A bottom-up, detailed assessment of operational and maintenance (O&M) variables (such as specific crew salaries, detailed insurance premiums, and exact hangar fees) is outside the immediate scope of the structural production phase. However, a macro-level estimation of key O&M expenses (including fuel, battery lifecycles, and routine maintenance) is introduced later in Subsection 11.2.1 solely to gauge the final concept's commercial feasibility and per-passenger ticketing viability.

Research, development, test, and evaluation costs (which will depend on the number of test aircraft developed; for this study, it was assumed that there would be two: $N_t = 2$) can be broken down into different factors: Airframe engineering, Development support, Engine and avionics, Manufacturing labor, Manufacturing materials, Tooling, Quality control, Flight test operations and Profit.

In addition, acquisition costs (in this case, dependent on the number of aircraft produced for sale; N_p) can also be expressed as the sum of various components: Airframe engineering, Engine and avionics, Manufacturing labor, Manufacturing materials, Tooling, Quality control and Profit.

It is therefore important to define a set of cost estimating relationships (CERs) that allow us to forecast the aforementioned costs [37], with the following variables: structural weight ($W_s = 3815.61 \text{ lbs}$), maximum speed ($V_{max} = 165.23 \text{ kts}$), and the number of aircraft produced (N_t or N_p).

11.1 Cost Estimating Relationships

11.1.1 Airframe engineering (RDT&E and A)

This applies to all engineering activities comprising the two design phases under consideration. According to empirical data, the airframe engineering cost is first expressed as the total number of hours, which can then be converted into a monetary value using a second equation (an appropriate hourly labor rate based on the year in which the study is being conducted - $Y = 2026$):

$$H_E = 4.860 \cdot W_s^{0.777} \cdot V_{max}^{0.894} \cdot N^{0.163} \quad (11.2)$$

$$CR_E = -5058 + 2.576 \cdot Y \quad (11.3)$$

11.1.2 Development Support Cost (RDT&E)

This portion relates to additional costs associated with the project development process, the amount of which can be estimated by:

$$C_D = 45.42 \cdot W_s^{0.630} \cdot V_{max}^{1.300} \quad (11.4)$$

11.1.3 Engine and Avionics Cost (RDT&E and A)

The costs associated with the propulsion and avionics systems were estimated based on average values for certain components of the propulsion system. The price of the selected EMRAX motor was not listed, so a value of \$20000 was assumed. In addition, the cost of the fuel cell (\$43680) and the battery (\$12253.86) was estimated based on studies that have already been conducted [38]. Thus, an equation can be derived to estimate these costs:

$$C_{EN} = 173993.86 \cdot N \quad (11.5)$$

11.1.4 Manufacturing Labor Cost (RDT&E and A)

This represents the number of hours required to manufacture and assemble the aircraft's structural frame, which can, in turn, be converted into a monetary value:

$$H_{ML} = 7.370 \cdot W_s^{0.820} \cdot V_{max}^{0.484} \cdot N^{0.641} \quad (11.6)$$

$$CR_{ML} = -4552 + 2.316 \cdot Y \quad (11.7)$$

11.1.5 Tooling Cost (RDT&E and A)

Costs related to equipment used in the construction of the aircraft may initially be reported in hours, but can be converted as described in the preceding points.

$$H_T = 5.99 \cdot W_s^{0.777} \cdot V_{max}^{0.696} \cdot N^{0.263} \quad (11.8)$$

$$CR_T = -5666 + 2.883 \cdot Y \quad (11.9)$$

11.1.6 Quality Control (RDT&E and A)

An important aspect of aircraft manufacturing is the assurance of quality. The cost associated with quality control can be expressed in hours (later converted to a monetary value) depending on the type of aircraft under consideration:

$$\begin{cases} H_{QC} = 0.076 \cdot H_{ML} \text{ (cargo aircraft)} \\ H_{QC} = 0.13 \cdot H_{ML} \text{ (others aircraft)} \end{cases} \quad (11.10)$$

Given the unconventional nature of the design under study (tilt-duct), we opted for the estimate with a ratio of 0.13.

11.1.7 Flight Test Cost (RDT&E)

Another important consideration is the test flights that must be conducted during the preliminary phase of the project, which also represent a cost that must be taken into account.

$$C_{FT} = 1243.03 \cdot W_s^{0.325} \cdot V_{max}^{0.822} \cdot N^{1.210} \quad (11.11)$$

Once estimates have been established for all relevant costs, it is important to set a target profit margin (15% was adopted). This allows us to calculate the cost of selling each unit:

$$C_U = \frac{1.15 \sum (C_{RDT\&E} + C_A)}{N_p} \quad (11.12)$$

11.2 Cost Assessment and Market Comparison

Having established this method for determining the unit cost (which depends on the number of aircraft produced for sale), we can then analyze (using a MATLAB code - A.8) how this cost varies as the number of aircraft produced increases.

In addition, we sought to understand how the design resulting from this concept would perform in the market. Thus, we gathered (in the Table 11.1) the prices of some aircraft currently on the market that, in some way, share similar characteristics with the concept developed.

Table 11.1: Acquisition cost per aircraft for each competitor.

Competitor	Acquisition Cost
Cessna 206 Turbo Stationair HD	\$795000 [39]
Pilatus PC 12 NGX	\$6772500 [40]
Piper M350	\$1217500 [41]
Piper M600	\$3207000 [42]
Airbus ACH130	\$5000000 [43]
Archer Midnight	\$5000000 [44]
Lilium Jet	\$10000000 [45]

These two analyses are summarized in the plot shown in the Figure 11.1.

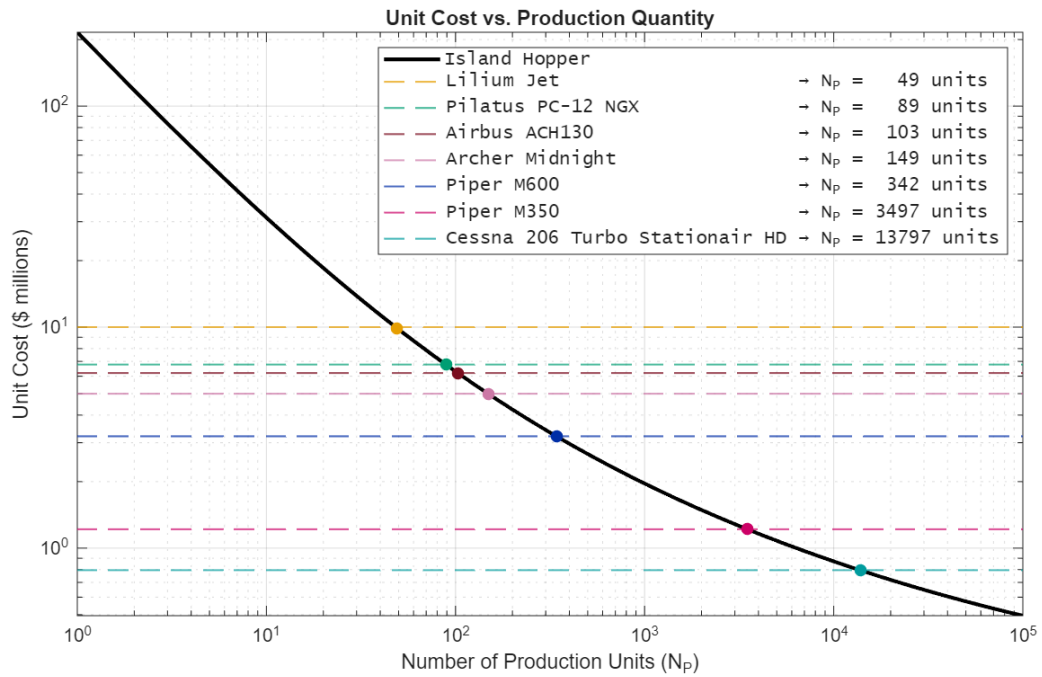


Figure 11.1: Acquisition cost per aircraft produced.

It can therefore be said that an increase in the number of aircraft produced results in a clear decrease in the unit price. A production of one aircraft will represent approximately \$215M. Thus, to compete commercially with other Urban Air Mobility projects, at least 49 units (Lilium Jet) or 149 units (Archer Midnight) would need to be produced. Even so, it is important to note that to compete with the Cessna 206 Turbo Stationair HD (a six-seat, single-engine piston aircraft), more than 13,797 units would need to be produced - a truly impressive number, since few aircraft have been produced on such a large scale (examples include the Cessna 172 or the Piper J-3 Cub [46]).

It is important to acknowledge that the cost estimates obtained through the aforementioned CERs yield a high value for the initial prototype. This is an inherent limitation of applying historical cost models-which are heavily derived from military aircraft and commercial production programs [8] to small-batch experimental UAM projects. These models significantly penalize the first units due to the full amortization of non-recurring engineering and heavy industrial tooling costs.

In practice, initial UAM development phases often rely on leaner prototyping methodologies, where costs are mitigated through digital twin simulation, modular testing, and sub-scale validation [47], rather than the "hard-tooling" approach assumed by mass-production regressions [37]. Therefore, the \$215M figure represents a worst-case theoretical scenario for a first-of-its-kind prototype, and should be viewed as an upper bound rather than an operational reality for this UAM concept.

Furthermore, given the iterative nature of the design process for a new aircraft, it was deemed prudent to analyze how changes in structural weight and maximum speed affect the unit price (Figure 11.2). To this end, it was assumed that 10,000 units would be produced.

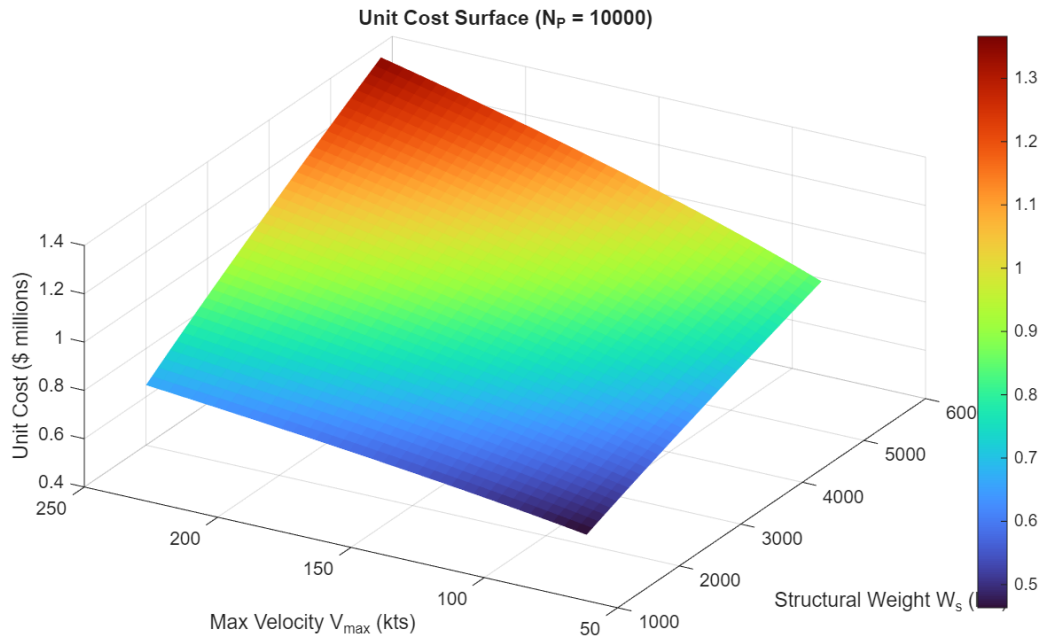


Figure 11.2: Impact of changes in structural weight and maximum speed on the purchase price.

This second analysis shows that the unit price will decrease as both the structural weight and the maximum speed are reduced. Thus, although reducing the maximum speed may not yield any benefits, future iterations could aim to reduce the weight instead.

11.2.1 Operational Feasibility and Per-Passenger Cost

To evaluate the economic feasibility of the proposed concept, its mass-production unit cost was benchmarked against the Archer Midnight (Table 11.1), yielding an estimated acquisition cost of approximately \$5M per aircraft. Regarding vehicle longevity, literature suggests a conservative lifecycle estimate of 10000 flight hours for UAM vehicles [48]. Given our standard flight duration of 33 minutes, a single vehicle could theoretically complete approximately 18,181 flights over its lifespan, translating to a baseline asset amortization of \$275.01 per flight, or \$45.84 per passenger.

However, the actual operational expenditure must account for critical recurring variables. As a preliminary O&M approximation, it is estimated that the batteries will need to be replaced at least every 1000 flight hours. Furthermore, additional recurring expenses (including hydrogen fuel, electricity, and routine maintenance) are projected to double the base amortization cost of the asset. Taking these operational variables into account, the total reference cost per passenger amounts to approximately \$91, providing a baseline for commercial viability.

Ultimately, even when applying a robust 50% profit margin for service implementation, the final ticket price per passenger (\$136.5) remains significantly below initial estimates, provided the concept successfully transitions into mass production.

12 Conclusion

This report developed and assessed a conceptual proposal for a premium Urban Air Mobility system for inter-island transportation in the Balearic Archipelago. The market and operational analyses confirmed that the Ibiza-Palma de Mallorca and Palma de Mallorca-Menorca routes represent a relevant use case for a fast, low-noise, and low-emission VTOL service.

The main achievement of the work was the definition of a technically plausible aircraft concept whose design point remains within the feasible design space as per the latest run on the *Aircraft Design Tool*. The selected tilt-duct configuration satisfies the reference mission while remaining compatible with the main operational and regulatory constraints considered throughout the report. The aircraft is able to perform the mission with one pilot and six passengers, while preserving a configuration that is consistent with distributed electric propulsion and future low-emission aviation objectives.

A central outcome of the project is the adoption of a hybrid electric-fuel-cell architecture. Although a fully electric aircraft was investigated, its feasibility was found to depend strongly on optimistic battery assumptions. The retained solution combines a Li-S battery pack for VTOL, hover, and transient power with a hydrogen PEMFC subsystem for cruise and climb. This architecture remains aligned with the initial objective of eliminating direct in-flight CO₂ emissions. The remaining environmental impacts are shifted mainly to hydrogen production, energy acquisition, and battery replacement, which surpasses the aviation domain to the economical and technological ones.

The hybrid architecture also showed promising scalability. The estimated additional hydrogen-storage mass for range extension is approximately 2.5 kg per extra 10 km of cruise range under the assumptions used in the report. This value is preliminary and will likely increase when secondary electrical loads, integration penalties, and refined reserves are included, but it still indicates one of the strongest advantages of the hybrid solution compared with a purely battery-based aircraft.

The noise and emissions studies also support the relevance of the concept. The estimated noise levels are within a promising range for a UAM aircraft, although the current assessment relies on simplified approximations and conventional rotor assumptions. Since the final aircraft uses ducted rotors, the actual acoustic behavior may differ and could potentially improve if the duct geometry is properly optimized. This reinforces the need for a dedicated aeroacoustic study in future iterations.

The cost analysis produced another important conclusion. Initial unit costs are high, as expected for a first-of-a-kind aircraft using historical cost-estimating relationships, but the cost per passenger becomes more competitive when larger production volumes and high utilization are considered. The results suggest that the concept may not need to remain exclusively in a luxury segment and could potentially target a medium-high market segment, provided that operating costs, infrastructure costs, and maintenance requirements remain controlled.

The main limitation of the current design is geometric compactness. Although the tilt-duct architecture offers advantages in safety and integration, the resulting aircraft still has significant dimensions, particularly because of the wing and duct arrangement required to satisfy the mission and VTOL power requirements. This means that the initial objective of a highly compact VTOL aircraft was only partially achieved, and compatibility with existing heliports or constrained vertiport infrastructure must be studied further.

Overall, the aircraft can perform the reference mission under the assumptions adopted in this conceptual design.

The project therefore supports the feasibility of a hybrid tilt-duct UAM aircraft for Balearic inter-island transport, while also identifying the main areas where higher-fidelity analysis is required before the concept can be considered mature.

12.1 Future Work

The difficulties encountered during this conceptual development provide a clear direction for future work. The first priority should be to improve the aerodynamic and propulsive fidelity of the design. The ducted rotors should be assessed with higher-order aerodynamic tools, including CFD, to verify hover performance, transition behavior, rotor-duct interaction, tip-clearance effects, and acoustic emissions. The rotor geometry, blade profiles and duct shape should then be optimized together rather than treated as independent components.

A second priority is structural and mass validation. Several subsystem masses were necessarily estimated using simplified methods, and these assumptions should be replaced by preliminary structural sizing. The wing, canard, tail, fuselage, ducts, tilting mechanisms, and landing-gear attachments require dedicated analysis to confirm that the assumed mass distribution is realistic. Particular attention should be given to the concentrated loads introduced by the tilt-duct units.

The hybrid powertrain also requires further refinement. Future work should define the hydrogen tank geometry, pressure-vessel installation, fuel-cell cooling system, battery C-rate capability, high-voltage protection, wiring layout, and failure-isolation strategy in greater detail. A mission-level energy-management simulation should also be implemented to evaluate battery state of charge, fuel-cell operating points, reserve margins, and off-design scenarios such as diversions, missed approaches, or degraded propulsion operation.

The economic and environmental assessments should also be expanded. The cost model should include operation and maintenance, hydrogen or electricity supply, infrastructure, crew, battery replacement, and fleet utilization. These factors are essential to estimate the real unit cost, cost per passenger, and market positioning of the aircraft. The environmental analysis should also be extended to include hydrogen production, battery manufacturing, component replacement, and end-of-life treatment.

Finally, future configuration studies should investigate whether a smaller aircraft could provide a better balance between compactness, cost, and operational flexibility. One possible direction is to reduce the capacity from six passengers to four passengers, which could reduce the overall dimensions, simplify the fuselage, and potentially remove the canard as a structural dependency. Alternative propulsion layouts should also be considered, such as four ducts or two larger and more powerful ducted units mounted primarily on the main wing. These alternatives emerged from the main design challenges identified during the project and may lead to a more compact and operationally flexible second iteration of the concept.

A Appendix

A.1 AHP Method - Full Tables

Main Criteria Relative Scores					
C/C	Performance	Emissions	Costs	Comfort	Maintenance
Performance	1	2	3	1	3
Emissions	0.5	1	1.5	0.5	1.5
Costs	0.333	0.667	1	0.333	1
Comfort	1	2	3.003	1	3
Maintenance	0.333	0.667	1	0.333	1
Final Percentages (%)	0.316	0.158	0.105	0.316	0.105

Table A.1: Pairwise comparison matrix for the main design criteria.

Subcriteria Relative Scores					
Performance	Range	Cruise Speed	Costs	Fixed	Variable
Range	1	2	Fixed	1	7
Cruise Speed	0.5	1	Variable	0.143	1
Final (%)	0.667	0.333	Final (%)	0.875	0.125

Emissions	Pollutant	Acoustic	Comfort	Stability	Cabin Noise
Pollutant	1	0.5	Stability	1	1
Acoustic	2	1	Cabin Noise	1	1
Final (%)	0.333	0.667	Final (%)	0.5	0.5

Table A.2: Detailed sub-criteria assessment matrices arranged by category.

Alternative Designs Relative Scores															
Range					Variable Costs					Acoustic					
TR	TW	TD	FW		TR	TW	TD	FW		TR	TW	TD	FW		
Tilt-rotor	1	2	3	2	Tilt-rotor	1	1	2	3	Tilt-rotor	1	1	0.5	4	
Tilt-wing	0.5	1	1.5	1	Tilt-wing	1	1	2	3	Tilt-wing	1	1	0.5	4	
Tilt-duct	0.333	0.667	1	0.667	Tilt-duct	0.5	0.5	1	1.5	Tilt-duct	2	2	1	8	
Fixed-wing	0.5	1	1.499	1	Fixed-wing	0.333	0.333	0.667	1	Fixed-wing	0.25	0.25	0.125	1	
Ratio	0.429	0.214	0.143	0.214	Ratio	0.353	0.353	0.176	0.118	Ratio	0.235	0.235	0.471	0.059	
Cruise Speed					Maintenance					Stability					
TR	TW	TD	FW		TR	TW	TD	FW		TR	TW	TD	FW		
Tilt-rotor	1	1	0.5	3	Tilt-rotor	1	2	2	0.5	Tilt-rotor	1	1	0.5	0.25	
Tilt-wing	1	1	0.5	3	Tilt-wing	0.5	1	1	0.25	Tilt-wing	1	1	0.5	0.25	
Tilt-duct	2	2	1	6	Tilt-duct	0.5	1	1	0.25	Tilt-duct	2	2	1	0.5	
Fixed-wing	0.333	0.333	0.167	1	Fixed-wing	2	4	4	1	Fixed-wing	4	4	2	1	
Ratio	0.231	0.231	0.461	0.077	Ratio	0.25	0.125	0.125	0.5	Ratio	0.125	0.125	0.25	0.5	
Fixed Costs					Pollutants					Cabin Noise					
TR	TW	TD	FW		TR	TW	TD	FW		TR	TW	TD	FW		
Tilt-rotor	1	3	2	0.5	Tilt-rotor	1	1	2	3	Tilt-rotor	1	1	0.5	4	
Tilt-wing	0.333	1	0.667	0.167	Tilt-wing	1	1	2	3	Tilt-wing	1	1	0.5	4	
Tilt-duct	0.5	1.499	1	0.25	Tilt-duct	0.5	0.5	1	1.5	Tilt-duct	2	2	1	8	
Fixed-wing	3	5.988	4	1	Fixed-wing	0.333	0.333	0.667	1	Fixed-wing	0.25	0.25	0.125	1	
Ratio	0.207	0.087	0.13	0.522	Ratio	0.353	0.353	0.176	0.118	Ratio	0.235	0.235	0.471	0.059	

Table A.3: Alternative design assessment matrices for the Balearic UAM mission.

Final Scores	
Tilt-rotor	0.265
Tilt-wing	0.195
Tilt-duct	0.279
Fixed-wing	0.256

Weighting Criteria	Range	C. Speed	Fixed C.	Var. C.	Pollut.	Acoust.	Stab.	C. Noise	Maint.
Weight	0.2108	0.1052	0.0919	0.0131	0.0526	0.1054	0.1580	0.1580	0.1050
Weight (%)	21.1%	10.5%	9.2%	1.3%	5.3%	10.5%	15.8%	15.8%	10.5%

Table A.4: Final AHP performance scores and global weighting distribution.

A.2 Energy Budget Estimation Methodology

This section presents the analytical framework employed to estimate the total energy consumption of the hybrid-electric aircraft across all phases of a representative mission profile. The mission is decomposed into five distinct segments: hover, vertical take-off and landing (VTOL), climb, cruise, and descent (with regenerative energy recovery). Each segment is analysed independently, and the results are subsequently summed to yield the total energy budget.

A.2.1 Hover and VTOL Phase

Lift and Thrust Requirements

For steady, level hover, the total lift generated by the rotor system must equal the aircraft weight:

$$L_{\text{hover}} = W \quad (\text{A.1})$$

A structural and aerodynamic margin of 20% is applied to define the maximum lift condition:

$$L_{\text{max}} = 1.2 L_{\text{hover}} \quad (\text{A.2})$$

Assuming the thrust is distributed equally among the n rotors, the thrust per rotor in hover and at maximum load is, respectively:

$$T_{\text{hover}} = \frac{L_{\text{hover}}}{n} \quad (\text{A.3})$$

$$T_{\text{max}} = \frac{L_{\text{max}}}{n} \quad (\text{A.4})$$

Ideal Rotor Power

The ideal (actuator-disc) power required per rotor is derived from momentum theory. For a rotor of disc area A_{rotor} operating in air of density ρ , the ideal power associated with a given thrust T is:

$$P_{\text{ideal}} = \sqrt{\frac{T^3}{2\rho A_{\text{rotor}}}} \quad (\text{A.5})$$

Accordingly, the total ideal power for the full rotor system in hover and at maximum thrust conditions is:

$$P_{\text{hover, ideal}} = n \sqrt{\frac{T_{\text{hover}}^3}{2\rho A_{\text{rotor}}}} \quad (\text{A.6})$$

$$P_{\text{max, ideal}} = n \sqrt{\frac{T_{\text{max}}^3}{2\rho A_{\text{rotor}}}} \quad (\text{A.7})$$

where ρ is evaluated at sea level under International Standard Atmosphere (ISA) conditions, representing the worst-case scenario for rotor efficiency.

Real Power with Safety and Non-ideality Margins

A combined correction factor k is applied to account for a safety margin and for aerodynamic non-idealities (tip losses, induced losses, blade profile drag, etc.):

$$P_{\text{hover, real}} = k P_{\text{hover, ideal}} \quad (\text{A.8})$$

$$P_{\text{max, real}} = k P_{\text{max, ideal}} \quad (\text{A.9})$$

Energy Consumed in Hover and VTOL

The energy consumed in each phase is the product of the (real) power demand and the duration of that phase. Both the take-off and landing VTOL segments are accounted for (factor of 2), and similarly both hover segments:

$$E_{\text{hover}} = 2 P_{\text{hover, real}} \cdot t_{\text{hover}} \quad (\text{A.10})$$

$$E_{\text{VTOL}} = 2 P_{\text{max, real}} \cdot t_{\text{VTOL}} \quad (\text{A.11})$$

A.2.2 Cruise Phase

During level cruise, the required thrust equals the aerodynamic drag:

$$T_{\text{cruise}} = D = \frac{1}{2} \rho_{\text{cruise}} v_{\text{cruise}}^2 S C_D \quad (\text{A.12})$$

where ρ_{cruise} is the air density at cruise altitude, v_{cruise} is the true airspeed, S is the reference wing area, and C_D is the drag coefficient. The shaft power delivered to the propulsion system, accounting for the propulsive efficiency η_{prop} , is:

$$P_{\text{cruise}} = \frac{D v_{\text{cruise}}}{\eta_{\text{prop}}} \quad (\text{A.13})$$

The energy consumed over the cruise segment of duration t_{cruise} is:

$$E_{\text{cruise}} = P_{\text{cruise}} \cdot t_{\text{cruise}} \quad (\text{A.14})$$

A.2.3 Climb Phase

During the climb segment, power must overcome both aerodynamic drag and the rate of change of gravitational potential energy. Denoting the climb angle as γ and the climb true airspeed as v_{climb} , the gravitational and drag power components are:

$$P_{\text{grav}} = m g v_{\text{climb}} \sin \gamma \quad (\text{A.15})$$

$$P_{\text{drag}} = \frac{1}{2} \rho_{\text{climb}} v_{\text{climb}}^3 S C_D \quad (\text{A.16})$$

where ρ_{climb} is the average air density over the climb segment. The total mechanical power required, corrected for propulsive efficiency, is:

$$P_{\text{climb}} = \frac{P_{\text{grav}} + P_{\text{drag}}}{\eta_{\text{prop}}} \quad (\text{A.17})$$

The energy consumed during the climb phase of duration t_{climb} is:

$$E_{\text{climb}} = P_{\text{climb}} \cdot t_{\text{climb}} \quad (\text{A.18})$$

A.2.4 Descent Phase with Regenerative Energy Recovery

During a powered glide descent at angle γ and airspeed v_{desc} , gravitational power exceeds drag power, yielding a net surplus that can be partially recovered through regenerative braking:

$$P_{\text{grav, desc}} = m g v_{\text{desc}} \sin \gamma \quad (\text{A.19})$$

$$P_{\text{drag, desc}} = \frac{1}{2} \rho_{\text{desc}} v_{\text{desc}}^3 S C_D \quad (\text{A.20})$$

The net power available for regeneration (a negative value, indicating energy returned to the system) is:

$$P_{\text{desc}} = P_{\text{drag, desc}} - P_{\text{grav, desc}} \quad (\text{A.21})$$

Applying the regeneration efficiency η_{reg} , the recovered energy over the descent segment of duration t_{desc} is:

$$E_{\text{reg, desc}} = \eta_{\text{reg}} \cdot P_{\text{desc}} \cdot t_{\text{desc}} \quad (\text{A.22})$$

Since $P_{\text{desc}} < 0$, $E_{\text{reg, desc}}$ is negative, representing energy fed back into the storage system.

A.2.5 Total Mission Energy

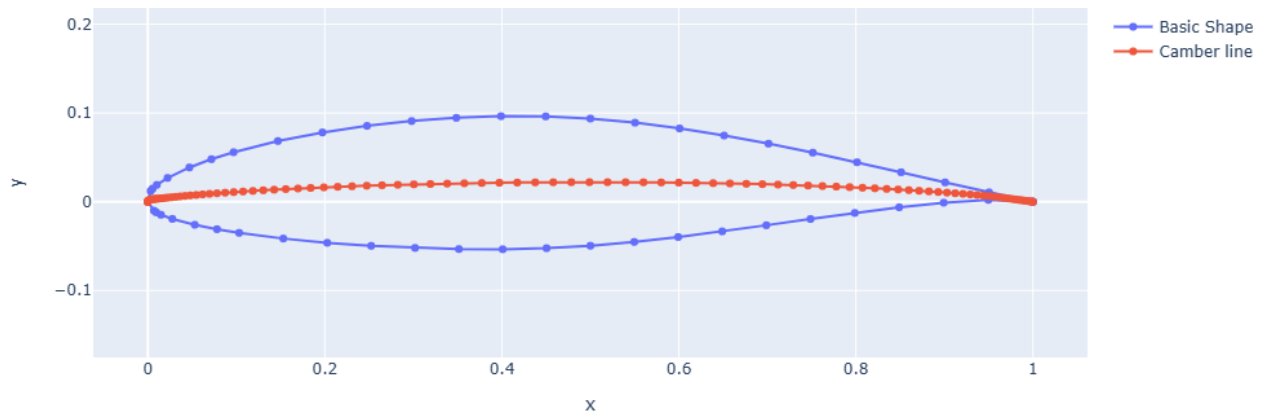
The total electrical energy demand for the complete mission profile is the algebraic sum of the contributions from each segment:

$$E_{\text{total}} = E_{\text{hover}} + E_{\text{VTOL}} + E_{\text{climb}} + E_{\text{cruise}} + E_{\text{reg, desc}} \quad (\text{A.23})$$

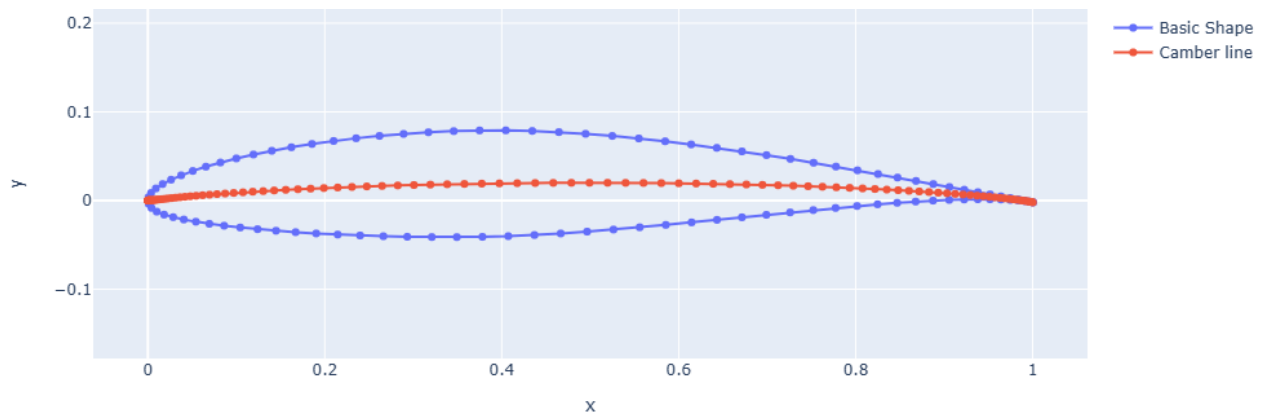
This total constitutes the minimum energy capacity that the hybrid-electric energy storage system must supply, and serves as the primary sizing criterion for the battery and/or fuel-cell components of the hybrid subsystem.

A.3 Profiles Considered for the Airfoil Selection Study

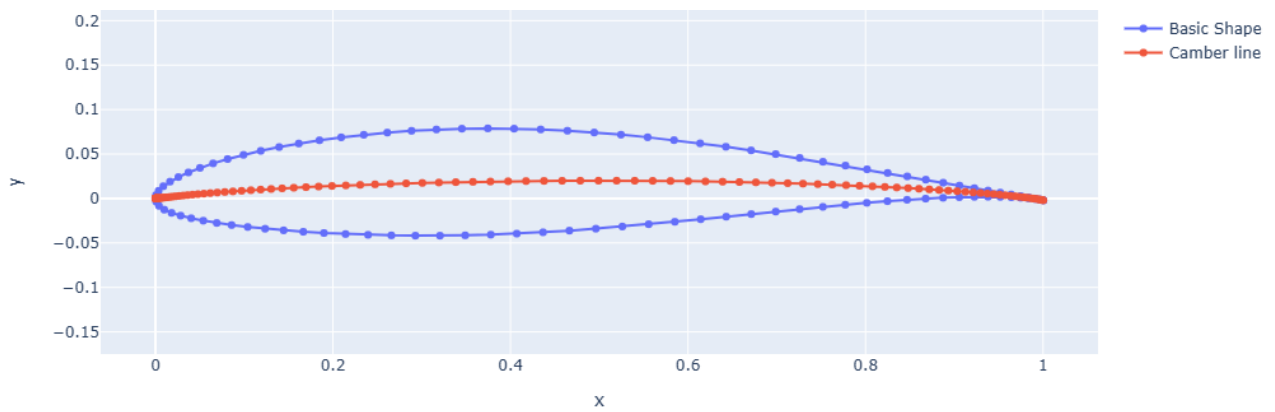
A.3.1 NACA 65₂-415



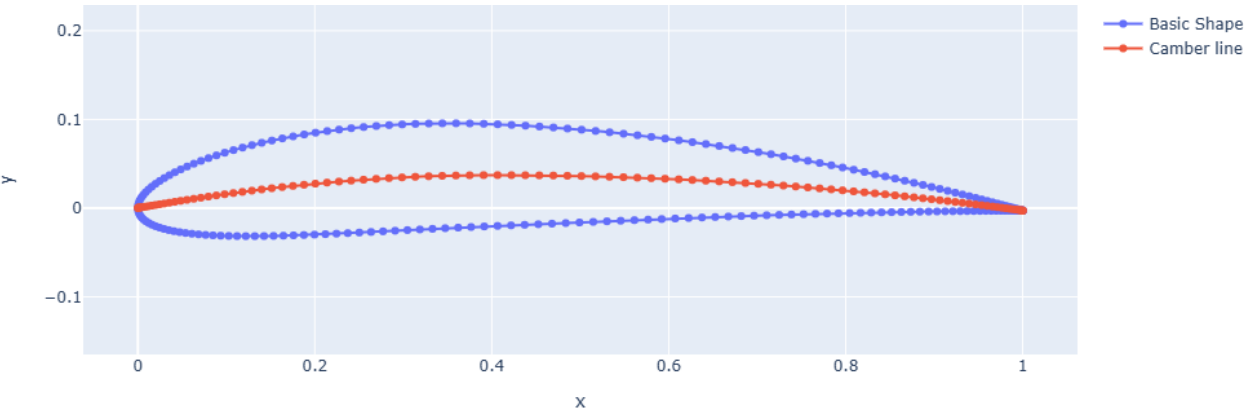
A.3.2 NACA 64₁-412



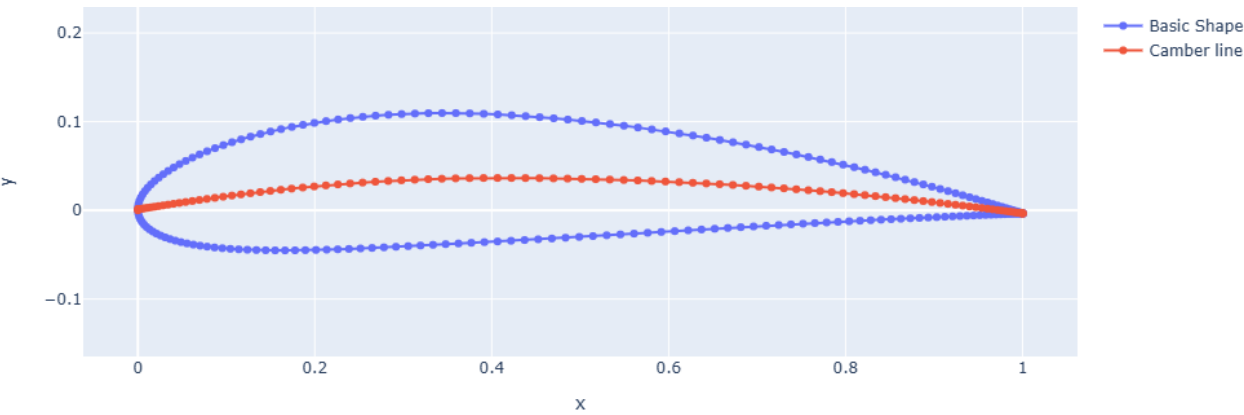
A.3.3 NACA 63₁-412



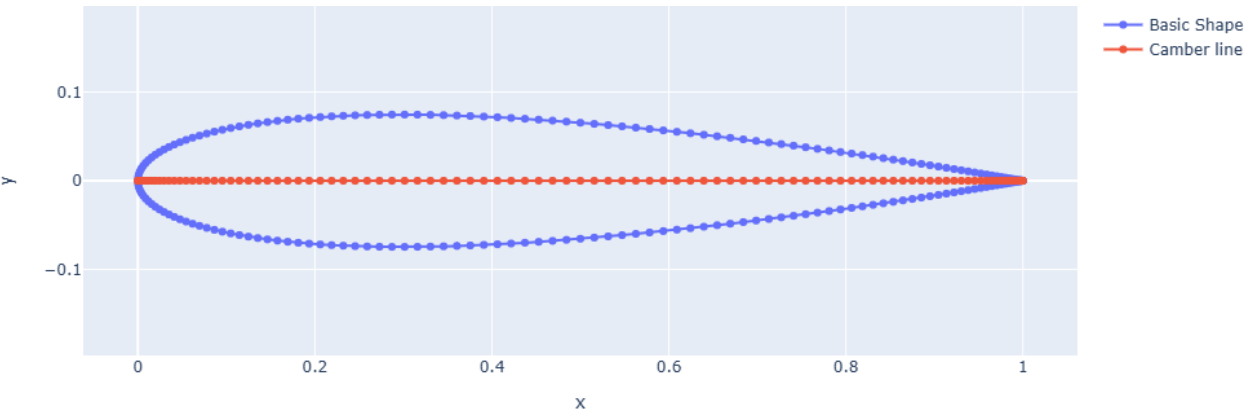
A.3.4 NACA 4412



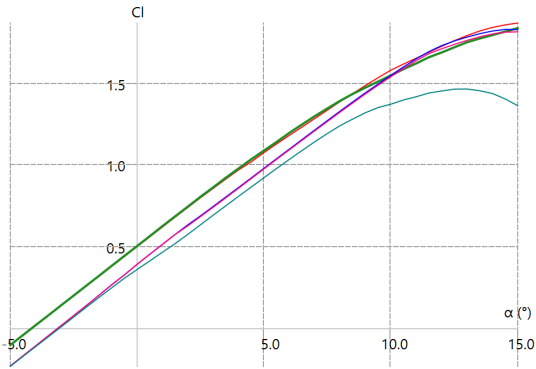
A.3.5 NACA 4415



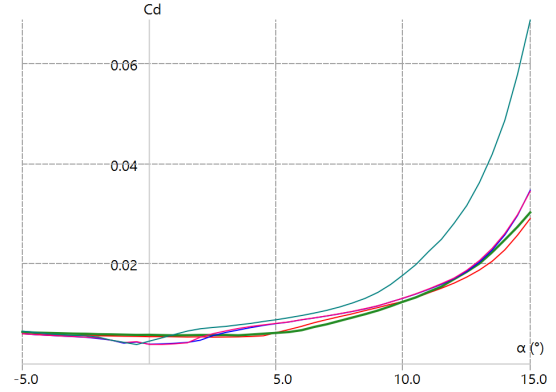
A.4 Profile Selected for the Canard and Tail (NACA 0015)



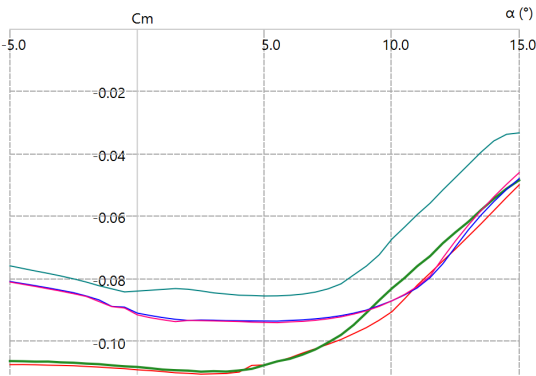
A.5 2D Aerodynamic Polars for Candidate Airfoils



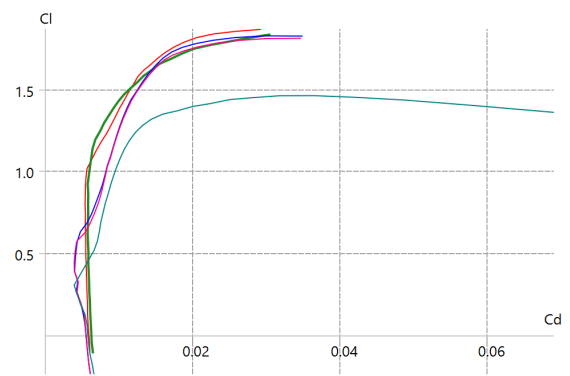
(a) C_l vs. α



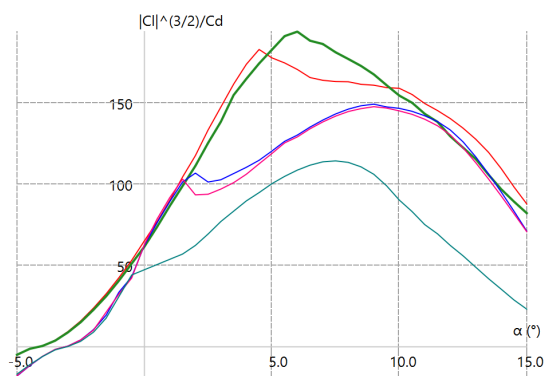
(b) C_d vs. α



(c) C_m vs. α



(d) C_l vs. C_d

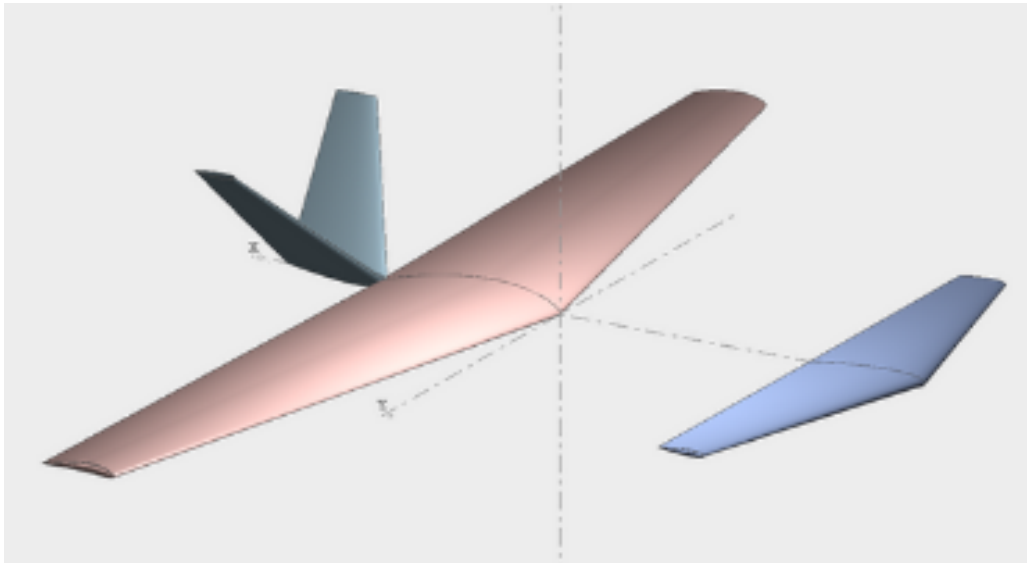


(e) $|C_l|^{3/2}$ vs. α

NACA 4412 (Red), 4415 (Green), 63₁-412 (Blue), 64₁-412 (Pink), and 65₂-415 (Teal).

$$\alpha \in [-5^\circ, 15^\circ]$$

A.6 XFLR5 Model



A.7 SPL - MATLAB Code

```
1 clear; clc; close all;
2 %% Noise Emissions - SPL Calculations
3 %% Input Parameters
4 m = 1;
5 NB = 5;
6 Omega = 293.2; % rad/s (Max Load Angular Speed)
7 a = 343; % m/s
8 rho = 1; % kg/m^3
9 c = 0.185; % m
10 t_max = 0.02; % m
11 T = 4735; % N (from estimation)
12 Q = 750; % N.m (between 500 (continuous) and 1000 (max))
13 r_e = 0.8; % m
14 p_ref = 20e-6; % Pa
15 NR = 6;
16
17 % Order of the Bessel function (n = m * NB)
18 bessel_order = m * NB;
19
20 %% Loop Setup for Different Distances and Angles
21 distances = 100:50:2500; % Vector of distances (m)
22 thetas_deg = [30, 45, 60, 90, 120, 135, 150,]; % Angles in degrees
23 thetas_rad = deg2rad(thetas_deg);
24
25 p_m_L_mat = zeros(length(distances), length(thetas_rad));
26 p_m_T_mat = zeros(length(distances), length(thetas_rad));
27 SPL_mat = zeros(length(distances), length(thetas_rad));
28
29 %% Nested Loop Calculation
30 for j = 1:length(thetas_rad)
31     theta = thetas_rad(j);
32
33     % Bessel calculations
34     bessel_arg = ((m * NB * Omega) / a) * r_e * sin(theta);
35     J_val = besselj(bessel_order, bessel_arg);
36
37     % Loading Noise term 2 only depends on theta
38     term2_L = T * cos(theta) - Q * (a / (Omega * r_e^2));
39     numerator_T = -rho * (m * NB * Omega)^2 * NB * c * t_max * r_e * J_val;
40
41     for i = 1:length(distances)
42         delta_S = distances(i);
```

```

43
44     % Loading Noise Calculation
45     term1_L = (m * NB * Omega) / (2 * sqrt(2) * pi * a * delta_S);
46     p_m_L_mat(i, j) = term1_L * term2_L * J_val;
47
48     % Thickness Noise Calculation
49     denominator_T = 3 * sqrt(2) * pi * delta_S;
50     p_m_T_mat(i, j) = numerator_T / denominator_T;
51
52     % Sound Pressure Level (SPL) Calculation
53     p_squared_sum = p_m_L_mat(i, j)^2 + p_m_T_mat(i, j)^2;
54     SPL_mat(i, j) = 10 * log10( NR * (p_squared_sum / p_ref^2) );
55 end
56 end
57
58 %% Display Results for specific points (Optional example for the first angle)
59 fprintf('Results for Angle: %d degrees\n', thetas_deg(end));
60 fprintf('Distance (m) | SPL (dB)\n');
61 fprintf('-----\n');
62 fprintf('%12.1f | %6.2f\n', distances(1), SPL_mat(1, 1));
63 fprintf('%12.1f | %6.2f\n', distances(round(end/2)), SPL_mat(round(end/2), 1));
64 fprintf('%12.1f | %6.2f\n', distances(end), SPL_mat(end, 1));
65
66 %% Results Plot
67 figure;
68 hold on;
69
70 for j = 1:length(thetas_deg)
71     plot(distances, SPL_mat(:, j), 'LineWidth', 1.5, ...
72          'DisplayName', sprintf('\theta = %d^\circ', thetas_deg(j)));
73 end
74
75 grid on;
76 xlabel('Distance (m)');
77 ylabel('Overall Sound Pressure Level (dB)');
78 title('SPL during Cruise Flight vs Distance from Source');
79 legend('show', 'Location', 'best');
80 hold off;
81
82 %% SPL Vortex
83 clear; clc;
84
85 T = 4735;
86 Omega = 293.2;

```

```

87 NB = 5;
88 c = 0.185;
89 NR = 6;
90 rho = 1.225;
91
92 R_rotor = 1.0;
93 A_rotor = pi * R_rotor^2;
94 V_tip = Omega * R_rotor;
95 sigma = (NB * c) / (2 * pi * R_rotor);
96 DL = T / A_rotor;
97 k2 = 1.206e-2;
98 k2_metric = k2 * (3.28084)^3;
99 vortex_load_root = sqrt(NR * (T / sigma)) * DL;
100
101 delta_S = 100:50:2500;
102 SPL_vortex = zeros(length(delta_S), 1);
103
104 for i = 1:length(delta_S)
105     vortex_core_ratio = V_tip / (rho * delta_S(i));
106     SPL_vortex(i) = 20 * log10(k2_metric * vortex_core_ratio * vortex_load_root);
107 end
108
109 figure;
110 plot(delta_S, SPL_vortex, 'LineWidth', 1.5);
111 xlabel('Distance (m)');
112 ylabel('Overall Sound Pressure Level (dB)');
113 title('SPL regarding vortex noise vs Distance from Source');
114 grid on;
115
116 Cl_bar = (6 * T) / (rho * V_tip^2 * NB * c * R_rotor);
117 fprintf('Cl_bar: %f\n', Cl_bar);

```

A.8 Costs Analysis - MATLAB Code

```
1 clear; clc; close all;
2
3 %% Parameters
4 W_s = 3815.61; % lbs
5 V_max = 165.23; % kts
6 N_t = 2; % test/RDT&E units
7 profit = 1.15;
8 y = 2026; % year
9
10 cessna_206_turbo_stationair_hd = 795000; % $
11 pilatus_pc_12_ngx = 6772500; % $
12 piper_m350 = 1217500; % $
13 piper_m600 = 3207000; % $
14 airbus_ach130 = 6200000; % $
15 archer_midnight = 5000000; % $
16 lilium_jet = 10000000; % $
17
18 %% Hourly Rates ($/hr)
19 CR_E = -5058 + 2.576 * y; % Engineering
20 CR_ML = -4552 + 2.316 * y; % Manufacturing Labor
21 CR_T = -5666 + 2.883 * y; % Tooling
22 CR_QC = -5112 + 2.600 * y; % Quality Control
23
24 %% FUNCTION DEFINITIONS
25
26 % Airframe Engineering Hours (RDT&E and A)
27 function C_E = airframe_eng(N, W_s, V_max, CR_E)
28     H_E = 4.860 * W_s^0.777 * V_max^0.894 * N^0.163; % hr
29     C_E = H_E * CR_E; % $
30 end
31
32 % Development Support Cost (RDT&E)
33 function C_D = dev_support(W_s, V_max)
34     C_D = 45.42 * W_s^0.630 * V_max^1.300; % $
35 end
36
37 % Engine and Avionics Cost (RDT&E and A)
38 function C_EN = engine_avionics(N)
39     C_EN = 20000*6*N + 43680*N + 12253.86*N; % Engine + Full Cell + Battery, $
40 end
41
42 % Manufacturing Labor Cost (RDT&E)
```

```

43 function [C_ML, H_ML] = manufac_labor(N, W_s, V_max, CR_ML)
44     H_ML = 7.370 * W_s^0.820 * V_max^0.484 * N^0.641; % hr
45     C_ML = H_ML * CR_ML; % $
46 end
47
48 % Manufacturing Materials Cost (RDT&E and A)
49 function C_MM = manufac_mat(N, W_s, V_max)
50     C_MM = 11.00 * W_s^0.921 * V_max^0.621 * N^0.799; % $
51 end
52
53 % Tooling Cost (RDT&E and A)
54 function C_T = tooling(N, W_s, V_max, CR_T)
55     H_T = 5.99 * W_s^0.777 * V_max^0.696 * N^0.263; % hr
56     C_T = H_T * CR_T; % $
57 end
58
59 % Quality Control Cost (RDT&E and A)
60 function C_QC = qual_control(H_ML, CR_QC)
61     H_QC = 0.13 * H_ML; % hr
62     C_QC = H_QC * CR_QC; % $
63 end
64
65 % Flight Test Cost (RDT&E)
66 function C_FT = flight_test(N, W_s, V_max)
67     C_FT = 1243.03 * W_s^0.325 * V_max^0.822 * N^1.210; % $
68 end
69
70 % RDT&E Total
71 function C_RDTE = rdte(N, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC)
72     [C_ML, H_ML] = manufac_labor(N, W_s, V_max, CR_ML);
73     C_RDTE = airframe_eng(N, W_s, V_max, CR_E) + dev_support(W_s, V_max) +
        engine_avionics(N) + C_ML + manufac_mat(N, W_s, V_max) + tooling(N, W_s,
        V_max, CR_T) + qual_control(H_ML, CR_QC) + flight_test(N, W_s, V_max);
74 end
75
76 % Acquisition Total
77 function C_A = acquisition(N, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC)
78     [C_ML, H_ML] = manufac_labor(N, W_s, V_max, CR_ML);
79     C_A = airframe_eng(N, W_s, V_max, CR_E) + engine_avionics(N) + C_ML + manufac_mat
        (N, W_s, V_max) + tooling(N, W_s, V_max, CR_T) + qual_control(H_ML, CR_QC);
80 end
81
82 % Program Cost with Profit
83 function C_P = prof(N_T, N_P, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC, profit)

```

```

84     C_P = profit * (rdte(N_T, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC) + acquisition(N_P
      , W_s, V_max, CR_E, CR_ML, CR_T, CR_QC));
85 end
86
87 % Unit Cost
88 function C_U = unit(N_T, N_P, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC, profit)
89     C_U = prof(N_T, N_P, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC, profit) / N_P;
90 end
91
92 %% PLOTS
93
94 N_P_range = 1:100000;
95
96 C_U_range = arrayfun(@(N_P) unit(N_T, N_P, W_s, V_max, CR_E, CR_ML, CR_T, CR_QC,
      profit), N_P_range);
97
98 figure;
99 loglog(N_P_range, C_U_range / 1e6, 'k-', 'LineWidth', 2, 'DisplayName', 'Island_
      Hopper');
100 hold on;
101 names = {'Cessna_206_Turbo_Stationair_HD', 'Piper_M350', 'Piper_M600', 'Airbus_
      ACH130', 'Pilatus_PC-12_NGX', 'Archer_Midnight', 'Lilium_Jet'};
102 prices = [cessna_206_turbo_stationair_hd, piper_m350, piper_m600, airbus_ach130,
      pilatus_pc_12_ngx, archer_midnight, lilium_jet];
103 colors = {[0.90 0.62 0.00], [0.00 0.62 0.45], [0.48 0.05 0.12], [0.80 0.47 0.65],
      [0.00 0.18 0.65], [0.80 0.00 0.40], [0.00 0.61 0.62]};
104 [prices, sort_idx] = sort(prices, 'descend');
105 names = names(sort_idx);
106 for i = 1:length(prices)
107     idx = find(C_U_range <= prices(i), 1, 'first');
108     if ~isempty(idx)
109         label = sprintf('%-30s->N_P=%4dunits', names{i}, N_P_range(idx));
110     else
111         label = sprintf('%-30s->out of range', names{i});
112     end
113     yline(prices(i)/1e6, '--', 'Color', colors{i}, 'LineWidth', 1, 'DisplayName',
      label);
114     if ~isempty(idx)
115         loglog(N_P_range(idx), C_U_range(idx)/1e6, 'o', 'Color', colors{i}, '
      MarkerSize', 6, 'MarkerFaceColor', colors{i}, 'HandleVisibility', 'off');
116     end
117 end
118 lg = legend('Location', 'northeast');
119 lg.FontName = 'LucidaConsole';

```

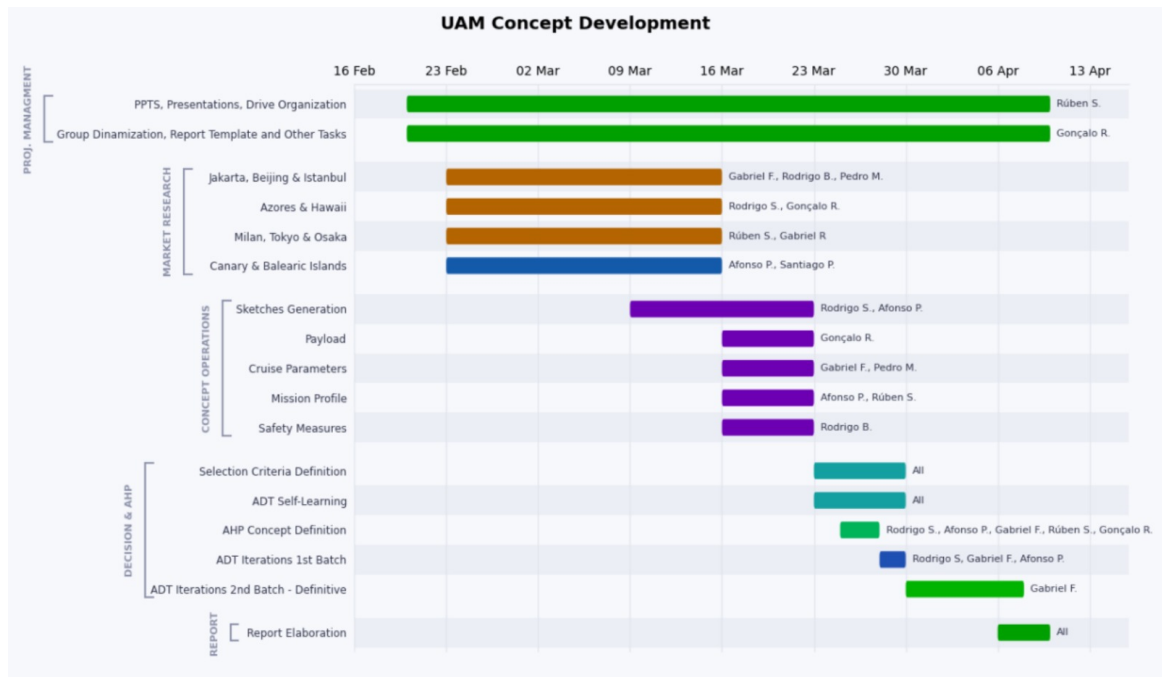
```

120 lg.FontSize = 10;
121 hold off;
122 xlabel('Number of Production Units (N_P)');
123 ylabel('Unit Cost ($ millions)');
124 title('Unit Cost vs. Production Quantity');
125 theme 'light';
126 grid on;
127
128 figure;
129 N_P_fixed = 10000;
130 ws_vec = W_s * linspace(0.5, 1.5, 40);
131 v_vec = V_max * linspace(0.5, 1.5, 40);
132 [WS, VV] = meshgrid(ws_vec, v_vec);
133 CU = zeros(size(WS));
134 for i = 1:numel(WS)
135     CU(i) = unit(N_t, N_P_fixed, WS(i), VV(i), CR_E, CR_ML, CR_T, CR_QC, profit);
136 end
137 surf(WS, VV, CU / 1e6, 'EdgeColor','none')
138 colormap turbo;
139 colorbar;
140 xlabel('Structural Weight W_s (lbs)');
141 ylabel('Max Velocity V_{max} (kts)');
142 zlabel('Unit Cost ($ millions)');
143 title(sprintf('Unit Cost Surface (N_P = %d)', N_P_fixed));
144 view(300, 45);
145 theme light;
146 grid on;

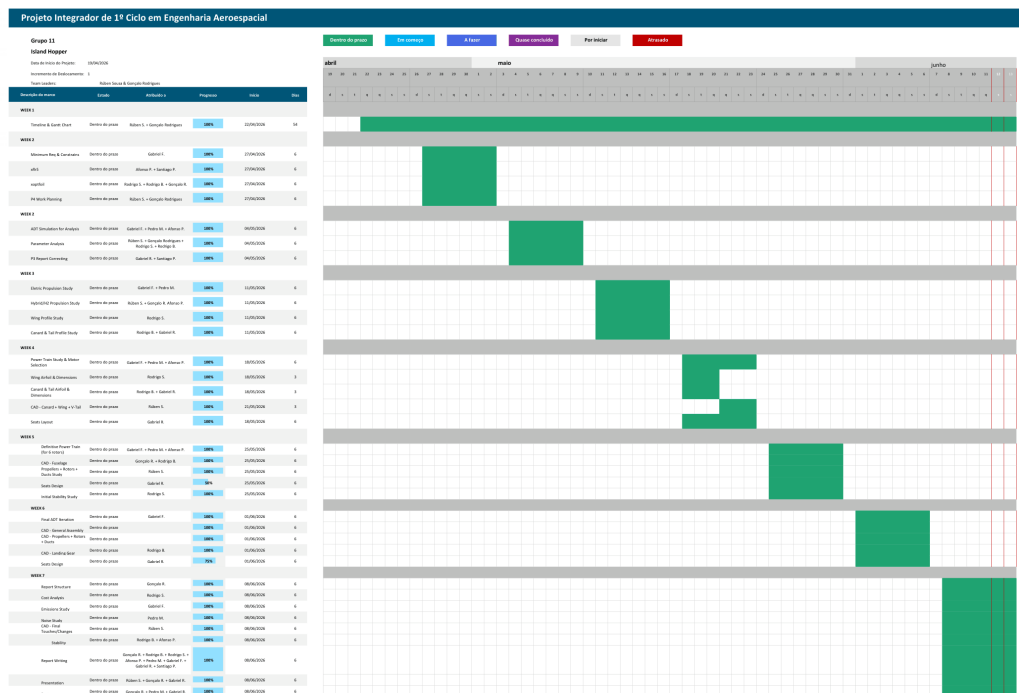
```


A.9 Gantt Chart

A.9.1 Gantt Chart - P3



A.9.2 Gantt Chart - P4



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